

The Dynamic Efficiency of Policy Uncertainty: Evidence from the Wind Industry

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Abstract

This paper examines the dynamic efficiency of policy uncertainty in the U.S. wind energy industry. Policy expiration induced uncertainty for wind farm investors and expedited investment. I document timing misalignment among wind farm investment, turbine technology advancement, and evolving demand for wind energy. I then develop a dynamic entry model under policy uncertainty that incorporates long-term contract negotiation and buyer choice. Model estimates reveal that a policy lapse reduced the perceived likelihood of renewal to 39%. Eliminating policy uncertainty increases the social surplus by 3.2 billion dollars and could reduce fiscal expenditure without compromising social welfare.

Keywords: Wind energy, policy uncertainty, dynamic model, firm belief

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1 Introduction

Over the past two decades, the U.S. renewable electricity market has been transformed, and wind power has been at the center of that transformation. Wind energy grew from less than 1% of U.S. electricity generation in 2000 to the largest renewable electricity source by 2019, and supplied about 10% of U.S. generation in 2024. This expansion coincided with substantial technological progress: turbine prices fell sharply, while taller towers and longer rotor blades increased the productivity of new installations. As a wind farm requires a large irreversible upfront investment, takes years to develop, and typically operates for over 20 years, waiting is valuable due to future cheaper and more productive turbines.

Yet the growth of wind capacity was anything but smooth. The industry has been heavily supported by federal tax incentives, known as the Production Tax Credit (PTC). The PTC provides qualified projects with inflation-adjusted tax credits for each unit of output over a ten-year period. During the key years of the sample, qualification was tied to policy expiration deadlines. Although the PTC has been in place since 1992, Congress implemented it through a sequence of short policy windows with set expiration dates.¹ Moreover, occasional lapses between expiration and renewal exacerbated uncertainty about future eligibility. Since waiting could mean forgoing ten years of tax credits, investors had an incentive to expedite investment before expiration. This incentive led to substantial bunching, most notably in December 2012, just before the PTC expired and briefly lapsed before renewal.

This paper studies how uncertainty about policy renewal affects investment timing and welfare in the U.S. wind industry, where investors made large irreversible investments while turbine costs fell, turbine productivity rose, and the main federal subsidy repeatedly approached expiration. The welfare effect of this uncertainty is theoretically ambiguous. Earlier entry brings wind generation online sooner and accelerates environmental benefits. However, earlier entry may also be inefficient because projects use older and more expensive technology. Moreover, the demand for wind power is itself dynamic: utilities procure wind capacity to satisfy state Renewable Portfolio Standards, so their compliance demand declines as procurement accumulates. Consequently, policy uncertainty can misalign investment with technology and buyer demand. The main contribution of this paper is a

¹As noted in [Bistline, Mehrotra, and Wolfram \(2023\)](#), the continual expiration and extension of the PTC in the wind industry created an “on-again/off-again” status of the policy and resulted in a boom-bust cycle of wind development. The industry calls for “strong long-term policy support” according to the [Union of Concerned Scientists \(February 9, 2015\)](#).

quantitative empirical study of how policy uncertainty and technological progress jointly shape investment timing and welfare in the U.S. wind industry.

I first provide empirical evidence for this welfare trade-off. I assemble project-level data on investment, production, turbine technology, and long-term contracts from 2003 to 2018 and document three key stylized facts. First, wind farm investment bunched at PTC expiration dates, especially in 2012, when the credit lapsed before being renewed retroactively. Second, this bunching was misaligned with turbine technology. Projects entering in earlier deadline years used less productive and more expensive turbines than projects entering later in the sample. Third, the timing distortion affected the allocation of projects across buyers. Utilities are major buyers of wind capacity through long-term contracts, but their renewable demand shrank as they procured more wind energy. Policy uncertainty therefore moved older-technology projects into periods when utility demand was high, while later projects with better technology faced buyers with lower wind demand.

Motivated by these facts, I develop a dynamic entry model for wind farm investors.² Potential entrants are projects that have obtained interconnection agreements and have not yet entered. Each year, a potential entrant decides whether to enter or wait. Entry secures current subsidy eligibility, while waiting preserves the option to adopt future turbines with higher productivity and lower prices. The continuation value depends on the perceived probability of PTC renewal. A lower renewal belief reduces the value of waiting and raises the probability of a deadline-year rush.

The model also links the dynamic entry decision to the institutional details of wind contracting. Conditional on entry, wind farms choose between two sales channels. They may sell capacity to utilities through long-term Power Purchase Agreements, in which prices and capacities are determined by Nash bargaining. Alternatively, they may sell to non-utility buyers, including corporations, or use merchant hedge contracts. These contracting components identify the flow profits entering the dynamic problem, while the dynamic states, including policy status, turbine productivity, turbine prices, renewable credit prices, and utility demand, drive the optimal entry timing decision.

The main identification challenge is to separate wind investors' beliefs about policy renewal from unobserved entry costs. The identification strategy uses the temporal structure of the PTC. After the 2013 safe harbor rule, eligibility no longer depended as tightly on

²“Investment” and “entry” are used interchangeably and defined as the decision to start building a wind farm. During the sample period, the frequency of wind farm retrofitting was low.

being placed in service by the expiration date. After the 2015 Consolidated Appropriations Act, the subsidy path was announced through 2019. I use this later period of more certain policy renewal to identify the entry cost distribution. Conditional on entry costs, flow profits, and the transition of technology and demand, the excess bunching in earlier deadline years identifies investors' perceived renewal probabilities. The argument follows from the monotonicity of the dynamic problem: lower renewal beliefs reduce the option value of waiting, and thus increase entry rates in deadline years.

The estimates imply that the PTC was central to wind farm profitability. Utilities capture about 60% of the gains from trade in bargaining, and the subsidy is largely retained by wind farms. Without the PTC, 18% of wind farms would fail to reach an agreement or earn non-positive profits; among projects that remain profitable, average profits would be 61% lower. This result highlights the potential cost of missing deadlines and losing the qualification of the PTC. It also explains the rushed entry when there is a perception of a lower probability for the PTC renewal. Moreover, the estimated demand elasticity for non-utility buyers is around -1.8, and the probability of choosing a utility buyer decreases with geographic distance. Most importantly, the dynamic estimates imply substantial uncertainty for the 2011 cohort: the average perceived probability of PTC renewal is about 39%, consistent with the investment spike before the 2012 expiration.

The counterfactual analysis evaluates investment and welfare under alternative policy designs. The main counterfactual removes renewal uncertainty by setting the perceived probability of PTC renewal to one. This reduces the number of new wind projects in 2011 by 67% and shifts entry to later years, increasing the number of new projects between 2012 and 2018 by 24%. Delayed projects postpone entry by 4.1 years. The key welfare trade-off of policy uncertainty stands out in the counterfactual analysis: although environmental benefits arrive later after removing renewal uncertainty, the improved investment timing aligns better with technology and demand. Overall, removing policy renewal uncertainty raises the social surplus from wind investment between 2009 and 2018 by 3.2 billion dollars, or 9.8% from the baseline. Thus, in this market, the efficiency gains from waiting for better and cheaper turbines outweigh the loss from delaying renewable generation.

Two further counterfactual analyses translate the result into lessons for subsidy design. First, under full renewal certainty, the subsidy level could be reduced by \$2/MWh, around 9%, without compromising social welfare. This comparison measures the fiscal cost of pol-

icy uncertainty. Second, resolving uncertainty before investment decisions are made recovers one quarter of the welfare gain from fully eliminating renewal uncertainty, holding the expected subsidy fixed. This finding shows that the timing of policy announcements matters alongside the probability of renewal. Overall, longer authorization windows, renewal decisions announced before expiration, and eligibility rules that weaken hard deadline incentives can improve the efficiency of the PTC.

The mechanism studied here is relevant beyond wind. Temporary policy is common: many subsidies and tax provisions are enacted with an expiration date and then renewed, lapsed, or reinstated, often at the last minute. Whenever the investment a policy supports is irreversible and the return to waiting is positive, an expiration date with uncertain renewal pulls investment forward and bunches it before the deadline, at the cost of a worse match between when projects are built and when building them is most valuable. The welfare cost of short policy commitment is therefore largest where the return to waiting is high due to falling costs, improving technology, or evolving demand. Solar, energy storage, and electric vehicles are the clearest current cases, since each involves a trade-off between accelerating deployment today and adopting a cheaper, better technology later.

This paper contributes to three strands of literature. The first is the empirical literature on renewable energy markets and clean energy subsidies. Recent work studies the value of wind energy, upstream innovation in turbines, and the design of renewable subsidies, including their timing, generosity, and choice of instrument (Cullen, 2013; Novan, 2015; Johnston, 2019; Aldy et al., 2023; Petersen et al., 2024; Ricks and Kay, 2025; Covert and Sweeney, 2026; Elliott, 2026).³ This paper develops a new structural model of the U.S. wind industry that connects dynamic entry, turbine technology, long-term contracting, and buyer choice, and it identifies the continuity of a subsidy, as distinct from its level, as a design margin with first-order welfare consequences.

The second strand studies policy uncertainty and the option value of waiting (Dixit and Pindyck, 1994; Collard-Wexler, 2013; Kellogg, 2014; Baker et al., 2016; Handley and Limão, 2017; Dorsey, 2019; Gowrisankaran et al., 2025). Closest to this paper, Gowrisankaran et al. (2025) measure the welfare cost of uncertainty about the Mercury and Air Toxics

³Related work on renewable energy markets also examines transmission congestion (Fell et al., 2021), contract risk (Fabra and Llobet, 2025), risk sharing (Hara, 2023), interconnection (Gonzales et al., 2023; Johnston et al., 2026), time discounting (De Groote and Verboven, 2019), and dynamic subsidies (Langer and Lemoine, 2022).

Standards for coal power plants. I study the uncertainty stemmed from the renewal of a subsidy with a fixed expiration date, in an industry where waiting brings cheaper and more productive technology, and I show that the welfare cost runs through the misalignment of investment with technology and demand.⁴ The third strand is empirical dynamic models with firm beliefs ([Doraszelki, Lewis, and Pakes, 2018](#); [Jeon, 2022](#)), to which the paper contributes a strategy for recovering cohort-specific beliefs from the bunching of investment at policy deadlines.

The remainder of the paper proceeds as follows. Section 2 describes the institutional background: the U.S. wind industry and the federal and state subsidies that support it. Section 3 presents descriptive evidence on the timing of investment and its misalignment with technological advancement and demand. Section 4 develops the dynamic entry model and the static contracting components. Section 5 discusses identification and estimation. Section 6 reports the estimation results. Section 7 presents the counterfactual analysis and subsidy-design implications, and Section 8 concludes.

2 Institutional Background

2.1 The U.S. Wind Industry

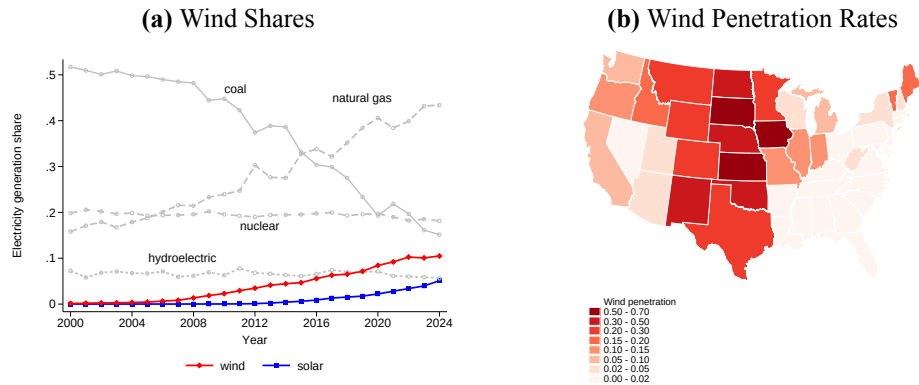
Wind energy has become America's largest renewable energy source, providing 10% of total electricity generation in 2024. As shown in Panel (a) of Figure 1, wind energy grew from a very marginal share of total electricity generation in 2000 to the fourth most important energy source in the U.S. in 2024. Geographically, wind energy is concentrated in Texas, the Midwest, and the Plains, as shown in Panel (b). Texas enjoyed the largest wind generation, taking up 28% of total wind power generation nationwide in 2024.

A wind farm requires enormous upfront investment. For example, investors had to spend more than 100 million dollars to construct an average-sized wind farm in 2019 just for turbine procurement, not to mention the additional costs for turbine transportation, wind farm construction, land lease, permits, and grid access.⁵ It also takes a long time to plan and construct a wind farm as summarized in Supplemental Appendix Figure SA7. First,

⁴Uncertainty about the Production Tax Credit itself has been documented by [Barradale \(2010\)](#) and [Johnston and Yang \(2019\)](#).

⁵In 2019, an average wind farm had 65 turbines with an average turbine nameplate capacity of 2,550 kW. The market price of wind turbines was \$700/kW, and thus the turbine cost alone would be \$116 million.

Figure 1: Shares and Penetration Rates of Wind Energy



Notes: This figure shows electricity generation shares and penetration rates of wind energy. Panel (a) presents shares of electricity generation between 2000 and 2024 by different energy sources based on data from EIA-906, EIA-920, and EIA-923. Red diamonds denote shares of electricity generation from wind farms, while blue squares denote shares using solar energy. Panel (b) presents wind penetration rates in 2024 for each contiguous state. Wind penetration rate is defined as the share of electricity produced by wind compared to total generation.

investors need to sign a land lease, acquire government permits, and apply for an interconnection agreement after lengthy waiting in the interconnection queue. Next, investors negotiate with upstream turbine manufacturers for equipment procurement, negotiate with utilities or corporations to sell outputs, and seek financing from banks. Finally, with contracts secured, investors can start the construction process. A typical wind development process takes three to four years in total, and the construction process alone takes six to nine months. Once a wind farm starts operation, it will typically be in service for 20-30 years. Large sunk costs, coupled with a long time to build, highlight the importance of dynamic incentives in wind investment.

There are two types of investors in the market: independent power producers and utilities, and they together own over 99% of wind energy. On the one hand, independent power producers own approximately 80% of total capacity. They typically sign a long-term wind procurement contract with utilities or non-utility buyers (e.g., corporations). These contracts are known as the Power Purchase Agreements (PPAs). Negotiating and signing a PPA is critical for project financing as it secures a long-term revenue stream. A typical PPA specifies the procurement price, procured capacity, duration of the agreement, and other details. Moreover, independent power producers could also sign merchant hedge contracts.⁶

⁶One of the most common forms of merchant hedge contracts in ERCOT is a physical fixed-volume hedge. Under this contract, a wind project owner sells its actual energy generated at the floating price at

As shown in Appendix Figure A1, utility PPAs are the most common channel to sell wind power, while more non-utility PPAs emerged after 2015.

On the other hand, utilities directly own the remaining 20% of wind capacity. Since they can either own wind farms or procure wind energy from independent power producers, endogenizing wind capacity under direct utility ownership requires modeling their make-or-buy choices and is beyond the scope of this paper. Instead, this paper focuses on wind farms invested by independent power producers due to their dominant market shares.

2.2 Federal and State Subsidies

The U.S. wind industry crucially relies on federal support, as well as numerous state-level policies. This section describes three sets of subsidies that shape the investment incentives modeled in this paper: the federal Production Tax Credit and its recurring expiration schedule, the Section 1603 Grant available between 2009 and 2012, and state-level policies led by the Renewable Portfolio Standards. The most important federal policy is the Production Tax Credit (PTC), which provided qualified wind farms with a 10-year inflation-adjusted tax credit for wind power generation and stood at \$24/MWh in 2018. Although the PTC has been in effect since 1992, PTC incentives were segmented into smaller policy windows with set expiration dates. Conditions to qualify for the PTC are tied to these expiration dates: before 2013, a wind farm was required to start operation before the policy expiration, while after 2013, a wind farm was required to demonstrate that five percent or more of its total investment cost has been incurred before the policy expiration, with a two-year safe harbor to start operation (extended to four years after 2016).⁷ As shown in Figure 2, the PTC was extended through successive acts during the sample period, including the American Recovery and Reinvestment Act (2009), the American Taxpayer Relief Act (2013), the Tax Increase Prevention Act (2014), and the Consolidated Appropriations Act (2015).

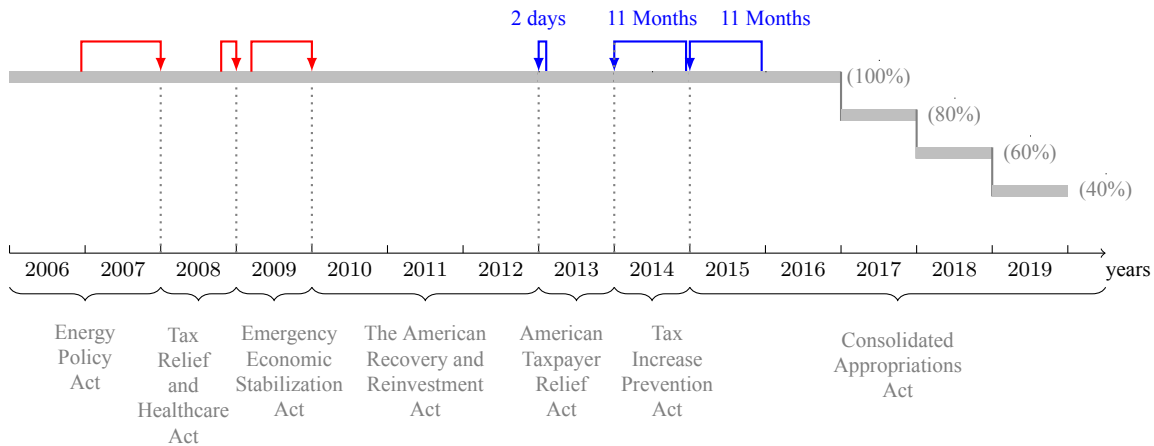
Since 2005, there have been seven different acts implementing the PTC sequentially, which segmented the policy into windows of one to five years. Before 2009, PTC renewals in the next act were announced several months before expiration. However, at the end of 2012, 2013, and 2014, PTC renewals were announced after the deadlines had passed. Although the lapse between policy expiration and renewal could be as short as two days at the

the node, and hedging counterparty pays the wind project owner for fixed signed energy amount at the price difference between pre-negotiated fixed price and the floating price at the node (Bartlett, 2019).

⁷A more recent change in the safe harbor rule can be referenced at the [National Law Review](#), July 8, 2021.

beginning of 2013, it still disturbed market incentives and created policy discontinuities. With a lack of government commitment, wind investors were faced with policy uncertainty before expiration about whether the PTC would be extended or not. The delayed policy action from Congress and political debates about renewable subsidies exacerbated the uncertainty in the market.⁸

Figure 2: Timeline of the Production Tax Credit



Notes: This figure shows the timing of the Production Tax Credit. Starting points of arrows indicate announcement time of policy renewal in the next act, while endpoints represent start time of the new act. There were 2-day, 11-month, and 11-month lapses between expiration of the previous act and announcement of the next act at the end of 2012, 2013, and 2014, respectively, though the policy was retroactive.

The 2011 Wind Technologies Market Report ([Wiser and Bolinger, 2012](#)), published by the Department of Energy in August 2012, suggested that investors were uncertain about the PTC renewal, and tended to rush into the market to qualify for the tax credit. According to the report, “...the wind energy sector is currently experiencing serious federal policy uncertainty, and therefore rushing to complete projects by the end of the year. Moreover, 2011 saw another year pass without any concrete Congressional action on what are seemingly the wind power industry’s two highest priorities – a longer-term extension of federal tax (or cash) incentives and passage of a federal renewable or clean energy portfolio standard...”

Concerns about the expired PTC were ultimately proven to be unnecessary, as it was extended again only two days after expiration through the American Taxpayer Relief Act

⁸For example, Republican U.S. presidential candidate Mitt Romney declared that he would let wind power tax credits expire (see [The Guardian, August 2, 2012](#)). The American Taxpayer Relief Act of 2012 was introduced in the House on July 24, 2012, as a partial resolution to the U.S. fiscal cliff. The passage of the bill involved days of negotiations between Senate leaders and the Obama administration (see [Star Tribune, January 2, 2013](#)).

with the tax credit applied retroactively. Similar events occurred again in 2014 and 2015. Although the latter two lapses were much longer, wind farms only needed to incur five percent of their total investment cost before deadlines to qualify for the PTC thanks to the safe harbor period. Starting in 2015, incentives provided by the PTC were stabilized, despite the decreasing amount of the tax credit.⁹

Along with the PTC, there was also the Section 1603 Grant, which provided an upfront investment subsidy covering 30 percent of total investment cost. Between 2009 and 2012, investors could choose either the PTC or the Section 1603 Grant. The Section 1603 Grant was announced to expire after 2012 and has been terminated ever since.¹⁰

Apart from federal policies, there are also various state-level policies. One important state-level policy is the Renewable Portfolio Standards (RPS). RPS stipulates the minimum share of electricity generation using qualified renewable energy for utilities. If utilities fail to satisfy the requirement, they have to buy renewable credits from the credit market. Otherwise, they can also sell credits for profit. RPS provides important incentives to utilities to procure wind energy.¹¹ States could also have corporate/sales tax incentives, property tax incentives, feed-in tariffs, bond/loan programs, and other industry recruitment policies for wind farms. As shown in Supplemental Appendix Figure SA8, states with RPS are also more likely to have other kinds of state-level incentives for wind energy.

3 Data and Descriptive Evidence

3.1 Data

I compile several data sets in the U.S. wind industry. The first two data sets come from the United States Wind Turbine Database (USWTDB) maintained by the United States Geological Survey (USGS) and the EIA-860 maintained by the Department of Energy's Energy Information Administration, respectively. These two data sets provide universal information about investment and characteristics of the utility-scale wind farms that were online

⁹The solar Investment Tax Credit (ITC) was extended in October 2008 under the Energy Improvement and Extension Act, preserving the 30% credit through 2016. As a result, solar developers faced no solar-specific federal deadline in 2011.

¹⁰Johnston (2019) and Aldy, Gerarden, and Sweeney (2023) study the selection and efficiency consequences of having both production tax credits and investment subsidies in the market.

¹¹Abito, Flores-Golfín, Hinchberger, van Benthem, and Vasey (2025) study the consequences of cross-state trading restrictions and state-specific interim annual targets under RPS.

between 2003 and 2019. USWTDB has more comprehensive coverage and detailed wind turbine characteristics, while EIA-860 also includes information about the owners of wind farms and rich data for other energy sources. Moreover, I supplement these two data sets with EIA-923, which covers the monthly electricity generation and enables me to measure production efficiency of wind projects.

Both USWTDB and EIA-860 record the month when a wind farm starts to supply electricity. However, as illustrated in Supplemental Appendix Figure SA7, there is a lag between finalizing investment decision and starting operation, including a construction period of six to nine months. I follow Johnston and Yang (2019) to use the information from the Federal Aviation Administration (FAA) Obstruction Evaluation/Airport Airspace Analysis (OE/AAA) database. The FAA data reports the scheduled dates of starting construction. I match the FAA data with EIA-860 and measure the time of investment as the time when a wind farm starts construction.¹²

The next is the Power Purchase Agreement (PPA) data from the American Clean Power Association (formerly American Wind Energy Association). The PPA data includes long-term contract information such as names of the wind farms and buyers, the amount of capacity, negotiated price, and contract duration.

Moreover, I obtain renewable credit price data from a financial service platform Marex. I also use retail electricity price data from EIA-861, agricultural land price data from the USDA National Agricultural Statistics Service, the annual turbine price and interconnection queue data from Lawrence Berkeley National Laboratory. The state-level policies including Renewable Portfolio Standards were hand-collected from the Database of State Incentives for Renewables & Efficiency (DSIRE).¹³

3.2 Stylized Facts

3.2.1 The timing of investment: bunching at policy deadlines

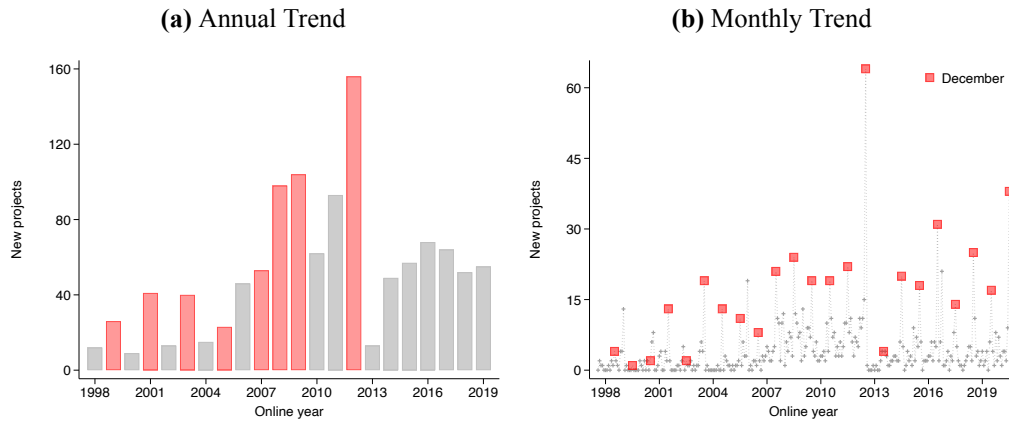
I first investigate the time trend of wind farm investment. Figure 3 presents the annual and monthly numbers of wind farms that are newly online. Significant bunching in wind farm

¹²FAA data started in 2008 and many projects did not report the scheduled time to begin construction. Around 42% of wind farms online between 2003 and 2018 from EIA-860 can be matched with the FAA data. For unmatched wind farms, I impute the scheduled construction start date by subtracting the average construction period for their commissioning year from their commissioning date.

¹³For more detailed data processing, please refer to the Supplemental Appendix Sections SA.1, 2, and 3.

investment occurred whenever the policy was scheduled to expire. A substantial number of wind farms started operation between 2008 and 2012, especially in 2012. Following the large investment spike in 2012, new investment dropped significantly in 2013. It was only after 2015 that the annual level of investment recovered, and the time trend stayed stable afterward. I plot the time trend of the total new wind capacity in Appendix Figure A2, and find a similar bunching pattern. Moreover, the sizes of wind farms were stable until 2012 and have shown an increasing trend over time.

Figure 3: Time Trend for Wind Projects Newly Online



Notes: This figure shows the annual and monthly numbers of wind projects that are newly online. I construct the annual and monthly time trends based on the data from EIA-860. Red bars in Panel (a) represent years with policy expiration, while red squares in Panel (b) represent the new projects that are online in December.

This time pattern aligns well with the timing of policy implementation. During the years between 2009 and 2012, in addition to the Production Tax Credit, there was also the Section 1603 Grant, which provided extra funding flexibility to investors and partly explained the surge of wind projects during this period. By the end of 2012, there was significant uncertainty about the PTC extension due to the time lapse in renewal. Consequently, there was a rushed inflow of new wind projects before the policy expiration, as wind farm investors hoped to secure subsidies for the next ten years of operation, resulting in the bunching of new investment in 2012. The impacts of the PTC expiration are more pronounced when examining the monthly trend of new wind projects. As shown in Panel (b), the bunching in 2012 was mainly driven by a massive entry in December 2012, which was the exact month of expiration of the PTC. Although the PTC was renewed shortly after its expiration in 2013, the investment flow did not recover immediately, as it takes a relatively long time to build new wind farms. After 2015, the PTC was promised for a longer window, resulting in a

steady time trend of new wind projects between 2015 and 2019.

There are alternative explanations for the bunching in the online timing. First, wind farms might shorten the construction process to meet the expiration dates of the PTC. However, as shown in Panel (a) of Appendix Figure A3, the average construction time remains stable at around nine months across different online years. There is suggestive evidence in Panel (b) that wind projects starting construction in 2012 were more likely to have a shorter construction period to meet the end-of-year expiration date. However, this difference is relatively small in magnitude. Second, the massive entry in 2012 might reflect the expedited waiting process in the interconnection queue. However, as shown in Panel (a) of Appendix Figure A4, the total years spent between entering into the interconnection queue and starting construction are also stable across years when wind farms start construction. Moreover, Panel (b) shows that many projects that started construction in 2011 entered the interconnection queue before 2006. Therefore, the bunching in the online years is achieved mainly through the expedited investment decision, instead of merely reflecting the shortened construction time or the interconnection approval time.

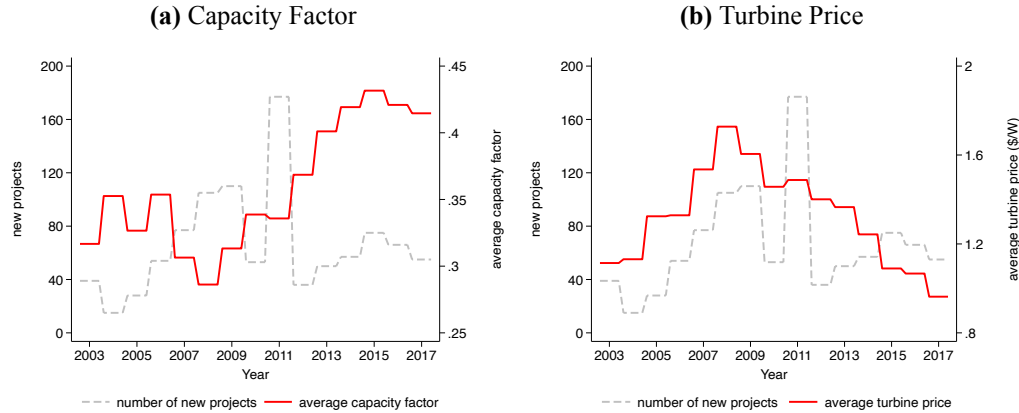
3.2.2 Misalignment with technological advancement

In contrast to the bunched investment timing, the technology of wind turbines is continuously improving over time. There are three key components of a typical horizontal-axis wind turbine: a tower, a nacelle, and three rotor blades. The potential of wind power generation crucially depends on the height of the tower and the length of the rotor blades. Taller towers enable the turbine to access better wind resources up in the air, while longer rotor blades lead to larger swept areas and capture more wind energy inputs. As shown in Appendix Figure A5, the hub heights and rotor diameters of new wind farms are getting larger, almost following linear trends after 2009. The average hub heights and rotor diameters of wind farms invested between 2014 and 2019 are 6.5% and 24.6% larger than those invested between 2008 and 2013.

Policy uncertainty induces a misalignment between the timing of investment and technological advancement. Figure 4 presents the contrast between the bunched investment timing and improving turbine technology. I plot the number of new wind farms according to their construction start years. Moreover, I measure the technological efficiency using capacity factor at the age of one, defined as the ratio of average power output to maximum

power capability.¹⁴ Newly invested wind farms between 2008 and 2013 had an average capacity factor of 0.32, while that number between 2014 and 2018 rose to 0.41, increasing by 27.2%. While the investment bunched in earlier years, the turbine technology is continuously and quickly improving, and thus there were many wind farms equipped with less productive turbines as a result of policy uncertainty.¹⁵

Figure 4: Investment, Turbine Technology, and Turbine Price



Notes: This figure shows the time trend of wind farm investment, as well as average capacity factor and turbine price for newly installed wind projects. Panel (a) shows the time trend of capacity factor, measured as the ratio of total annual output to the maximum power output (nameplate capacity scaled by 24×365), based on the data from EIA-923. I plot the investment time trend as the gray dashed line for comparison. Panel (b) shows the time trend of turbine price using data from Lawrence Berkeley National Laboratory.

Moreover, the average turbine prices are also decreasing over time. As shown in Panel (b), since peaking at around 1,700 dollars per kilowatt between 2008 and 2009, the average turbine price has declined and fallen below 1,000 dollars per kilowatt since 2015. Therefore, early investment between 2008 and 2011 largely forgoes later cheaper turbines.

Decreasing turbine procurement prices and increasing turbine production efficiency together indicate a substantial option value of delaying entry into the market for better and cheaper technology. However, policy uncertainty expedited wind farm entry so that investors partially missed the benefits of technological improvement and potentially led to

¹⁴According to EIA, capacity factor is defined as “the ratio of the electrical energy produced by a generating unit for the period of time considered to the electrical energy that could have been produced at continuous full-power operation during the same period.” A detailed description of capacity factors can be found in Supplemental Appendix Section SA.4.

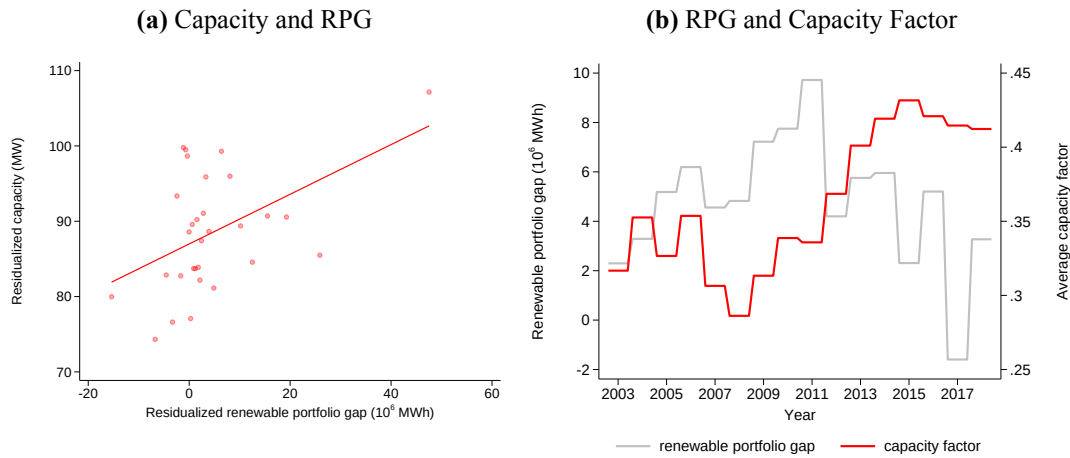
¹⁵One concern is that the average productivity of a wind farm is also affected by the wind resources of its location, and later entrants might be faced with locations with worse wind resources. However, as shown in Appendix Figure A6, the average wind speed for each cohort is generally stable over time. The wind resources are less volatile for later entrants, as the standard deviation of daily average wind speed is lower.

inefficient investment timing.

3.2.3 Misalignment with demand: matching between wind farms and utilities

Utilities are important buyers of wind power, as they procure wind capacity through long-term contracts from wind farms. One crucial incentive for them to procure wind capacity is to meet the state Renewable Portfolio Standards, which require utilities to have a certain share of total electricity generation from renewable energy. I construct a variable, renewable portfolio gap (RPG), which is defined as the difference between the amount stipulated by the Renewable Portfolio Standards and renewable energy generation. It measures the demand of each utility for renewable energy in order to meet the Renewable Portfolio Standards.¹⁶

Figure 5: Matching Efficiency between Utilities and Wind Farms



Notes: This figure provides descriptive evidence about the matching efficiency between utilities and wind farms. Panel (a) shows the binned scatter plot of wind capacity to the renewable portfolio gap (RPG) of utilities. Renewable portfolio gap measures the demand of each utility for renewable energy in order to meet the Renewable Portfolio Standards. I control for electricity prices, turbine productivity, and time trends. Panel (b) shows the time trend of the average RPG of utilities that procured wind capacity each year, as well as the mean turbine capacity factor for each new cohort of wind farms.

I find that utilities with a larger renewable portfolio gap, and thus more compliance demand, are more likely to procure a larger amount of wind capacity through long-term contracts as shown in Panel (a) of Figure 5. This relationship is robust to conditioning on a set of controls including electricity prices, turbine productivity, and linear yearly trends. I further plot the time trend for average renewable portfolio gaps of utilities in Panel (b). The

¹⁶The details of how this variable is constructed and estimated can be found in Section 4 and Supplemental Appendix Section SA.4.

average renewable portfolio gaps of utilities increased before 2011 as more states implemented Renewable Portfolio Standards. However, with the addition of more wind capacity, the average renewable portfolio gaps for utilities have decreased sharply since 2011, in contrast to the ongoing increase in turbine productivity over time. Consequently, a mass of wind farms rushed into the market due to policy uncertainty in the early years of the industry, equipped with turbines of lower productivity but matched with utilities of larger compliance demand. For more recently entered wind farms, although the turbine technology has improved significantly, they could only sell capacity to utilities with smaller compliance demand. This misalignment leads to a loss in matching efficiency between the utilities and wind farms.

Motivated by industry background and stylized facts, I build an empirical model of wind farms' dynamic entry decisions under evolving technology, demand, and policy uncertainty. I explore the key determinants of wind farm profits, and how policy beliefs held by investors evolve over time.

4 Model

I develop a dynamic model of wind farms' entry problem. Wind farm investors form beliefs about the PTC renewal probability and decide whether to enter or wait each period. Under the policy uncertainty, wind farm investors secure a flow of future federal subsidies if they enter before the PTC expires, but they forgo better and cheaper technology in the future. Upon entry, there are two channels for wind farms to sell their capacity. First, they can negotiate a long-term contract with a utility, jointly determining the power purchase price and the procured capacity. Second, they can sell capacity to non-utility buyers, such as corporations, or enter into financial arrangements, such as merchant hedge contracts.

I assume time t is discrete at a yearly level, and denote a wind farm as i and a utility as j . I divide the model into three modules: bilateral bargaining with utilities, demand from non-utility buyers and buyer choice, and dynamic entry under policy uncertainty. For the model, identification and estimation, as well as results, I present each component in parallel.¹⁷

¹⁷The timing assumption about the model can be found in Appendix Figure A7.

4.1 Bilateral Bargaining with Utilities

Profit function for utilities By procuring wind capacity k_{ij} from wind farm i , utility j earns revenues from electricity generation and obtains renewable energy credits, and it pays procurement costs at the price p_{ij} in the power purchase agreement to the wind farm. A market m is a U.S. state, and I assume both the electricity market and the renewable credit market to be competitive.¹⁸ Therefore, utility j takes retail electricity price p_{mt}^e , renewable credit price p_{mt}^c , and Renewable Portfolio Standards requirement z_{mt} as given. If the share of electricity generation using renewable energy falls short of z_{mt} , utilities need to buy renewable credits to fulfill the requirement; otherwise, they could sell renewable credits to earn revenues. I suppress subscript m for the remainder of the section.

Suppose utility j begins a power purchase agreement with wind farm i in year t for a duration of T years.¹⁹ The profit function for utility j from this contract is

$$\pi_t^U(p_{ij}, k_{ij}) = \sum_{s=t+1}^{t+T} \mathbb{E}_t \beta^{s-t} \left\{ \underbrace{(p_s^e - p_{ij}) \alpha_i k_{ij}}_{\text{profit from elec. gen.}} + \underbrace{p_s^c [(1 - z_s) \alpha_i k_{ij} - Q_{js}^{gap}] - h_{js}(k_{ij})}_{\text{profit from renewable credits}} \right\}.$$

Profit flow starts from year $t + 1$ as it takes one year on average between finalizing the investment decision and starting production for wind farms, and β denotes the annual discount factor. I assume the production function for wind farm i as $\alpha_i k_{ij}$, where α_i is the annualized capacity factor.²⁰ Moreover, Q_{jt}^{gap} is calculated as the difference between renewable energy generation required by the state ($z_{mt} \times$ utility j 's total output) and existing renewable energy generation, which measures the compliance demand to meet the standard. I assume utilities have perfect foresight of the state-level Renewable Portfolio Standards z_t , and they hold rational expectations with respect to the transition dynamics of electricity price p_t^e and renewable credit price p_t^c .²¹

¹⁸According to Appendix Figure A8, more than 85% of the utilities own wind capacity less than 5% at the REC tracking system level, and more than 65% own wind capacity less than 5% at the state level.

¹⁹I use t to denote the year when negotiation happens, which I assume to be determined in the dynamic entry decision. Consequently, for each bargaining pair, t is predetermined.

²⁰The linear functional form fits data well as shown in Appendix Figure A9. Moreover, I find that capacity factors evolve systematically with the cohort, but display limited variation with respect to the age of wind farms. Therefore, I treat the annualized capacity factor to be constant as a wind farm ages and calculate it at the age of one for the best data coverage.

²¹Even when Renewable Portfolio Standards bind, certificate banking and advance targets leave flexibility in investment timing, allowing PTC deadlines to accelerate investment without changing long-run procurement requirements. Moreover, interstate certificate trading and purchases by buyers without compliance obligations further weaken the link between state requirements and investment.

The hassle cost, denoted as $h_{js}(k_{ij})$ for year s within the contract, captures frictions in the renewable credit market, dynamic credit banking incentives, and utilities' limited appetite for procuring wind capacity beyond their compliance needs. Define the *gap-filling scale* $\bar{k}_{ijs}$ as the project size that would exactly close utility j 's renewable portfolio gap. I specify the hassle cost as a quadratic function of the deviation of the contracted capacity from this scale, weighted by the project's productivity, which fits the data pattern that utilities further away from the state-level goal tend to procure more wind capacity as illustrated in Panel (a) of Figure 5.

$$h_{js}(k_{ij}) = \alpha_i \frac{\delta}{2} [k_{ij} - \bar{k}_{ijs}]^2, \quad \bar{k}_{ijs} \equiv \frac{Q_{js}^{gap}}{(1 - z_s) \alpha_i}. \quad (1)$$

Profit function for wind farms The profit that wind farm i receives equals the sum of total revenues from power purchase agreements and total subsidies from the government, minus turbine costs.

$$\pi_t^W(p_{ij}, k_{ij}) = \underbrace{\frac{\beta(1 - \beta^T)}{1 - \beta} p_{ij} \alpha_i k_{ij}}_{\text{revenue from PPA}} + \underbrace{\frac{\beta(1 - \beta^{10})}{1 - \beta} d_t \alpha_i k_{ij}}_{\text{PTC}} - \underbrace{c_{ijt} \alpha_i k_{ij}}_{\text{turbine cost}}.$$

Wind farm i receives the tax credit for the first ten years of production. I denote the value of the tax credit per unit of wind energy generation by d_t and, following Johnston (2019), I value tax credits at 85% of their face value to account for transaction costs, asymmetric information, and market power in the tax credit market. I also account for the availability of the Section 1603 Grant between 2008 and 2012. A wind farm that opts for the Section 1603 Grant receives a subsidy equal to 30% of its investment cost. Because the choice between the two subsidies has a relatively small effect on profits, I present the model with the PTC as the sole subsidy for ease of exposition. Appendix Section B presents detailed empirical evidence on wind farms' subsidy choice and explains how it is incorporated into the profit estimates. Unless otherwise noted, references to the PTC in analyses of the 2008–2012 investment cohorts refer to the federal subsidy bundle, including the Section 1603 Grant.

Moreover, I parameterize the turbine cost per unit of annualized output c_{ijt} such that $c_{ijt} = \mathbf{X}'_{it} \gamma_1 + \frac{\gamma_2}{2} k_{ij} + \xi_{it}$. Cost shifters $\mathbf{X}_{it}$ collect average turbine prices p_t^{tb} and turbine brand dummies. The parameter γ_2 allows flexible convexity of total turbine cost in capacity, and ξ_{it} denotes unobserved cost shocks to wind farms.

Bilateral bargaining Wind farm i and utility j participate in the bilateral bargaining pro-

cess to negotiate over the procured capacity k_{ij} and the contracted price p_{ij} simultaneously. The optimization problem is formulated as follows.

$$\max_{k_{ij}, p_{ij}} [\pi_t^U(p_{ij}, k_{ij}) - \pi_t^U(p_{ij} = \infty)]^\rho \times [\pi_t^W(p_{ij}, k_{ij}) - \pi_t^W(p_{ij} = \infty)]^{1-\rho}.$$

The bargaining weight of utilities is denoted by ρ . $\pi_t^U(p_{ij} = \infty)$ represents the profits that utilities would obtain with their current energy portfolios if the negotiation fails, and $\pi_t^W(p_{ij} = \infty)$ represents the payoffs that wind farms would earn from waiting for another year to enter. Under the assumption of Nash bargaining, the optimal capacity maximizes the gains from trade, while the optimal price divides the gains between the two parties.

The optimal capacity follows equation (2). If wind energy is more valuable due to either higher electricity prices (p_s^e), or higher renewable credit prices (p_s^c), or larger gap-filling scales ($\bar{k}_{ijs}$, defined in equation (1)), utilities are willing to pay more for additional wind capacity. The optimal wind capacity equalizes the marginal value of wind energy and its marginal cost net of the PTC.

$$\underbrace{\sum_{s=t+1}^{t+T} \mathbb{E}_t \beta^{s-t} [p_s^e + p_s^c(1 - z_s) + \delta(\bar{k}_{ijs} - k_{ij}^*)]}_{\text{willingness to pay}} = \underbrace{\mathbf{X}_{it}' \gamma_1 + \gamma_2 k_{ij}^*}_{\text{turbine cost}} - \underbrace{\sum_{s=t+1}^{t+10} \beta^{s-t} d_t + \varepsilon_{ij}^k}_{\text{PTC}}. \quad (2)$$

Moreover, the optimal price follows equation (3). If the utility has a larger bargaining power ρ , the negotiated price will be low enough to only cover turbine costs net of government subsidies. If the wind farm has a bigger bargaining power, the negotiated price will be closer to the willingness to pay for utilities. A higher outside option $\pi_t^W(p_{ij} = \infty)$ gives wind farms better bargaining positions and increases the negotiated price.

$$\begin{aligned} \frac{\beta(1 - \beta^T)}{1 - \beta} p_{ij}^* &= (1 - \rho) \times \underbrace{\sum_{s=t+1}^{t+T} \mathbb{E}_t \beta^{s-t} [p_s^e + p_s^c(1 - z_s) + \delta(\bar{k}_{ijs} - \frac{1}{2} k_{ij}^*)]}_{\text{willingness to pay}} \\ &+ \rho \times \left[\underbrace{\mathbf{X}_{it}' \gamma_1 + \frac{\gamma_2}{2} k_{ij}^*}_{\text{average turbine cost}} - \underbrace{\sum_{s=t+1}^{t+10} \beta^{s-t} d_t}_{\text{PTC}} + \underbrace{\frac{\pi_t^W(p_{ij} = \infty)}{\alpha_i k_{ij}^*}}_{\text{bargaining leverage}} \right] + \varepsilon_{ij}^p. \end{aligned} \quad (3)$$

4.2 Demand of Non-Utility Buyers and Buyer Choice

Demand of non-utility buyers An alternative channel for selling wind capacity is to sell to non-utility buyers such as corporations or to sign merchant hedge contracts. Due to a

lack of data on the characteristics of both corporate buyers and these financial contracts, I model this second channel using a linear demand curve. I assume non-utility buyers demand capacity $k_{i(t)}^{nu}$ at wind energy price $p_{i(t)}^{nu}$ from wind farm i . The demand function is

$$k_{i(t)}^{nu} = \zeta_1 p_{i(t)}^{nu} + \zeta_2 \alpha_{i(t)} + \zeta_3 \mathbf{X}_{i(t)} + \zeta_4 \mathbf{Z}_{i(t)}^{nu} + v_{i(t)}. \quad (4)$$

The subscript $i(t)$ indicates that a variable takes its value in wind farm i 's investment year t . Similar to the turbine cost shifters in equation (2), $\mathbf{X}_{i(t)}$ here includes the average turbine price as well as the turbine brand dummies. $\mathbf{Z}_{i(t)}^{nu}$ denotes a set of demand shifters including dummies for balancing authorities and contract types (long-term contracts with corporate buyers, or merchant hedge contracts). $v_{i(t)}$ represents unobserved demand shifters.

Buyer choice Wind farms sell capacity through one of the two aforementioned channels: a power purchase agreement with a utility, or the non-utility channel described above. I define the potential utility buyers $\mathcal{J}_i$ as those within 300 miles of wind farm i that signed a power purchase agreement during the sample period, together with the utility that wind farm i contracts with if it is located farther away. This modeling choice reflects the fact that contracts are concentrated within state and balancing authority boundaries and over short distances, as selling across a boundary involves additional transmission and compliance arrangements, and a distant utility is costly to reach.²² Therefore, I model wind farm i as considering each potential buyer $j \in \mathcal{J}_i$ with probability q_{ij} , independently across potential buyers. I assume that the consideration probability does not vary over time and depends only on whether wind farm i and utility j are in the same state ($\mathbb{1}\{m_i = m_j\}$), whether they are in the same balancing authority ($\mathbb{1}\{BA_i = BA_j\}$), and how far away they are ($dist_{ij}$). The consideration probability is formulated as below and $\Lambda(\cdot)$ denotes the logistic function.

$$q_{ij} = \Lambda(\phi_0 + \phi_s \mathbb{1}\{m_i = m_j\} + \phi_b \mathbb{1}\{BA_i = BA_j\} - \phi_d dist_{ij}). \quad (5)$$

Among the considered potential buyers, wind farm i signs with the one that yields the highest bargaining profit $\pi_t^W(p_{ij}^*, k_{ij}^*)$, evaluated at the Nash bargaining outcome in equations (2) and (3). If no potential buyer is considered, wind farm i sells capacity through the non-utility channel. Consequently, the probability $P_{it}^u(j)$ that wind farm i matches with utility j and the probability P_i^{nu} that it sells through the non-utility channel are given as

²²Among the matched pairs, 77 percent are in the same state and 84 percent are in the same balancing authority, and the median distance is 51 miles. Among the unmatched potential pairs within 300 miles, the shares are 30 percent and 51 percent, and the median distance is 188 miles.

follows.

$$P_{it}^u(j) = q_{ij} \prod_{j' \in \mathcal{J}_i: \pi_t^W(p_{ij}^*, k_{ij}^*) > \pi_t^W(p_{ij'}^*, k_{ij'}^*)} (1 - q_{ij'}), \quad P_i^{nu} = \prod_{j \in \mathcal{J}_i} (1 - q_{ij}). \quad (6)$$

The consideration draws determine whether wind farm i matches with a utility at all, and if so, which utility it matches with. The realized profit of wind farm i is $\pi_t^W(p_{ij}^*, k_{ij}^*)$ if it matches with utility j , and $\pi_t^{nu}(p_i^{nu}, k_i^{nu})$ if it sells through the non-utility channel.²³ I define the *ex-ante* profit π_{it} of wind farm i entering in year t as the expectation of its realized profit over the consideration draws, which is formulated as follows.

$$\pi_{it} = \sum_{j \in \mathcal{J}_i} \pi_t^W(p_{ij}^*, k_{ij}^*) P_{it}^u(j) + \pi_t^{nu}(p_i^{nu}, k_i^{nu}) P_i^{nu}. \quad (7)$$

4.3 Dynamic Entry under Policy Uncertainty

4.3.1 Potential Entrants and Dynamic Entry Decision

Wind farm investors need to enter the interconnection queue, get approved by several studies, and sign interconnection agreements before they are eligible to enter the market.²⁴ Therefore, I define projects that have been in the interconnection queue for two or more years as the set of potential entrants and model their optimal entry decisions.²⁵

At the beginning of year t , potential entrant i faces entry cost $\psi_{it} = \kappa_{0,r(i)} + \kappa_1 W_{it} + \nu_{it}$, where W_{it} denotes the observed entry cost shifter, measured by the state-level land price, and $\kappa_{0,r(i)}$ is a fixed component specific to wind farm i 's wholesale market region $r(i)$, capturing regional differences in development costs. ν_{it} is an i.i.d. entry cost shock that follows a logistic distribution with scale σ_ν , which captures idiosyncratic project-level shocks to the cost of entering this year relative to waiting. If potential entrant i decides to enter, I define the net profit of entry as $\Pi_{it} = \pi_{it} - \kappa_{0,r(i)} - \kappa_1 W_{it}$, where π_{it} is the *ex-ante* profit from

²³This is consistent with the data pattern that the profit under a utility Power Purchase Agreement is higher than in the non-utility buyer channel for 65% of the wind farms in the sample. The median PPA profit is 93 million dollars against 71 million in the non-utility channel.

²⁴As pointed out by [Fan and Xiao \(2015\)](#), it is crucial to model potential entrants as long-run players and incorporate the identities of potential entrants to recover the distribution of the entry cost in the optimal stopping problem, like the one studied in this paper.

²⁵In the project-level interconnection queue data of Lawrence Berkeley National Laboratory ([Rand et al., 2026](#)), the 1,247 wind projects that signed an interconnection agreement in 2003-2018 spent a median of 2.2 years in the queue between the request and the agreement, with an inter-quartile range of 1.2 to 3.6 years. Waiting times have lengthened and a typical project completed in 2008 had spent fewer than two years in the queue, compared with three years for a project completed in 2015 ([Utility Dive, 2023](#)).

equation (7), and the realized payoff of entering is $\Pi_{it} - \nu_{it}$.

Potential entrant i conditions on a vector of state variables $\mathbf{s}_{it}$ for the dynamic decision, including shifters for buyers' willingness to pay, turbine technology, turbine cost, subsidy level, and entry cost shifter W_{it} . Another important state variable is the policy status ω_t , which is a dummy variable that equals one if the federal subsidy is present in year t , and zero if the federal subsidy is absent in year t . Ex post, $\omega_t = 1$ in all periods because the PTC was always extended. The availability of the subsidy affects the negotiated price and capacity (p_{ij}^*, k_{ij}^*) and thus wind farm profits Π_{it} . Potential entrant i enters in year t when the net profit of entry exceeds the option value of waiting, and waits for one more year otherwise. I denote $V_t(\mathbf{s}_{it}, \omega_t)$ as the value function of wind farm i in year t before the entry cost shock is realized, defined as follows.

$$V_t(\mathbf{s}_{it}, \omega_t) = \mathbb{E}_\nu \max \left\{ \underbrace{\Pi(\mathbf{s}_{it}, \omega_t) - \nu_{it}}_{\text{net profit of entering today}}, \underbrace{\beta \mathbb{E}_t[V_{t+1}(\mathbf{s}_{it+1}, \omega_{t+1}) | \mathbf{s}_{it}, \omega_t]}_{\text{option value of waiting}} \right\}. \quad (8)$$

The option value of waiting depends on the shock scale σ_ν and the transition dynamics of state variables $G(\mathbf{s}_{it+1} | \mathbf{s}_{it})$ such that

$$\mathbb{E}_t[V_{t+1}(\mathbf{s}_{it+1}, \omega_{t+1}) | \mathbf{s}_{it}, \omega_t] = \int \mathbb{E}_t[V_{t+1}(\mathbf{s}_{it+1}, \omega_{t+1}) | \omega_t] dG(\mathbf{s}_{it+1} | \mathbf{s}_{it}).$$

Moreover, the option value of waiting depends crucially on an *ex-ante* belief for the policy evolution, denoted by $b_t(\omega_{t+1} | \omega_t) \in [0, 1]$. I allow the policy to be extended only year by year, and $b_t(\omega_{t+1} | \omega_t)$ can vary by time to capture that wind farm investors form different policy beliefs depending on the actions taken by the government as well as other political and economic shocks.²⁶ The option value of waiting can be further expanded as follows, with $b_t(\omega_{t+1} | \omega_t)$ being the source of non-stationarity in this dynamic problem.

$$\begin{aligned} \mathbb{E}_t[V_{t+1}(\mathbf{s}_{it+1}, \omega_{t+1}) | \omega_t] &= V_{t+1}(\mathbf{s}_{it+1}, \omega_{t+1} = 1) \times b_t(\omega_{t+1} = 1 | \omega_t) \\ &\quad + V_{t+1}(\mathbf{s}_{it+1}, \omega_{t+1} = 0) \times b_t(\omega_{t+1} = 0 | \omega_t). \end{aligned}$$

I denote the entry decision as a dummy variable E_{it} and the entry probability as $P_t^E(\mathbf{s}_{it}, \omega_t)$. Potential entrant i enters if and only if the entry cost shock ν_{it} falls below the difference between the net profit of entering today and the option value of waiting. Since the PTC shifts up firm values such that $V_{t+1}(\mathbf{s}_{it+1}, \omega_{t+1} = 1) > V_{t+1}(\mathbf{s}_{it+1}, \omega_{t+1} = 0)$, a lower perceived

²⁶In reality, the PTC could be announced with a window longer than one year, such as between 2010 and 2012. However, considering different lengths of policy windows introduces another layer of uncertainty and further complicates the model. Therefore, I allow the policy to be extended only year by year for simplicity.

probability of renewal reduces the option value of waiting, making potential entrants more likely to enter in the current year.

$$P_t^E(\mathbf{s}_{it}, \omega_t) = \Pr(E_{it} = 1 | \mathbf{s}_{it}, \omega_t) = \Lambda\left(\frac{\Pi(\mathbf{s}_{it}, \omega_t) - \beta \mathbb{E}_t[V_{t+1}(\mathbf{s}_{it+1}, \omega_{t+1}) | \mathbf{s}_{it}, \omega_t]}{\sigma_\nu}\right). \quad (9)$$

4.3.2 Belief Structure

Since the dynamic problem extends over an infinite horizon, potential entrants need to form beliefs about the entire sequence of future policy status $\{b_t(\omega_{t+s+1} | \omega_{t+s})\}_{s \geq 0}^\infty$. Moreover, if policy belief is permitted to be fully flexible, including examples in which perceived future subsidies switch between on and off, infinite streams of policy beliefs could rationalize one single investment decision. Therefore, I impose two assumptions to discipline policy beliefs and make the problem feasible for estimation.

Assumption 1 (absorbing state) $b(\omega_{t+1} = 0 | \omega_t = 0) = 1$.

Assumption 1 indicates that the policy is perceived as terminated once paused. If the policy is absent in year t , wind farm investors will hold the belief that the policy is terminated forever. This assumption is consistent with the reality that the Section 1603 Grant was discontinued after 2012 and has not been reinstated ever since. Under this assumption, $\mathbb{E}_t[V_{t+1}(\mathbf{s}_{it+1}, \omega_{t+1}) | \omega_t = 0] = V_{t+1}(\mathbf{s}_{it+1}, \omega_{t+1} = 0)$. Consequently, the value function $V_t(\mathbf{s}_{it}, \omega_t = 0)$ does not depend on time-varying beliefs and can be simplified as a stationary function $V^0(\mathbf{s}_{it})$. I further denote $\Pi(\mathbf{s}_{it}, \omega_t = 0)$ as $\Pi^0(\mathbf{s}_{it})$, which leads to the following Bellman equation.

$$V_t(\mathbf{s}_{it}, \omega_t = 0) = V^0(\mathbf{s}_{it}) = \sigma_\nu \log \left(\exp [\Pi^0(\mathbf{s}_{it}) / \sigma_\nu] + \exp [\beta \mathbb{E}[V^0(\mathbf{s}_{it+1}) | \mathbf{s}_{it}] / \sigma_\nu] \right). \quad (10)$$

Assumption 2 (simple forecast) $b_t(\omega_{t+s+1} = 1 | \omega_{t+s} = 1) = b_t(\omega_{t+1} = 1 | \omega_t = 1) = b_t, s \geq 0$.

Assumption 2 indicates that the perceived likelihood of a one-year policy extension will be applied to future years, which precludes cases in which wind investors have more information about future policy extensions beyond the next year. However, I allow the beliefs to change across years and the investors to revise their beliefs according to new information. Consequently, $b_t(\omega_{t+s+1} = 1 | \omega_{t+s} = 1)$ and $b_{t+s}(\omega_{t+s+1} = 1 | \omega_{t+s} = 1)$ could be different to reflect unanticipated shocks realized in year $t + s$. $b_t(\omega_{t+1} | \omega_t)$ is henceforth an index that

summarizes the policy uncertainty faced by wind farm investors in year t . Instead of imposing Assumption 2, the belief evolution could be parameterized as a first-order Markov process, but a relatively short time series makes such an approach infeasible. An alternative policy belief model could use a mixture distribution as described in the Supplemental Appendix Section SA.7. However, without underlying time-varying beliefs, such a model cannot rationalize the jumping bunching patterns in the investment time trend; with underlying time-varying beliefs, that alternative model is essentially isomorphic to the baseline model.

I denote $\Pi(\mathbf{s}_{it}, \omega_t = 1)$ as $\Pi^1(\mathbf{s}_{it})$ and $V_t(\mathbf{s}_{it}, \omega_t = 1)$ as $V^1(\mathbf{s}_{it}; b_t)$. Under Assumption 2, $V^1(\mathbf{s}_{it}; b_t)$ solves the following Bellman equation (11), which nests $V^0(\mathbf{s}_{it})$ solved from Bellman equation (10).

$$V^1(\mathbf{s}_{it}; b_t) = \sigma_\nu \log \left(\exp \left[\frac{\Pi^1(\mathbf{s}_{it})}{\sigma_\nu} \right] + \exp \left[\beta \{ \mathbb{E}_t[V^1(\mathbf{s}_{it+1}; b_t) | \mathbf{s}_{it}] \times b_t + \mathbb{E}[V^0(\mathbf{s}_{it+1}) | \mathbf{s}_{it}] \times (1 - b_t) \} / \sigma_\nu \right] \right). \quad (11)$$

4.3.3 State Space, Transition, and Equilibrium

I include three sets of variables in the state space $\mathbf{s}_{it}$ to capture the rich interactions among government policies, turbine technology, and utility demand in determining the profit of entry as well as the option values of waiting. The first set of state variables is time-varying and exogenous, including the annual average productivity of wind turbines $\bar{\alpha}_t$, the average turbine prices p_t^{tb} , the subsidy levels d_t , and the prevailing electricity price p_t^e .²⁷ I assume that potential entrants' beliefs about these state variable transitions follow AR(1) models, with the exception of the subsidy level d_t : its continuation is governed by the policy status ω_t in the years before 2015, and it follows the legislated phase-down schedule established after the Consolidated Appropriations Act of 2015. The second set of state variables are time-invariant and exogenous. I construct a predicted utility demand using time-invariant shifters in equation (2), and a buyer channel choice probability shifter in equation (5), i.e., the probability that no utility counterparty is considered, $\prod_{j \in \mathcal{J}_i} (1 - q_{ij})$. I project p_i^{nu} on Z_i^{nu} as in equation (4) to construct another time-invariant state variable for the demand of non-utility buyers.

²⁷I use the annual average productivity of wind turbines $\bar{\alpha}_t$ instead of realized productivity for each individual wind farm to ease concerns about selection.

The third set of state variables are time-varying and endogenous, including the renewable credit prices p_{mt}^c , the changing renewable portfolio gaps of utilities in the buyer pool, and cumulative wind capacity online K_{mt} in the state m . As each wind farm has about 13 candidate buyers on average in $\mathcal{J}_i$, keeping track of the renewable portfolio gap for each individual utility is computationally challenging. Motivated by [Hendel and Nevo \(2006\)](#) and [Gowrisankaran and Rysman \(2012\)](#), I use inclusive values for wind farms that can be attributed to the changing renewable portfolio gaps for buyers, denoted by $\Phi_{it} = \pi_{it}(\{Q_{jt}^{gap}\}_{j \in \mathcal{J}_i}) - \pi_{it}(\{Q_{jt}^{gap} = 0\}_{j \in \mathcal{J}_i})$. The inclusive value Φ_{it} is the difference in the *ex-ante* profit functions from equation (7) with and without the realized renewable portfolio gaps. The transition of Φ_{it} is endogenous in the model because the renewable portfolio gaps of utilities shrink after they procure additional new wind capacity. Therefore, more entry of wind farms today will reduce future values of Φ_{it} . I model its transition as an AR(1) process with the one-year lagged cumulative wind capacity online K_{mt-1} in the state m as an endogenous shifter. Similarly, the renewable credit price also follows an AR(1) process and carries the one-year lagged cumulative wind capacity in the state as a shifter, since installed wind supplies renewable credits and depresses their price.²⁸ Both inclusive value Φ_{it} and the renewable credit prices p_t^c capture preemptive incentives of wind farms such that they would like to enter early to access buyers with a higher willingness to pay, counteracting incentives to delay their entry for better and cheaper technology. I assume wind farms perceive the evolution of K_{mt} as the following model.

$$K_{mt} = \rho_1^K K_{mt-1} + \rho_0^K + \xi_{mt}^K. \quad (12)$$

I adopt an equilibrium concept similar to the moment-based Markov Equilibrium ([Ifrah and Weintraub, 2017](#)) and assume that each wind farm keeps track of its own states and moments of the industry states.²⁹ This equilibrium concept is widely used in recent empirical papers ([Jeon, 2022](#); [Barwick, Kalouptsi, and Zahur, 2025](#); [Vreugdenhil, 2026](#)). The entry decision satisfies equation (9), given the policy beliefs $b_t(\omega_{t+1}|\omega_t)$ and the perceived transitions of state variables $G(s_{it+1}|s_{it})$, which include the law of motion of cumulative capacity in equation (12). This perceived law of motion of cumulative capacity must be consistent with the entry decisions implied by the model. I therefore solve for a fixed point at which

²⁸The one-year lag captures the gap between the entry of a wind farm and the start of its production.

²⁹I assume that each wind farm is atomic and the impact of its action on the aggregate state variable is negligible due to a low level of market concentration.

expected growth in cumulative capacity equals the growth generated by wind farms' entry decisions under those expectations.

4.4 Model Discussion

I close the section with brief discussions of the model assumptions. First, I only endogenize the capacity of procured wind energy but abstract away from responses of other fuel sources. The wind penetration rate was low during my sample period in most states, and I treat the responses of other fuel sources as exogenous to keep the model tractable. However, my model still captures increasing gas power capacity, decreasing coal power capacity, and decreasing gas prices, mainly through utilities' willingness to pay for wind capacity which subsumes the evolution of electricity prices and renewable portfolio gaps.

Second, I assume that policy uncertainty does not directly influence bargaining terms. I find little evidence of discontinuity in the average capacity (Appendix Figure A2) or negotiated prices (Appendix Figure A10) for wind farms that began construction in 2011. This suggests that the impact of policy uncertainty on profits from investment is relatively minor compared to its effect on entry timing decisions.

Third, I model utilities' procurement behavior through the consideration probability in equation (5) rather than utilities' forward-looking decisions. Because wind turbine productivity and utilities' wind demand are complementary in generating total surplus, wind farms' buyer choice problem is sufficient to capture these complementarities, with the payoff from each potential match evaluated at the corresponding bargaining outcome.

Fourth, I assume that turbine technology and price are exogenous to investment in the U.S. wind industry. The cumulative wind power capacity in the U.S. represents 16% of the global wind power fleet, and there is no observed trend break in either turbine technological advancements or turbine prices following the surge of U.S. investments (Figure 4). I conduct a robustness check in Section 7.1 to quantify potential bias from ignoring the learning-by-doing effects, following Covert and Sweeney (2026).

5 Identification and Estimation

I discuss how data variations identify the model and how estimation procedures recover model parameters in this section. I start with flow-profit primitives including the turbine

cost function, the hassle cost, the bargaining weight of utilities, the demand function for non-utility buyers, and the buyer choice probability. Given the flow profits, I then discuss how to identify and estimate the dynamic parameters, including the entry cost function, the scale of the entry shocks, and the policy beliefs.

5.1 Bilateral Bargaining with Utilities

There are two estimation equations from the bilateral bargaining problem: the optimal capacity equation (2) and the optimal price equation (3), which I estimate jointly on the sample of utility power purchase agreements by nonlinear least squares.³⁰ The turbine cost shifters are the average turbine price p_t^{tb} and the turbine brand dummies (GE, Vestas, Siemens Gamesa, and others) in $\mathbf{X}_{it}$. I further control for a set of shifters, including dummies for the states of the utilities, utility types, contract duration intervals, and year groups.³¹ Conditional on the rich set of controls, error terms ε_{ij}^k and ε_{ij}^p mainly capture measurement errors in the willingness to pay and turbine costs, and are assumed exogenous to the observables.

There are two curvature parameters, unit hassle cost δ and cost convexity γ_2 , whose compound effect is identified by the effect of the unit subsidy on the negotiated capacity. Moreover, the optimal price equation separates these two, since the two curvatures enter the negotiated price with opposite signs. Cost convexity γ_2 raises the per-unit price of larger contracts, as wind farms must recover a higher average cost, while hassle cost δ lowers it, because the utility's average value per unit decreases with size. The dependence of negotiated prices on contracted capacity therefore identifies the decomposition of the total curvature into δ and γ_2 , which the capacity margin cannot separate.

In the optimal price equation, I define $\pi_t^W(p_{ij} = \infty)$ as the payoff that wind farms would have earned from waiting for another year to enter and selling capacity to a utility from the rest of the potential buyer pool. I find that conditional on all other observables in equation (3), the residual variation in negotiated prices is positively correlated with the average willingness to pay from nearby alternative utilities, which gives a better bargaining position for the wind farm (Panels (a) and (b) of Appendix Figure A10). Moreover, the average p_{ij}^* displays a large variation across cohorts (Panel (c)). Motivated by these data facts, I express $\pi_t^W(p_{ij} = \infty)$ as a flexible control function with quadratic bases of the

³⁰A more detailed description of the estimation can be found in Supplemental Appendix Section SA.4.

³¹The utility types include cooperative, investor-owned, or others. The contract duration intervals include three groups: 15 years or fewer, between 15 and 20 years, or more than 20 years.

average willingness to pay of nearby alternative utilities as well as year dummies.

The key parameter in the optimal price equation (3) is the bargaining weight ρ . The identification of ρ comes from the relative pass-through ratios of the utility's willingness to pay and the turbine cost net of subsidy on the negotiated price. If the utility has a larger bargaining weight ρ , the negotiated price tends to be low and co-moves closer to the net turbine cost, after flexibly controlling for bargaining leverage.

5.2 Demand of Non-Utility Buyers and Buyer Choice

Demand of non-utility buyers I estimate the linear demand function for non-utility buyers in equation (4) with instruments. As $v_{i(t)}$ captures unobserved demand shifters, it is correlated with the price $p_{i(t)}^{nu}$, which introduces bias to the price coefficient ζ_1 . I use three supply-side instruments to tackle this identification challenge. The first instrument is a dummy variable indicating whether the state has adopted an industry recruitment policy, which offers tax incentives to firms that manufacture or install renewable energy equipment in the state and thus lowers the cost of wind equipment and project development. The second instrument is the number of wind component manufacturing sites in the state, which lowers turbine transportation and procurement costs. The third instrument is the wind farm's distance to the nearest substation, which shifts the cost of interconnection. Each instrument shifts the price at which wind farms are willing to supply capacity, but is plausibly excluded from the demand shocks of non-utility buyers, who account for a minority of total wind capacity and whose payoffs these cost shifters do not enter directly other than shifting wind energy prices. I control for turbine productivity, annual turbine prices, and brand dummies, as well as fixed effects for balancing authorities and contract types. Robustness checks add the state-level land price and dummies for state tax incentives as additional instruments.

Buyer choice The parameters in buyer consideration probability $(\phi_0, \phi_s, \phi_b, \phi_d)$ are estimated by maximum likelihood on all wind farms jointly, with the two margins in equation (6) entering one likelihood function. A wind farm matched with utility j contributes the probability $P_{it}^u(j)$ that j was considered and no candidate offering a higher bargaining profit was, while a wind farm selling through the non-utility channel contributes the probability of the non-utility channel P_i^{nu} . The gradients of observed matching with respect to the same-state indicator, the same-balancing-authority indicator, and distance identify ϕ_s , ϕ_b , and ϕ_d . The overall frequency of utility matches relative to non-utility outcomes, given

the size of the potential buyer sets, pins down the intercept ϕ_0 .

5.3 Dynamic Entry under Policy Uncertainty

The primary identification challenge in the dynamic model is to separate the parameters governing entry costs and entry shocks ($\{\kappa_{0,r(i)}\}$, κ_1 , and σ_ν) from the policy belief parameters b_t . The main identification strategy is to exploit the temporal structure of the policy. The American Taxpayer Relief Act renewed the credit in January 2013 and introduced a safe harbor provision, under which a wind farm qualifies once five percent of its investment cost is incurred before the policy expires, with a two-year window to begin operation (extended to four years in 2016). The Consolidated Appropriations Act, announced in December 2015, further fixed the path of the credit through the end of the sample, including its phase-down schedule. Together, these provisions decoupled eligibility from fixed placed-in-service deadlines and effectively eliminated the renewal margin in the later years of the sample, consistent with the stable investment pattern after 2013 in Figure 3. Therefore, I assume that there is no renewal uncertainty from 2013 onward. Investors perceive $b_t = 1$ in 2013 and 2014, and from 2015 the credit follows its legislated schedule with no renewal margin. Entry behavior in these certainty years pins down the entry cost parameters, and conditional on those parameters as well as the transition process of the state variables, the bunching of investment in earlier deadline years identifies the belief parameters b_t . I provide a formal identification proof in Supplemental Appendix Section SA.8.

Estimation proceeds in three steps: constructing the flow-profit surfaces, solving the dynamic problem, and matching entry moments. First, I construct the flow-profit surfaces by simulating the estimated static model over a grid of states centered around the observed state-year panel. I generate this grid by adding Gaussian perturbations to the observed states, with each perturbation scaled by one quarter of the estimated standard deviation of the corresponding transition shock. At each grid point, I compute the *ex-ante* entry profit in equation (7) under both policy regimes. Specifically, I re-solve for negotiated prices and capacities with and without the subsidy. I then approximate the simulated profits, Π^1 and Π^0 , as well as expected contracted capacity, using a polynomial basis in the state variables.

Second, I solve the dynamic problem separately for each policy regime, and approximate the value function using the same basis (Sweeting, 2013; Barwick and Pathak, 2015). For years through 2014, I solve the stationary value function V^0 from the Bellman equation

(10) and the belief-weighted value function $V^1(\cdot; b_t)$ from the Bellman equation (11). For 2015–2018, when the tax credit follows its legislated phase-down, I solve the value functions by backward induction along the statutory path. The terminal value in 2019 is evaluated at 40% of the full PTC, which is the last step of the legislated schedule. Consistent with the equilibrium concept in Section 4.3, endogenous state variables evolve according to model-implied entry. Predicted entry increases cumulative wind capacity, which affects the transitions of the renewable credit price and the inclusive value Φ_{it} from renewable portfolio gaps, while reducing the remaining pool of potential entrants. I solve the perceived transition of cumulative capacity (equation (12)) as a fixed point, requiring investors’ beliefs about market evolution to be consistent with the entry behavior implied by the candidate parameter values.

Third, I estimate $\theta = (\{\kappa_{0,r}\}_{r=1}^5, \kappa_1, \sigma_\nu, \{b_t\}_{t=2009}^{2012})$ by matching entry moments. The estimation sample consists of fifteen states with sufficient wind capacity spanning five wholesale-market regions: MISO, SPP, PJM, CAISO, and ERCOT, and the pool of potential entrants is constructed from interconnection-queue data. Let $E_{mt} = \sum_{i \in m} E_{it}$ denote the observed number of entering wind farms in state m and year t , and let $\hat{E}_{mt}(\theta) = N_{mt} \hat{P}_t^E(\mathbf{s}_{mt}, \omega_t = 1; \theta)$ denote the corresponding model prediction. Here, $\mathbf{s}_{mt}$ contains the state variables measured at the state-year level, N_{mt} is the number of potential entrants, and $\omega_t = 1$ in every sample year because the tax credit was ultimately in force throughout the sample. I match three sets of moments: aggregate entry by year, aggregate entry by wholesale-market region, and market-specific entry in 2011. Appendix Section C provides additional details on the dynamic estimation procedure.

Robustness of the estimation procedure I implement six alternative estimation procedures to assess the robustness of the results. First, in the baseline model, entry costs vary over time only through changes in state-level land prices. I relax this restriction by adding a time trend to entry costs, allowing for gradual changes in development frictions that are not captured by land prices, such as increasing local resistance to siting or longer permitting processes. Second, the baseline imposes $b_t = 1$ in 2013 and 2014. I instead include these two years in the uncertainty window and estimate b_{2013} and b_{2014} , allowing the data to determine whether investors perceived renewal as certain in those years. Third, the baseline assumes that, after 2015, the credit follows its legislated phase-down with certainty. I relax this assumption by introducing an additional belief parameter governing the perceived

probability that the statutory phase-down continues, analogous to the renewal uncertainty in earlier years. Fourth, the baseline evaluates the 2019 terminal value at a residual credit equal to 40 percent of the statutory level. I vary this residual between 30 and 50 percent. Fifth, Assumption 1 treats expiration of the credit as an absorbing state. I relax this assumption by allowing an expired credit to be reinstated in the following year with a calibrated probability, while keeping the belief structure tractable. Sixth, I discuss how discount factors and risk aversion of wind investors affect model estimates.

6 Results

6.1 Bilateral Bargaining with Utilities

Panel A of Table 1 presents the estimation results for the bilateral bargaining model. The hassle cost parameter δ is positive and precisely estimated, indicating that utilities prefer contract sizes close to the capacity needed to meet their renewable portfolio requirements. This pattern captures frictions in the renewable credit market and utilities' limited incentives to procure wind capacity beyond their compliance needs. The estimated convexity parameter of turbine costs, γ_2 , is small and statistically indistinguishable from zero, suggesting that the curvature in negotiated capacity arises primarily from utilities' demand rather than wind farms' cost. Moreover, I estimate the utilities' bargaining weight ρ to be 0.61, implying that utilities capture approximately three-fifths of the gains from trade in the bargaining process. The estimate is significantly different from one, indicating that changes in the PTC are not fully passed through to the negotiated price. Thus, a take-it-or-leave-it model in which utilities fully extract the wind farm's rents would understate the importance of the PTC for wind farm investors.

Figure 6 compares negotiated prices and discounted wind-farm profits with and without the PTC. Removing the PTC reduces negotiated capacity and increases the negotiated price, as utilities absorb part of the lost subsidy through the price. Nevertheless, wind-farm profits decline for every contract. The discounted sum of profit π_{ij}^W is 105.7 million dollars on average, 146.1 million dollars at the 75th percentile, and 204.1 million dollars at the 90th percentile. Only 3.2% of wind farms fail to reach an agreement. When the PTC is removed, bilateral bargaining yields a lower negotiated capacity, but a higher negotiated price. The negotiated price without the PTC $p_{ij}^*(\omega_t = 0)$ is 16.3% higher compared with $p_{ij}^*(\omega_t = 1)$. I

assume that a negative negotiated capacity will lead to the failure of the bargaining. About 18.4% of wind farms will fail (I normalize as zero profit) without the PTC, underscoring the critical role of this federal incentive in supporting the industry. Even for wind farms that still reach an agreement, $\pi_{ij}^W(\omega_t = 0)$ on average is 61% smaller than $\pi_{ij}^W(\omega_t = 1)$. This result highlights the significant cost of missing deadlines and losing PTC eligibility, explaining the rushed entry given a low perceived likelihood of the PTC extension.

Table 1: Parameter Estimates for Bargaining, Non-utility Demand, and Buyer Choice

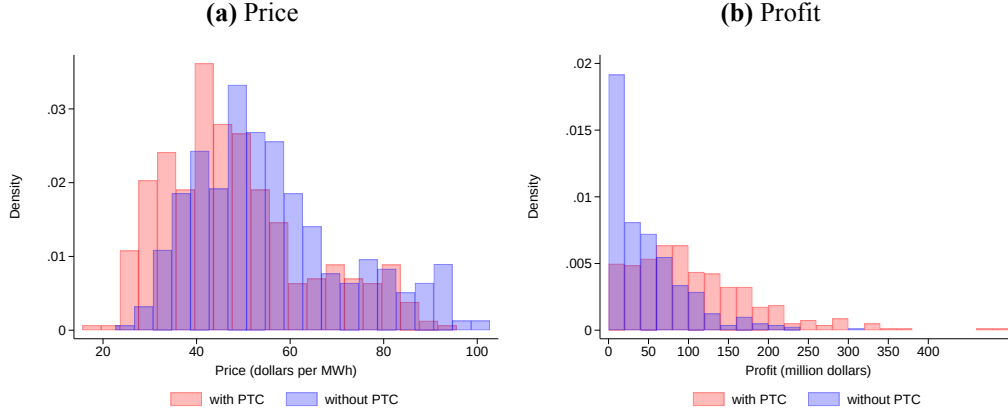
Coefficient	Parameter	Estimate	S.E.
<i>Panel A: Parameters of Bilateral Bargaining</i>			
Hassle Cost	δ	0.236	(0.052)
Turbine Price	γ_1	0.058	(0.019)
Unit Capacity Cost Convexity	γ_2	0.012	(0.510)
Bargaining Weight	ρ	0.606	(0.075)
<i>Panel B: Parameters of Demand for Non-Utility Buyers</i>			
Price Coefficient, OLS	ζ_1	-0.872	(0.193)
Price Coefficient, IV	ζ_1	-1.179	(0.393)
<i>Panel C: Parameters of Utility Matching and Buyer Type Choice</i>			
Utility Choice Probability, Constant	ϕ_0	-3.880	(0.202)
Utility Choice Probability, Same State	ϕ_s	1.646	(0.138)
Utility Choice Probability, Same Balancing Authority	ϕ_b	1.153	(0.146)
Utility Choice Probability, Distance	ϕ_d	2.041	(0.681)

Notes: This table shows the estimation results of the bilateral bargaining model (equations (2) and (3)), the linear demand curve for non-utility buyers (equation (4)), and the buyer choice model (equation (6)). Standard errors are in parentheses.

6.2 Demand of Non-Utility Buyers and Buyer Choice

Panel B of Table 1 presents the demand function for non-utility buyers. The OLS estimate of the price coefficient ζ_1 is -0.87 . Using the three supply-side instruments, the IV estimate is -1.18 , about one-third larger in magnitude than the OLS estimate. This difference is consistent with an upward bias in OLS caused by unobserved demand factors that increase both the price and the quantity of wind capacity sold to non-utility buyers. Supplemental Appendix Section SA.5.1 reports the estimates under alternative instrument sets, which are

Figure 6: Estimated Profit and Price with and without the PTC



Notes: This figure shows the distributions of negotiated prices and discounted profits of wind farms with and without the PTC, evaluated at the estimates in Table 1.

close to the baseline. There is a sparse reference for the demand elasticity in the wind capacity, but the magnitude roughly aligns with the previous estimates in the liquefied natural gas industry (Zahur, 2022) and solar panel industry (Gerarden, 2023). Panel C presents the buyer choice model. Geography dominates which utilities a wind farm considers. A utility in the same state as the wind farm is considered with much higher probability, and so is a utility in the same balancing authority: the estimates of ϕ_s and ϕ_b raise the odds of consideration by factors of 5.2 and 3.2, respectively, and every additional 100 miles between the pair lowers the odds by 18%.

6.3 Dynamic Entry under Policy Uncertainty

Table 2 presents the estimation results of the dynamic model. Column (1) is the baseline specification, which estimates the policy beliefs for 2009-2012 and matches the annual and market aggregate moments together with the market-by-2011 moments. Column (2) drops the market-by-2011 moments. Columns (3) and (4) extend the belief window to 2014 and estimate b_{2013} and b_{2014} instead of imposing $b_t = 1$ in these two years.

Panel A presents the entry cost parameters. The entry cost at the average land price of each market ranges from about 10 million dollars in SPP and ERCOT to 92 million dollars in PJM and 115 million dollars in CAISO. The ranking is consistent with the difficulty of siting and interconnecting wind farms across regions, with the sparsely populated wind belt at the low end and the congested PJM queue and California at the high end. The land price

coefficient κ_1 is positive and significant, so a higher land price raises the entry cost within a market over time. The scale of the entry cost shock σ_v is 23 million dollars, which is of the same order as the entry costs themselves and reflects the substantial project-level heterogeneity in development costs. The deterministic part of the entry cost averages 57 million dollars across the simulated 2009-2012 entrants, compared with an investment cost net of turbine cost of 44 million dollars reported by the Section 1603 awardees that came online in those years.

Panel B presents the policy beliefs. The perceived probability of a one-year policy renewal is close to one in 2010 and 2012, and 0.87 in 2009. In 2011, the year before the credit lapsed at the end of 2012, the perceived probability of renewal falls to 0.39.³² The estimate is consistent with the investment spike in 2011 documented in Section 3: investors who expected the credit to expire rushed to qualify before the deadline. The 2011 belief is estimated to range from 0.28 to 0.55 across different specifications. When b_{2013} and b_{2014} are estimated rather than imposed, they are 0.93–0.96, which supports treating these years as a certainty window in the baseline. Figure 7 compares the predicted and observed numbers of new wind farms by year. The model reproduces the 2011 spike and the total number of entrants over the sample period, and fits the aggregate time trend of wind farm entry well. Appendix Section C.5 examines the robustness of the dynamic parameter estimates to alternative model specifications. The estimates are broadly robust to allowing for a time trend in entry costs, estimating a common policy belief parameter for 2015 onward, and varying the terminal PTC value in 2019. Although the model cannot separately identify renewal beliefs from the probability of reinstatement following a PTC expiration, the discount factor, or investors’ risk aversion, the estimated beliefs remain broadly stable across a reasonable range of assumptions about these factors. More importantly, these alternative specifications also leave the welfare conclusions largely unchanged, as discussed in the next section.

7 Counterfactual Analysis

The counterfactual analysis addresses the paper’s central question: what was the cost of PTC renewal uncertainty, given that the subsidy was ultimately renewed each time? Inter-

³²This paper doesn’t disentangle the expiration of the 1603 grant with the uncertainty about PTC expiration. The estimated belief for 2011 should therefore be interpreted as the perceived likelihood of continued federal support as a whole.

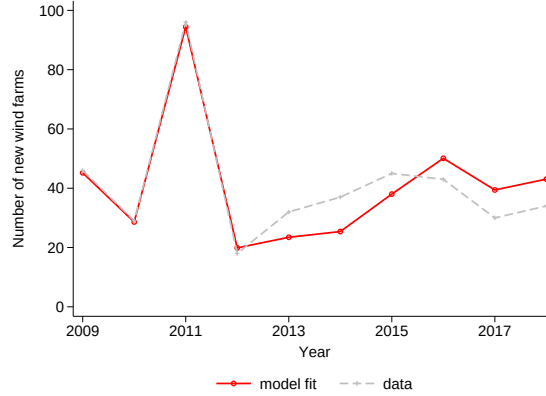
Table 2: Parameter Estimates for the Dynamic Model

Parameter and Notation	(1) Beliefs 2009–2012	(2) Beliefs 2009–2012	(3) Beliefs 2009–2014	(4) Beliefs 2009–2014
<i>Panel A: Entry Cost Parameters</i>				
Entry Cost, MISO	64.14 (18.51)	61.20 (19.11)	68.78 (23.86)	68.40 (18.35)
Entry Cost, SPP	10.11 (14.59)	9.36 (14.36)	28.46 (20.38)	27.08 (19.06)
Entry Cost, PJM	91.58 (20.39)	87.14 (21.64)	83.67 (18.42)	82.25 (19.90)
Entry Cost, CAISO	114.77 (56.74)	117.84 (54.45)	216.75 (81.02)	233.28 (83.75)
Entry Cost, ERCOT	9.43 (19.13)	12.18 (19.59)	56.43 (30.97)	59.42 (30.83)
Land Price	10.92 (6.63)	11.25 (6.74)	20.09 (9.66)	18.65 (8.80)
Scale of Entry Cost Shock	22.67 (4.97)	21.70 (5.76)	13.13 (5.47)	12.16 (6.09)
<i>Panel B: Belief Parameters</i>				
Policy Belief 2009	0.869 (0.127)	0.880 (0.134)	0.808 (0.242)	0.776 (0.209)
Policy Belief 2010	0.999 (0.011)	0.999 (0.007)	0.969 (0.031)	0.956 (0.027)
Policy Belief 2011	0.385 (0.282)	0.546 (0.293)	0.395 (0.233)	0.278 (0.239)
Policy Belief 2012	0.999 (0.018)	0.999 (0.025)	0.959 (0.033)	0.933 (0.053)
Policy Belief 2013	1	1	0.949 (0.050)	0.931 (0.061)
Policy Belief 2014	1	1	0.959 (0.063)	0.951 (0.082)
Annual aggregate moments	✓	✓	✓	✓
Market aggregate moments	✓	✓	✓	✓
Market $\times$ 2011 Moments	✓		✓	
Objective Value	597.98	550.96	181.54	79.58

Notes: This table shows the estimation results of the dynamic model. Entry costs are in million dollars and evaluated at the average land price of each market. Columns (1) and (2) estimate the policy beliefs for 2009–2012 and fix $b_{2013} = b_{2014} = 1$; columns (3) and (4) estimate the beliefs for 2009–2014. Columns (1) and (3) add the market-by-2011 moments to the annual and market aggregate moments. Standard errors are block-bootstrapped over states with 100 draws.

preting this exercise requires recognizing that policy uncertainty is not an instrument that policymakers can directly control. It arises from periodic reauthorization, political turnover,

Figure 7: Model Fit: New Wind Farms by Year



Notes: This figure compares the number of new wind farms predicted by the dynamic model at the baseline estimates in column (1) of Table 2 with the observed number in the fifteen sample states.

and fiscal constraints, some of which may be unavoidable. The exercises below therefore quantify the cost of uncertainty associated with the PTC’s institutional design. The results shed light on the potential value of institutional features that could reduce this uncertainty, including renewal decisions made before expiration, longer policy windows, and qualification rules that decouple PTC eligibility from fixed subsidy expiration dates.

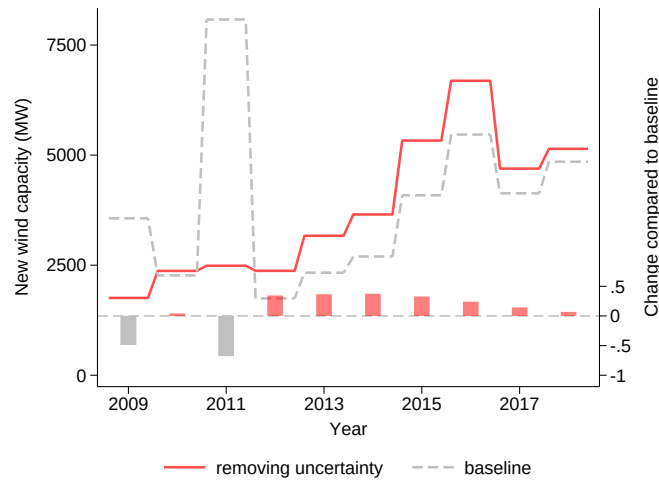
7.1 Investment and Welfare without Renewal Uncertainty

I simulate the baseline scenario using the estimated beliefs reported in column (1) of Table 2 and a counterfactual scenario without renewal uncertainty, in which $b_t = 1$ in every year. Because the PTC was renewed in every period ex post, the realized policy path is identical across the two scenarios; the only difference is investors’ beliefs about that path. In each scenario, wind farms solve the dynamic problem of Section 4.3. The endogenous state variables, including the renewable portfolio gaps of utilities, the renewable energy credit price, and the cumulative wind capacity in each state, evolve with the simulated entry. Importantly, setting $b_t = 1$ changes both the variance and the expected value of the subsidy stream: it eliminates renewal uncertainty while simultaneously increasing the subsidy that investors expect to receive. Section 7.2 separates these two effects through an exercise that holds the expected subsidy fixed.

Investment timing Figure 8 plots the annual new wind capacity under the two scenarios. Without renewal uncertainty, the 2011 spike disappears: 31 wind farms enter that year,

compared with 94 in the baseline. About 85 wind farms that enter between 2009 and 2011 in the baseline delay entry, and about two thirds of these enter between 2012 and 2018, when turbines were cheaper and more productive. The number of new projects between 2012 and 2018 rises by 24%. Among projects that postpone entry, the average delay is 4.1 years. Across all new projects entering between 2011 and 2018, the average entry year shifts later by 0.7 years. Over the full 2009–2018 period, the number of new wind farms falls by 6.6% and total capacity by 4.0%, as reported in Table 3. However, total output falls by only 0.5%, as wind farms that enter later have larger average sizes with higher capacity factors.

Figure 8: New Wind Capacity with and without Renewal Uncertainty



Notes: This figure shows the simulated capacity of new wind farms by year in the baseline scenario with the estimated policy beliefs and in the counterfactual scenario without renewal uncertainty ($b_t = 1$).

Social welfare Panels B and C of Table 3 further report profit and social surplus of wind farms entering between 2009 and 2018 under the two scenarios, discounted to 2009. I evaluate the benefits following Callaway, Fowle, and McCormick (2018).³³ Wind farms operate for 20 years, and their output displaces fossil generation, yielding three benefits: avoided carbon emissions, avoided fuel costs, and capacity value. I calculate avoided emissions using each state’s marginal operating emissions rate and value them at a social cost of carbon of \$80 per ton.³⁴ I value avoided fuel costs and capacity jointly using the wholesale

³³Callaway, Fowle, and McCormick (2018) find that regional average marginal operating emissions rates offer a useful means of “calibrating regional policy incentives to compensate for external emissions benefits.”

³⁴According to Brookings, the Obama administration estimated the social cost of carbon at \$43 per metric

market value of wind output in each market, obtained from Lawrence Berkeley National Laboratory’s wind market report data ([Wiser et al., 2025](#)).

Policy uncertainty induced earlier entry, bringing forward the benefits of wind energy and creating a potential welfare gain. Total benefits are \$4.5 billion (4.3%) higher in the baseline than in the scenario without renewal uncertainty, with environmental benefits accounting for about three quarters of the increase. However, uncertainty also encouraged investment at times that were less favorable in terms of technology and demand, reducing efficiency. As shown in Panel B, earlier entry required investors to purchase turbines at the higher prices prevailing in 2009–2011 and to enter when land prices were near their peak. Consequently, turbine costs are \$5.2 billion (13.1%) higher and entry costs \$1.6 billion higher in the baseline. The government also pays \$0.8 billion more in subsidies for the additional early output. Wind farm profits are \$1.7 billion (11.0%) lower under policy uncertainty, and industry profits net of subsidies are negative. Overall, renewal uncertainty reduces the social surplus from wind energy by \$3.2 billion (9.8%). The benefits of earlier deployment do not offset the costs of installing older technology at higher prices.

Robustness and heterogeneity I explore the robustness of the welfare impacts across different dynamic model specifications, including changing the policy belief window, allowing a time trend in entry costs, and alternative terminal subsidy levels. Across all specifications, removing renewal uncertainty raises social surplus. Moreover, the baseline counterfactual holds queue inflows and withdrawals fixed at their observed levels. This understates the pool of potential entrants without renewal uncertainty if uncertainty itself induced withdrawals, and I discuss it in Appendix Section [C.6](#).

A remaining concern is the assumption that turbine technology is exogenous, as discussed in Section [4.4](#). Removing renewal uncertainty delays the entry of the affected projects by a few years, and the turbine market is global, with the U.S. holding a 16% share of cumulative capacity in 2019. A back-of-the-envelope calculation using the upper-bound learning parameter from [Covert and Sweeney \(2026\)](#) indicates that even if slower U.S. deployment slowed the improvement of turbine technology, the social surplus gain would decrease by less than 10%.

ton globally, while the Trump administration only considered the effects of carbon emissions within the United States, estimating the number to be between \$3 and \$5 per metric ton. The Biden administration estimated the social cost of carbon to be \$51 per ton, but the EPA proposed a nearly fourfold increase to \$190 in November 2022.

Table 3: Outputs, Benefits, and Costs with and without Renewal Uncertainty

		Baseline	No Uncertainty	Difference	Percentage
<i>Panel A: Output</i>					
Number of Projects		407.6	380.8	-26.8	-6.6%
Total Capacity (10 ³ MW)		39.2	37.7	-1.6	-4.0%
Total Output (10 ⁶ MWh)		1615.4	1607.6	-7.8	-0.5%
<i>Panel B: Profit (Billion USD)</i>					
Turbine Cost	TC	39.8	34.6	-5.2	-13.1%
Entry Cost	EC	16.4	14.9	-1.6	-9.7%
Total Profit	TP	15.8	17.5	1.7	11.0%
<i>Panel C: Benefit and Cost (Billion USD)</i>					
Total Benefit	TB	104.7	100.2	-4.5	-4.3%
Environmental Benefit		74.0	70.7	-3.3	-4.4%
Others		30.7	29.5	-1.2	-3.9%
Subsidy	S	16.3	15.5	-0.8	-5.0%
Net Profit	TP-S	-0.5	2.0	2.5	
Social Surplus	TB-TC-EC-S	32.1	35.3	3.2	9.8%

Notes: This table shows the outputs, benefits, and costs of wind energy between 2009 and 2018 in the baseline scenario and in the counterfactual scenario without renewal uncertainty. All dollar values are discounted to 2009 with a discount factor of 0.95. Total profit is the discounted static profit of wind farms net of turbine and entry costs. The environmental benefit uses a social cost of carbon of \$80 per metric ton.

I examine how static parameters affect the welfare consequences of renewal uncertainty by varying the utility bargaining weight, hassle cost, non-utility price coefficient, and buyer choice parameters. Appendix Figure A11 reports the results. The welfare gain from removing uncertainty declines as utilities gain bargaining power and wind farms retain less of the subsidy. Lower hassle costs increase the gain by making contracts larger and more profitable, raising the subsidy at stake and strengthening wind investors' response to renewal uncertainty. Across a wide range of static parameters, removing renewal uncertainty consistently increases social surplus.

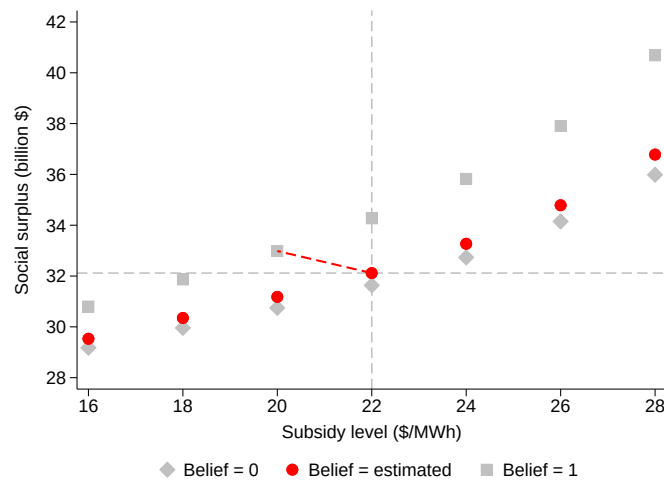
7.2 Lessons for Subsidy Design

The main counterfactual establishes that renewal uncertainty was costly. Two further exercises translate the finding into lessons for the design of subsidies.

Generosity versus commitment Figure 9 plots the social surplus of wind energy for PTC levels ranging from \$16 to \$28 per MWh under three renewal beliefs in 2011: the

estimated belief, certain renewal, and anticipated expiration. Social surplus rises with the subsidy under all three beliefs and is highest under certain renewal at every subsidy level. Comparing the subsidy levels needed to achieve the same social surplus provides a measure of the cost of uncertainty in subsidy terms. Under certain renewal, a subsidy of \$20 per MWh yields higher social surplus than the actual \$22 subsidy under the estimated belief. A credible commitment to renewal would therefore have allowed a roughly 9% reduction in the subsidy rate without sacrificing social surplus, while also reducing total subsidy payments. Conversely, under anticipated expiration, the \$22 subsidy delivers approximately the same social surplus as a \$21 subsidy under the estimated belief.

Figure 9: Social Surplus by Subsidy Level and Policy Belief

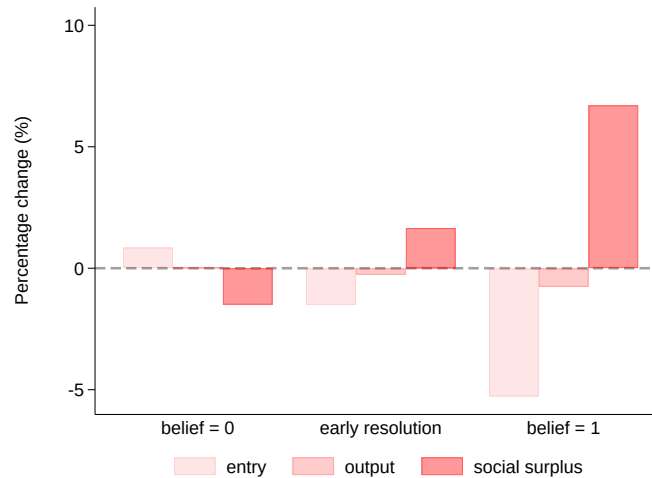


Notes: This figure shows the simulated social surplus of wind energy in 2009–2018 for PTC levels between \$16 and \$28 per MWh, under the estimated 2011 belief, under certain renewal ($b_{2011} = 1$), and under anticipated expiration ($b_{2011} = 0$). The beliefs in other years are at their estimates. Dollar values are in billions, discounted to 2009.

Early resolution The main counterfactual changes both the expected subsidy and uncertainty about renewal. To distinguish these effects, I consider a scenario in which the renewal decision for the subsidy scheduled to expire at the end of 2012 is announced before investors make their 2011 investment decisions. Renewal occurs with the probability implied by investors’ estimated belief. Investors thus learn whether the subsidy will be renewed ($b_{2011} = 1$) or allowed to expire ($b_{2011} = 0$), while the *ex-ante* expected subsidy remains unchanged. Figure 10 reports the expected percentage changes in entry, output, and social surplus relative to the baseline, together with the changes under certain renewal

and certain expiration. Early resolution reduces the number of new wind farms by 1.5% and total output by 0.3%, while increasing social surplus by 1.7%. By comparison, certain renewal increases social surplus by 6.7%. Resolving uncertainty while holding the expected subsidy fixed therefore recovers about one quarter of the welfare gain from certain renewal.

Figure 10: Early Resolution of Renewal Uncertainty



Notes: This figure shows the percentage changes in the number of new wind farms, total output, and social surplus in 2009–2018 relative to the baseline scenario under anticipated expiration ($b_{2011} = 0$), early resolution (the renewal decision announced before the 2011 investment decisions with the estimated probability of renewal), and certain renewal ($b_{2011} = 1$).

General lessons The findings offer three broader lessons for policy design. First, the value of commitment depends on the technology being subsidized. Uncertain renewal draws investment toward expiration deadlines, bringing forward environmental benefits but also locking long-lived projects into older, more expensive technology. In this setting, the additional costs outweigh the benefits of earlier deployment. Other things equal, credible commitment is therefore more valuable when technology improves rapidly and investment decisions are difficult to reverse.

Second, the timing of policy announcements matters alongside the probability of renewal. A subsidy that may expire is smaller in expectation, reducing the value of waiting. Announcing renewal decisions before investors commit allows them to respond to the realized policy rather than the threat of expiration. This can improve investment timing even without increasing the *ex-ante* expected subsidy. Early announcements can therefore deliver part of the welfare gain from certain renewal when a commitment to continued support is

not feasible.

Third, credible commitment can reduce the subsidy needed to achieve a given level of social surplus. In the 2011 exercise, a credit of \$20 per MWh under certain renewal delivers at least as much social surplus as a \$22 credit under the estimated belief. Greater policy certainty can thus partly substitute for a more generous subsidy. Together, these findings support longer authorization periods, earlier renewal decisions, and qualification rules that give investors more flexibility over when to build.

8 Conclusion

This paper quantifies how uncertainty about PTC renewal shaped investment timing and welfare in the U.S. wind industry. Repeated expiration deadlines encouraged investors to build sooner to secure subsidy eligibility, even as turbine prices fell and productivity improved. Earlier investment accelerated renewable generation and its environmental benefits, but it also locked projects into older, more expensive technology and changed their allocation across buyers with evolving demand.

To evaluate this trade-off, I develop and estimate a dynamic investment model that incorporates policy renewal beliefs, bilateral bargaining over long-term contracts, and buyer choice. The estimates imply a perceived renewal probability of about 39% for the 2011 cohort. In a full-commitment benchmark, where investors anticipate with certainty the renewals that ultimately occurred, entry in 2011 falls by 67%, and projects that delay entry do so by an average of 4.1 years. Social surplus rises by \$3.2 billion (9.8%). The gains from adopting cheaper and more productive turbines therefore outweigh the loss from postponing renewable generation.

The results also show that policy commitment can partly substitute for subsidy generosity. In the 2011 exercise, under full renewal certainty, the subsidy level could be reduced by \$2/MWh, around 9%, without compromising social welfare. Announcing renewal decisions before investment decisions also improves welfare, even when the *ex-ante* expected subsidy is unchanged. Both the credibility of continued support and the timing of policy announcements matter for investment incentives.

These findings measure the cost of renewal uncertainty, rather than the net benefit of eliminating it. The analysis does not value the government's flexibility to let a subsidy

expire as circumstances change. Nevertheless, it identifies a cost of short authorization periods that is especially relevant when capital is long-lived and technology improves rapidly. Longer policy windows, earlier renewal decisions, and more flexible qualification rules may allow investors to adopt better technology while preserving incentives for deployment. The broader lesson is that subsidy design should account not only for how much investment it induces, but also for when that investment occurs.

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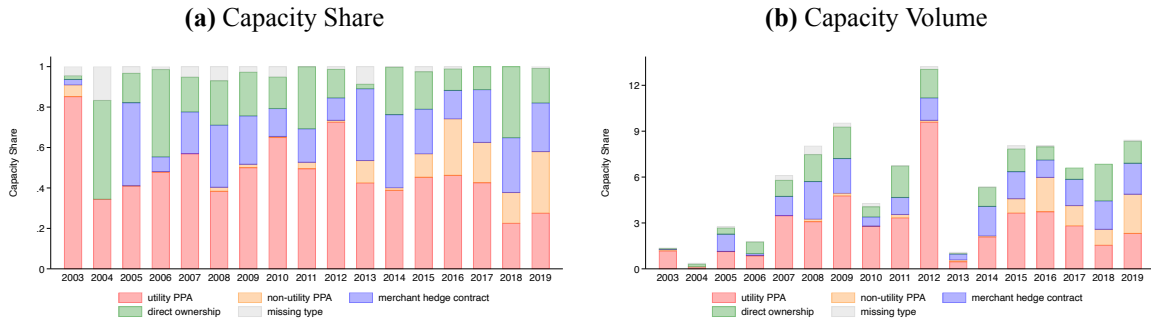
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Online Appendix for The Dynamic Efficiency of Policy Uncertainty: Evidence from the Wind Industry

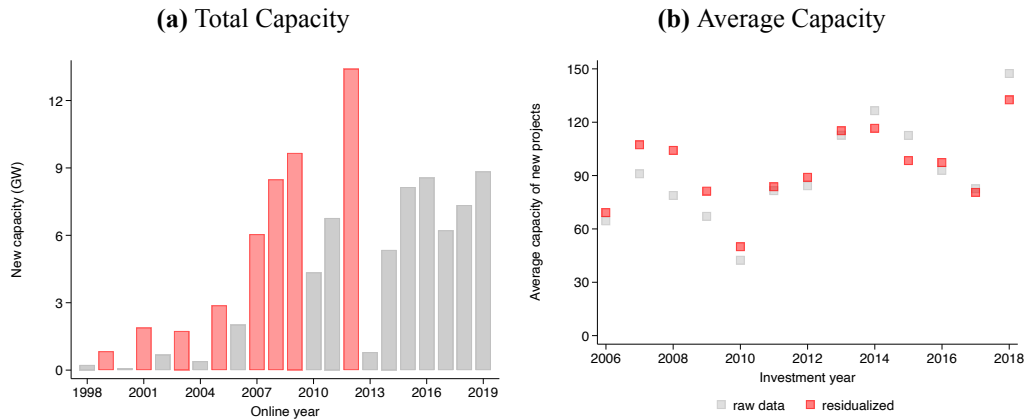
A Additional Figures

Figure A1: Capacity by Offtake Types



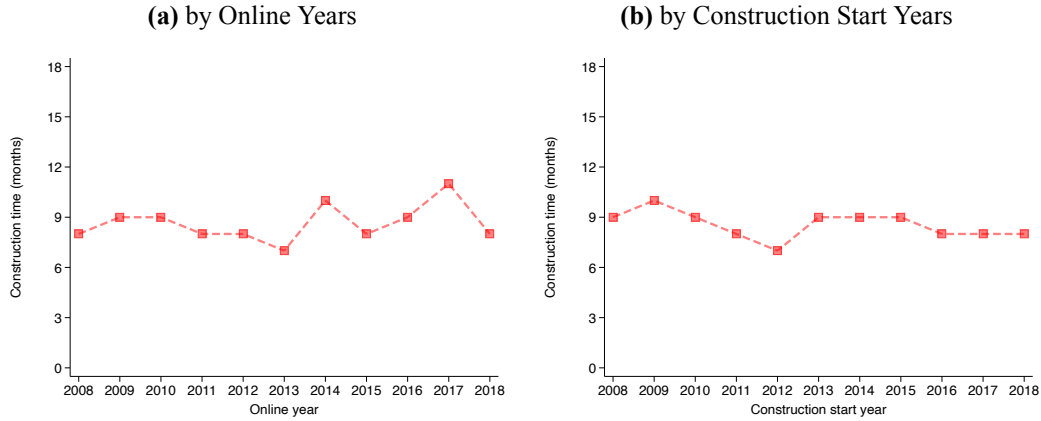
Notes: This figure shows the capacity distribution by offtake types across years. There are four offtake types: utility PPA, non-utility offtaker, merchant hedge contracts, and direct ownership. Panel (a) describes the share of capacity, while Panel (b) shows the volumes.

Figure A2: Time Trend for Investment: Capacity



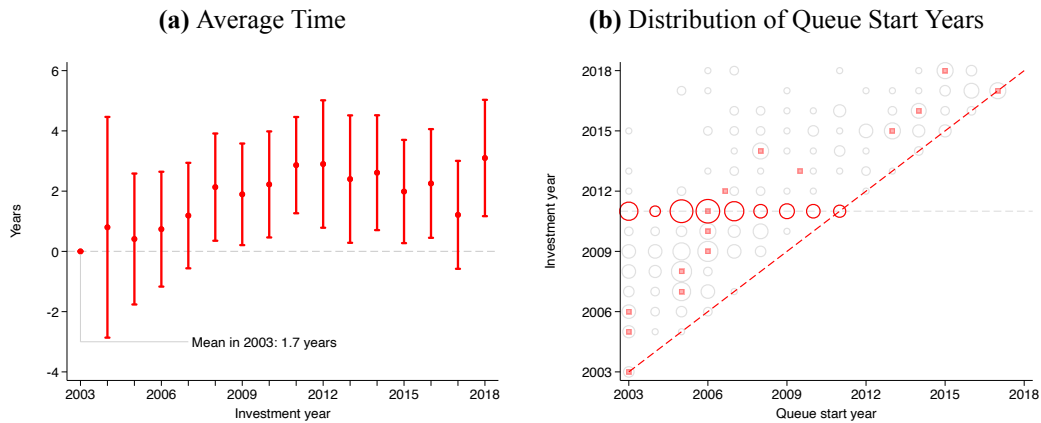
Notes: This figure shows the annual trends of the total capacity and average capacity of new wind projects. I construct the time series based on the data from EIA-860. The red bars in Panel (a) represent the years with policy deadlines.

Figure A3: Construction Time



Notes: This figure shows the time trends of the construction time for new wind projects by their online years (Panel (a)) and construction start years (Panel (b)). I construct the annual time trends of the average construction time from FAA data and EIA-860.

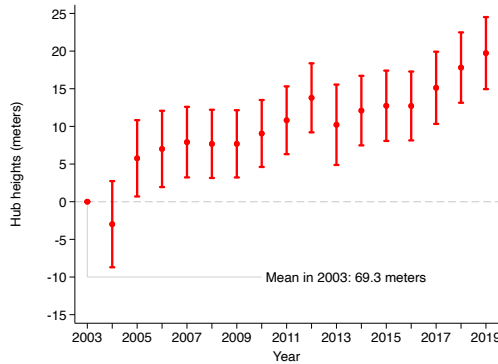
Figure A4: Interconnection Queues



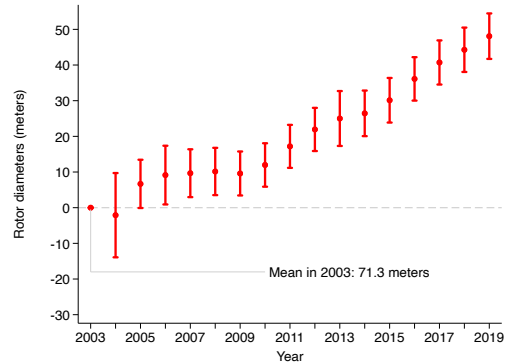
Notes: This figure shows the descriptive evidence for the interconnection queues. Panel (a) plots the average time spent between entering the interconnection queues and starting wind farm construction. Panel (b) plots the distribution of years to start construction and years to enter the queues, where the size of the circles represents the number of wind projects. The interconnection queue data is from Lawrence Berkeley National Laboratory including MISO, SPP, PJM, ISO-NE, NYISO, ERCOT, and CAISO.

Figure A5: Time Trend of Wind Turbine Technology

(a) Hub Heights



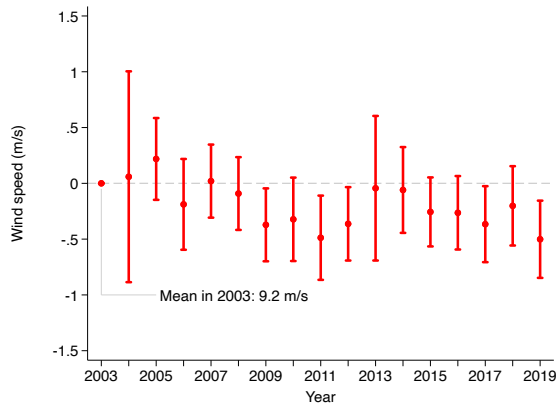
(b) Rotor Diameters



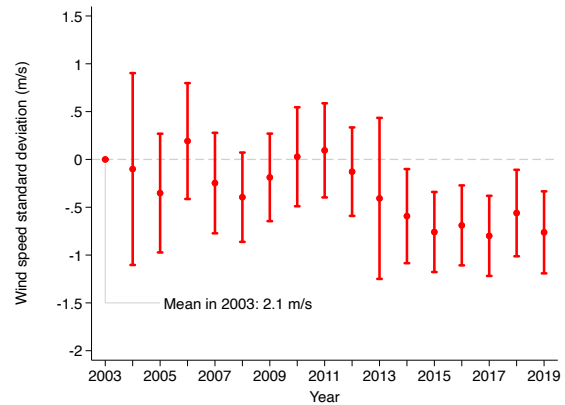
Notes: This figure shows the annual time trends of turbine technologies for new wind projects. I construct the annual time trends of hub heights and rotor diameters from the U.S. Wind Turbine Database (USWTDB) published by USGS.

Figure A6: Time Trend of Wind Speeds

(a) Mean

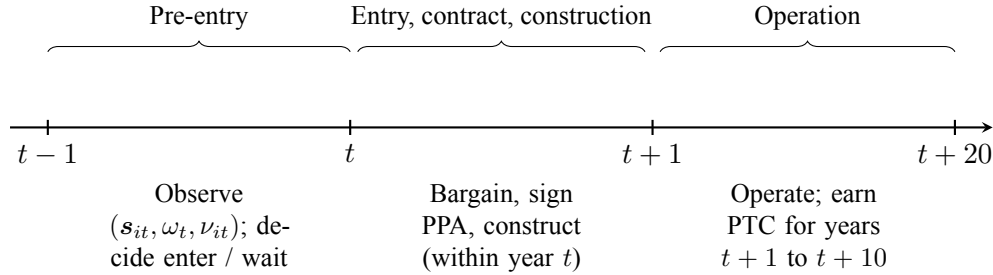


(b) Standard Deviation within Locations



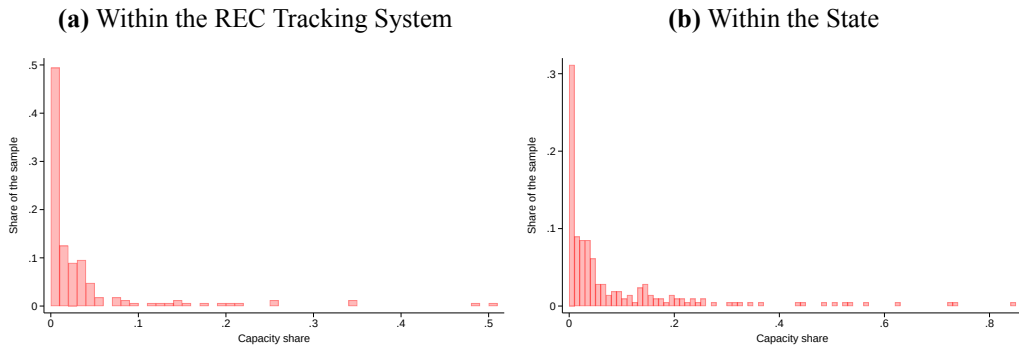
Notes: This figure shows the annual time trends of wind speed at locations of new wind projects. The wind speed is measured at 80 meters at sites nearest to the wind project location based on the Wind Toolkit Data from the National Renewable Energy Laboratory (NREL). The mean and standard deviation for each wind project are measured using hourly wind speed between 2007 and 2013.

Figure A7: Timing of Moves in the Dynamic Entry Model



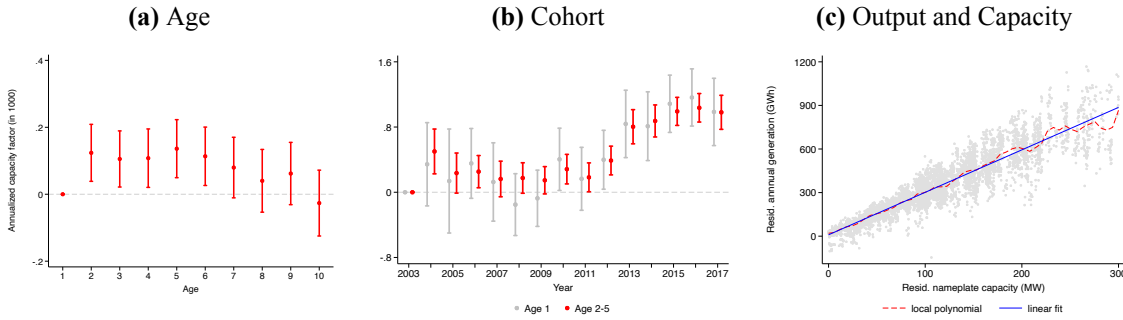
Notes: This figure plots the timeline of the model. Year t is the entry year, encompassing bargaining, contract signing, and construction. Year $t + 1$ is the first year of operation. The PTC is received in years $t + 1$ through $t + 10$ conditional on policy availability.

Figure A8: Utilities' Shares of Contracted Wind Capacity



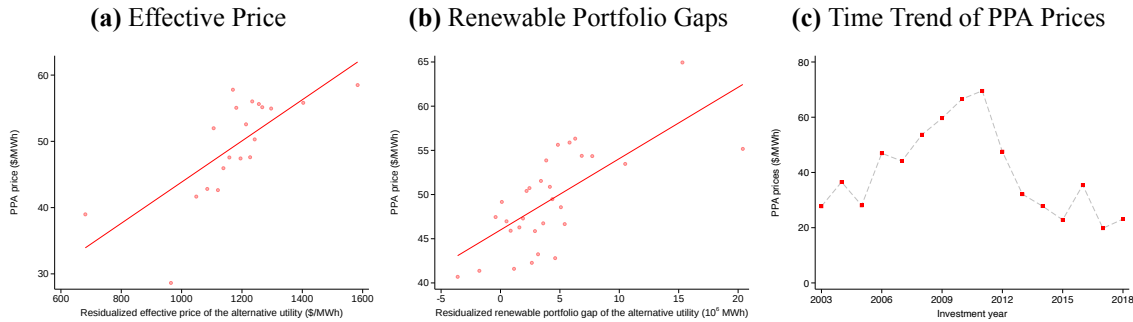
Notes: This figure shows the distribution of each utility's share of the wind capacity contracted by utilities between 2001 and 2019 in the AWEA data. Panel (a) computes the share within the renewable energy credit (REC) tracking system of the utility's state, and Panel (b) within the state. Several states share a tracking system, and the correspondence between states and tracking systems is in Supplemental Appendix Table SA1. Of the utilities, 85% hold less than 5% of the wind capacity contracted in their tracking system, and 65% hold less than 5% of the wind capacity contracted in their state.

Figure A9: Description of Annualized Capacity Factor



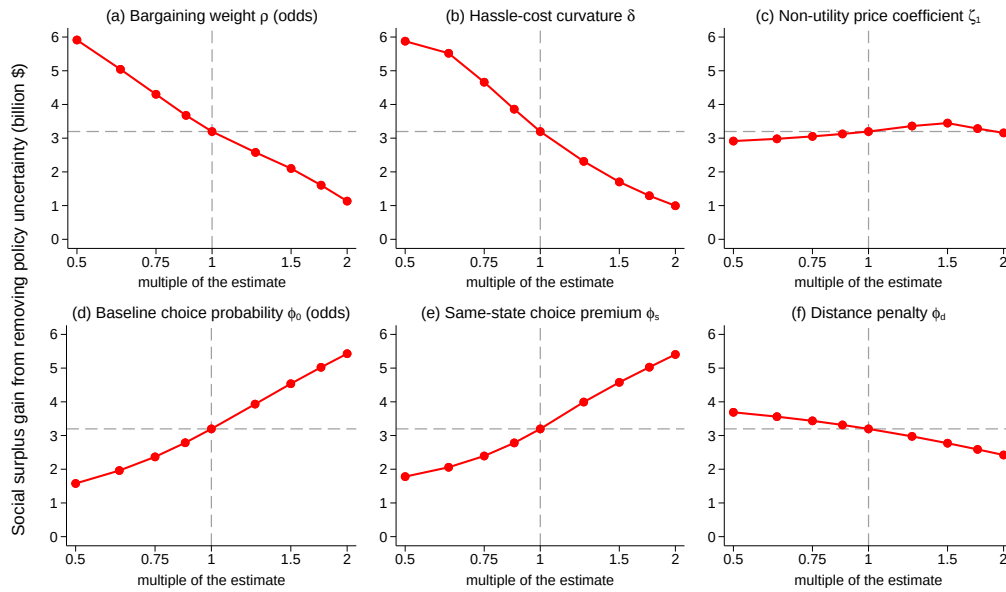
Notes: This figure presents the descriptive data patterns of the annualized capacity factor of wind farms. Panels (a) and (b) explore the relationship among the capacity factors, ages, and cohorts of wind farms. I rescale the annualized capacity factor and divide it by 1000. The average annualized capacity factor is 2.82 at the wind farm and year level. Panel (a) plots the coefficient estimates of β_a in equation (19), controlling for the entry cohort dummies. Panel (b) plots the coefficient estimates of β_c in equation (19), for the groups of wind farms of age 1 and age 2-5 separately. For both Panels (a) and (b), the 95% confidence intervals are constructed from the robust standard errors. Panel (c) shows the relationship between the annual output and the nameplate capacity of wind farms. I residualize both the annual output and the nameplate capacity on entry cohort dummies and age dummies. The scatter plot is at the wind farm and year level. The red solid line is the local polynomial approximation, while the gray dashed line is the linear fit between these two variables.

Figure A10: PPA Price and WTP of Alternative Utilities



Notes: This figure describes the basic pattern of the Power Purchase Agreement (PPA) prices. Panels (a) and (b) show the conditional relationship between PPA prices and willingness-to-pay shifters for the alternative utilities in the pool of potential buyers within 300 miles. Panel (a) shows the relationship between PPA prices and the average effective prices (retail electricity prices and renewable credit prices) for alternative utilities, while Panel (b) shows the relationship between PPA prices and the average renewable portfolio gaps for alternative utilities. Both Panels (a) and (b) control for the renewable portfolio gaps of the focal utility in the bargaining pair, effective market price, estimated unit capacity price, turbine cost, as well as the total capacity for the wind farm and the utility participating in the bilateral negotiation. Panel (c) plots the average time trend of the PPA prices.

Figure A11: Welfare Change from Removing Uncertainty under Various Static Parameters



Notes: This figure shows how the social surplus gain from removing policy uncertainty changes when one static parameter at a time is varied from half to twice its estimate, holding all other static parameters and the estimated dynamic parameters fixed. For each value, static profits are re-simulated and the counterfactual without policy uncertainty is recomputed. The six panels vary the bargaining weight, the hassle-cost curvature, the price coefficient of non-utility demand, the baseline choice probability, the same-state premium, and the distance penalty in buyer choice. The horizontal axis is on a log scale. The dashed horizontal line marks the gain at the estimated parameters, 3.2 billion dollars, and the dashed vertical line marks the estimate.

B Section 1603 Grant

Section 1603 of the American Recovery and Reinvestment Tax Act was implemented as part of the 2009 stimulus package, providing cash grants to qualified energy properties in lieu of tax credits. According to the program guidelines, qualified wind projects had to be “originally placed in service between January 1, 2009, and December 31, 2011, or placed in service after 2011 and before January 1, 2013, if construction of the property begins between January 1, 2009, and December 31, 2011.”³⁵ Wind projects that elected to receive the Section 1603 Grant in lieu of the Production Tax Credit (PTC) received an upfront cash grant equal to 30% of total investment costs. I obtained the list of Section 1603 awardees from the U.S. Department of the Treasury and manually matched the awardees to projects in the EIA data based on wind project names.³⁶ I then constructed a dummy variable indicating whether a project received the Section 1603 Grant. The list of Section 1603 awardees also reports the amount of each Section 1603 award. I first provide descriptive evidence on how subsidy choice varies across market and over time, how the award is associated with the wind farm capacity, and what determines subsidy choices. I then enrich the structural model in Section 4.1 to incorporate subsidy choices, and discuss how it changes the estimates of the bargaining model and the profits of wind farms.

B.1 Descriptive Evidence on Section 1603 Grant

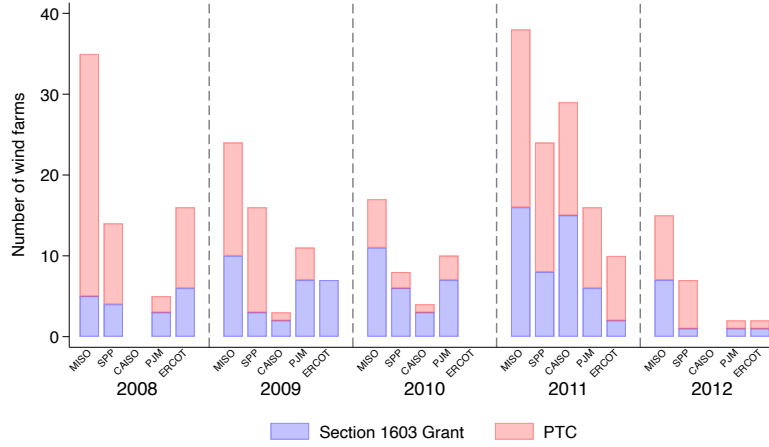
Subsidy choice across markets and years I summarize the subsidy choice patterns by market and by year. Wind farms that elected to receive the Section 1603 Grant account for 48% of all entrants between 2008 and 2012. The share was the highest in 2010 at 70%, and the lowest at 32% in 2008. For the rushed investment year in 2011, 40% of the wind farms in the five markets in Appendix Figure B1 opted into the Section 1603 Grant. The share was the highest in CAISO at 52%, and CAISO alone accounts for 16% of all the awardees in 2011.

Award and wind farm capacity I then investigate how the Section 1603 award is related to wind farm capacity. As shown in Panel (a) of Appendix Figure B2, the award amount is closely approximated by a linear function of total project capacity ($R^2 = 0.93$). I therefore model the total grant as ηk_{ij} , where η captures the award per unit of capacity. I calibrate η by regressing the total grant amount on project capacity without an intercept. The estimated

³⁵Detailed program guidelines are available in the [U.S. Treasury’s Section 1603 program guidance](#).

³⁶The full list of award recipients is available on the [U.S. Treasury’s Section 1603 program webpage](#).

Figure B1: Subsidy Choice across Markets and Years



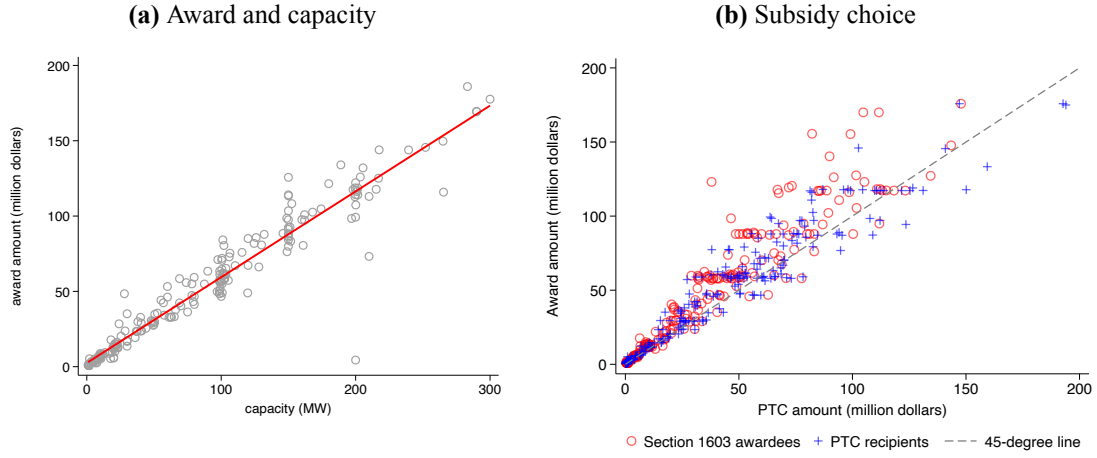
Notes: This figure shows subsidy choice by ISOs and years. The blue bars denote the number of wind projects that opted into the Section 1603 Grant, while the red bars denote the number of wind projects that opted into the PTC.

coefficient is approximately \$0.59 million per megawatt and is stable across years with negligible year-to-year heterogeneity.

Determinants of subsidy choice I further explore what determines the subsidy choice. I estimate a logit model of subsidy choice on the productivity and capacity of wind farms as shown in columns (1)-(2) in Appendix Table B1. More productive wind farms are more likely to select the output-based tax credit conditional on the size of the projects. Moreover, medium-sized wind farms are more likely to choose the Section 1603 Grant, as smaller wind farms have lower total investment costs to claim subsidies and larger wind farms might be less financially constrained and prefer tax credits for tax equity providers. Because wind farm size is an important determinant of the subsidy choice and is itself negotiated through bilateral bargaining, I model the subsidy choice as a joint decision of the two parties in the bilateral bargaining process.

I then calculate total subsidies for each wind farm under both subsidy types. On the one hand, I impute total subsidies under the Section 1603 Grant for each wind farm i that had chosen the PTC as $TS(k_{ij} \mid \text{Section 1603 Grant}) = \eta \times k_{ij}$. On the other hand, I calculate a 10-year discounted sum of total subsidy under the PTC for each wind farm i using its annualized capacity factor α_i , the amount of tax credit per unit of output d_t , and the observed capacity k_{ij} , such that $TS(k_{ij} \mid \text{PTC}) = \frac{\beta(1-\beta^{10})}{1-\beta} d_t \alpha_i k_{ij}$. I summarize the results in Panel (b) of Appendix Figure B2. Wind farms that chose the Section 1603 Grant on average re-

Figure B2: Section 1603 Award and PTC



Notes: This figure shows the data patterns of the Section 1603 Grant and the subsidy choice. Panel (a) displays a scatter plot between the Section 1603 award received by each wind farm and its total capacity. The red solid line denotes the linear fit. Panel (b) shows the total subsidy under the Section 1603 Grant and the PTC for each wind farm. I split the sample into Section 1603 Awardees (red circles) and PTC recipients (blue pluses) and the dashed gray line is the 45-degree line.

ceived a larger amount of federal subsidies under the Section 1603 Grant compared with the PTC. However, wind farms that chose the PTC do not seem better off, as many of them could have obtained a larger amount of federal subsidies if they had opted into the Section 1603 Grant. That wind farms selected the PTC even though there was a more profitable alternative available might be due to unobserved benefits to tax equity providers or behavioral inertia to stick to the default option. In columns (3)-(4) of Appendix Table B1, I include the difference in the total subsidies between these two choices in the logit model and find that on average, if the Section 1603 Grant yields a higher payoff, wind farms are more likely to choose it. However, the coefficient is small in magnitude, which implies the difficulty in explaining the subsidy choice merely through the payoff gaps. Therefore, I assume that wind farm investors choose the default option with some probability, regardless of the associated payoffs. Otherwise, they make a discrete subsidy choice based on the total surplus and an i.i.d. preference shock. This modeling approach not only allows me to partially rationalize the subsidy choice through the payoffs to the two parties in the bilateral bargaining but also allows unobserved preference shocks of wind farms to explain the residual variation.

Table B1: The Determinants of the Subsidy Choices

	Whether Opt to Section 1603			
	(1)	(2)	(3)	(4)
Productivity	-0.311** (0.137)	-0.418*** (0.144)		
log(Capacity)	0.609** (0.274)	0.668** (0.282)		
log(Capacity) ²	-0.091** (0.044)	-0.092** (0.046)		
Difference between Grant and PTC			0.031*** (0.008)	0.039*** (0.008)
Observations	453	453	453	453
Pseudo R^2	0.014	0.054	0.030	0.076
Year Dummies		✓		✓

Notes: This table shows the estimation results of subsidy choices. The regression model is logit and the dependent variable is defined as a dummy, which takes the value 1 if a wind farm chooses to receive the Section 1603 Grant. I calculate the difference in total subsidies (in million dollars) between these two choices. Standard errors are in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

B.2 Structural Model with Subsidy Choice

I incorporate the subsidy choice into the bargaining model in Section 4.1. I keep the notation of that section and describe only the changes.

Wind farms placed in service between 2009 and 2012 could elect the Section 1603 Grant in lieu of the PTC. I denote the subsidy choice by g_{ij} , which equals one for the PTC and zero for the Section 1603 Grant. As wind farms take approximately one year to finish construction, I allow the subsidy choice for wind farms with t between 2008 and 2012, the years in which the awardees are observed, and I set $g_{ij} = 1$ in the other years. Under the PTC, the wind farm receives d_t per unit of output for the first ten years of production. Under the Section 1603 Grant, the wind farm receives a cash grant equal to 30% of the investment cost. I model the grant as ηk_{ij} , where η is the award per unit of capacity calibrated from the data, so that the investment cost is $\eta k_{ij}/0.3$.

Wind farms may value both subsidies below their face values. On the one hand, wind farms typically partner with large investors that finance part of their investment costs in exchange for the tax credits generated by the project. Consequently, wind farms generally value tax credits less than cash subsidies because of transaction costs and asymmetric information arising in the partnership process, as well as the market power of large investors acting as tax equity providers (Johnston, 2019). I use τ to denote the value of one dollar of

tax credits in dollars of cash. On the other hand, wind farms can deduct their investment costs from their tax liability, but receiving a cash grant reduces the amount of investment costs that can be deducted and the associated tax benefits. I use C_1 to denote the present value of the forgone deductions per dollar of investment cost. It is a constant that combines the basis reduction, the marginal tax rate, the discount factor, and the depreciation schedule, and I calibrate it. The deductions are also claimed against tax liabilities, so I discount them by the same factor τ . In the baseline model in Section 4.1, the per-unit value of the PTC, d_t , already incorporates τ . Here, I show explicitly how τ enters the profit function. The total subsidy is therefore

$$TS_{ijt}(k_{ij}, g_{ij}) = g_{ij} \tau \frac{\beta(1 - \beta^{10})}{1 - \beta} d_t \alpha_i k_{ij} + (1 - g_{ij}) \beta \left(1 - \frac{\tau C_1}{0.3}\right) \eta k_{ij} \equiv \Omega_{it}(g_{ij}) k_{ij}, \quad (13)$$

where $\Omega_{it}(g_{ij})$ is the subsidy per unit of capacity. Outside 2008–2012, the total subsidy is the discounted sum of the PTC, scaled by τ .

The profit of wind farm i becomes

$$\pi_t^W(p_{ij}, k_{ij}, g_{ij}) = \frac{\beta(1 - \beta^T)}{1 - \beta} p_{ij} \alpha_i k_{ij} + \Omega_{it}(g_{ij}) k_{ij} - c_{ijt} \alpha_i k_{ij}, \quad (14)$$

where the subsidy term replaces the PTC term in the profit function in Section 4.1.

Wind farm i and utility j negotiate over the procured capacity k_{ij} , the contracted price p_{ij} , and the subsidy type g_{ij} . Under the assumption of Nash bargaining, the optimal capacity k_{ij} and the policy choice g_{ij} maximize the joint surplus, and the optimal price p_{ij} divides the joint surplus between two parties (Chipty and Snyder, 1999). For each subsidy type, the optimal capacity $k_{ij}^*(g_{ij})$ solves the capacity equation with the subsidy per unit of output in place of the PTC term,

$$\sum_{s=t+1}^{t+T} \mathbb{E}_t \beta^{s-t} [p_s^e + p_s^c(1 - z_s) + \delta(\bar{k}_{ijs} - k_{ij}^*)] = \mathbf{X}_{it}' \gamma_1 + \gamma_2 k_{ij}^* - \frac{\Omega_{it}(g_{ij})}{\alpha_i} + \varepsilon_{ij}^k, \quad (15)$$

and, as the outside option $\pi_t^W(p_{ij} = \infty)$ does not depend on the subsidy type, the optimal price follows

$$\begin{aligned} \frac{\beta(1 - \beta^T)}{1 - \beta} p_{ij}^* &= (1 - \rho) \times \sum_{s=t+1}^{t+T} \mathbb{E}_t \beta^{s-t} [p_s^e + p_s^c(1 - z_s) + \delta(\bar{k}_{ijs} - \frac{1}{2} k_{ij}^*)] \\ &\quad + \rho \times \left[\mathbf{X}_{it}' \gamma_1 + \frac{\gamma_2}{2} k_{ij}^* - \frac{\Omega_{it}(g_{ij})}{\alpha_i} + \frac{\pi_t^W(p_{ij} = \infty)}{\alpha_i k_{ij}^*} \right] + \varepsilon_{ij}^p, \end{aligned} \quad (16)$$

where k_{ij}^* is shorthand for $k_{ij}^*(g_{ij})$.

I use $S_{ij}(g_{ij})$ to denote the joint profit $\pi_t^U + \pi_t^W$ evaluated at the optimal capacity $k_{ij}^*(g_{ij})$. The outside options $\pi_t^U(p_{ij} = \infty)$ and $\pi_t^W(p_{ij} = \infty)$ do not depend on the subsidy type, so the comparison between the two options reduces to a comparison of $S_{ij}(0)$ and $S_{ij}(1)$. Motivated by the evidence provided above, I assume that wind farm investors keep the default option, the PTC, with probability ς regardless of the payoffs. Otherwise, the subsidy type maximizes the joint profit plus a preference shock, $g_{ij}^* = \operatorname{argmax}_{g \in \{0,1\}} \{S_{ij}(g) + \sigma_1 \epsilon_{ij}^g(g)\}$, where $\epsilon_{ij}^g(0)$ and $\epsilon_{ij}^g(1)$ are independent draws from the type-1 extreme value distribution. The preference shock captures determinants of the subsidy choice that are unrelated to the payoffs, such as the tax liability of the investors who finance the project and the credit constraints faced by the wind farm investor. Consequently, the probability of electing the Section 1603 Grant is

$$P_{ij}^g(g_{ij} = 0) = (1 - \varsigma) \Lambda\left(\frac{S_{ij}(0) - S_{ij}(1)}{\sigma_1}\right), \quad (17)$$

where $\Lambda(\cdot)$ denotes the logistic function.

Estimation I estimate equations (15), (16), and (17) jointly on the sample of utility power purchase agreements, with the same controls and the same treatment of the outside option as in Section 5.1. The residual of the third equation is the difference between the observed election of the Section 1603 Grant and its predicted probability, and it is defined only for the agreements signed between 2008 and 2012. I stack the residuals of the three equations and minimize their weighted sum of squares, with the weighting matrix given by the inverse of the residual covariance matrix from a first step. I calibrate τ at 0.85 following Johnston (2019) in the main specification, and I also report a specification that estimates it. Relative to the baseline model, the subsidy-choice model adds the probability of keeping the default option ς and the scale of the preference shock σ_1 . The payoff gap $S_{ij}(0) - S_{ij}(1)$ varies across wind farms because the grant is proportional to capacity while the tax credit is proportional to output, so it depends on the capacity factor α_i . The share of wind farms choosing the grant among those with a large payoff gap in favor of the grant identifies $1 - \varsigma$, and the response of the choice share to the payoff gap identifies σ_1 .

Results Appendix Table B2 reports the estimates. Column (1) is the baseline specification reported in Section 6: the credit is valued at $\tau = 0.85$, the subsidy choice is not part of the objective, and the default probability ς is fitted separately from the observed choices as the share of wind farms that kept the credit although the grant paid more. Column (2) drops the control for the outside option of wind farms. Column (3) estimates the

three equations jointly, and column (4) also estimates τ . The bargaining parameters are stable across columns (1), (3), and (4): the hassle cost curvature δ is 0.23–0.24, the turbine price coefficient is 0.058–0.064, and the bargaining weight ρ is 0.61–0.62, so modeling the subsidy choice leaves the pass-through of the credit unchanged. Dropping the control for the outside option lowers ρ to 0.44. The jointly estimated default probability is 0.18 (0.08), and the scale of the preference shock σ_1 is 40 million dollars (20), of the same order as the payoff gap between the two subsidies and consistent with the slope of the reduced-form logit in Appendix Table B1; the large scale reflects the many wind farms that kept the credit although the grant paid more. The estimated value of one dollar of tax credits, $\tau = 0.92$ (0.10), is close to the calibrated value of 0.85 from Johnston (2019).

Quantitative importance of the grant Appendix Figure B3 summarizes the simulated profits of wind farms in the bargaining sample under the two subsidies. Panel (a) plots, for each 2008–2012 contract, the profit under the grant against the profit under the credit. The two lie close to the 45-degree line for most wind farms, and the difference between the two subsidies for a given wind farm is an order of magnitude smaller than the variation in profits across wind farms. Panel (b) plots the average profit by year with both subsidies available, with the credit only, and with no subsidy. Averaged over the 2008–2012 contracts, the availability of the grant raises the profit of wind farms by 17% relative to the credit alone (100 against 86 million dollars), and the two subsidies together raise it from 37 to 100 million dollars relative to no subsidy. Profits rise over the sample period as turbine technology improves, and the gap between the profits with and without subsidies narrows as the credit phases down after 2016.

C Dynamic Model: Estimation and Simulation Details

This appendix details the implementation of the three estimation steps in Section 5.3: constructing the state space and flow-profit surfaces, estimating transition dynamics and solving the dynamic problem, and defining the moments, optimizing the estimation objective, and conducting the bootstrap. It then describes the forward simulation used for the counterfactual analysis in Section 7, presents robustness checks for the dynamic estimates, and examines the sensitivity of the counterfactual results to the treatment of withdrawals from the interconnection queue.

Table B2: Bilateral Bargaining Estimates with the Subsidy Choice

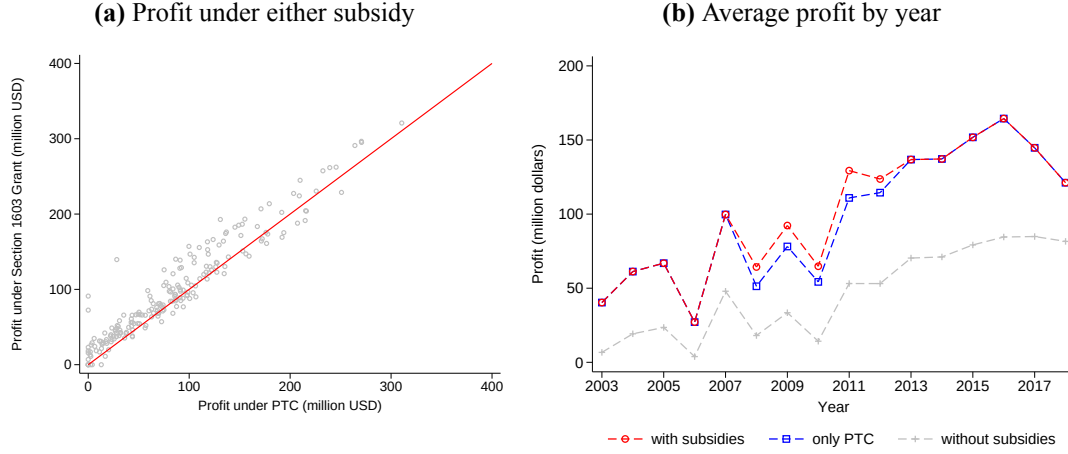
	(1)	(2)	(3)	(4)
<i>Panel A: Utility Willingness to Pay</i>				
Discounted Electricity and Credit Prices	0.018 (0.042)	-0.074 (0.033)	0.022 (0.045)	0.020 (0.045)
Hassle Cost, δ	0.236 (0.052)	0.192 (0.071)	0.237 (0.058)	0.230 (0.060)
<i>Panel B: Wind Farm Cost</i>				
Turbine Price, γ_1	0.058 (0.019)	0.071 (0.019)	0.063 (0.019)	0.064 (0.019)
Unit Capacity Cost Convexity, γ_2	0.012 (0.510)	2.034 (0.993)	0.320 (0.546)	0.311 (0.542)
GE	-11.302 (10.578)	-2.312 (10.347)	-11.619 (10.513)	-10.916 (10.524)
Siemens	3.082 (12.106)	8.668 (11.858)	4.881 (12.159)	3.084 (12.186)
Others	18.824 (12.495)	27.306 (11.877)	17.410 (12.424)	19.099 (12.361)
<i>Panel C: Bargaining and Policy Choice</i>				
Bargaining Weight, ρ	0.606 (0.075)	0.437 (0.066)	0.615 (0.069)	0.612 (0.068)
Default Probability, ς	0.395 (0.040)	0.395 (0.040)	0.180 (0.076)	0.072 (0.142)
Policy Choice, σ_1			0.040 (0.020)	0.055 (0.030)
Credit Valuation, τ				0.918 (0.095)
Observations	402	402	402	402
Calibrated τ	0.850	0.850	0.850	-
§1603 Choice Jointly Estimated			✓	✓
Control for $\pi^W(p_{ij} = \infty)$	✓		✓	✓
Utility-State, Term-Length, Utility-Type FE	✓	✓	✓	✓

Notes: This table shows the estimation results of the bilateral bargaining model with the subsidy choice (equations (15), (16), and (17)), in the parameterization of Appendix Section B.2. Columns (1) and (2) estimate the capacity and price equations with τ calibrated at 0.85 and the default probability ς fitted separately from the observed choices; column (2) drops the control for the outside option of wind farms. Columns (3) and (4) estimate the three equations jointly; column (4) also estimates τ . Standard errors are in parentheses.

C.1 State Space and Flow-Profit Surfaces

State variables The state vector s_{it} of a potential entrant in state m and year t consists of the following variables, measured at the state and year level: (1) the annual average

Figure B3: Simulated Profits under the Section 1603 Grant and the PTC



Notes: Panel (a) plots the simulated profit of each 2008–2012 wind farm in the bargaining sample under the Section 1603 Grant against its profit under the PTC; the red line is the 45-degree line. Panel (b) plots the average simulated profit by year with both subsidies available, with the PTC only, and with no subsidy, at the estimates in column (1) of Appendix Table B2.

productivity of new wind turbines $\bar{\alpha}_t$, measured as the average annualized capacity factor of the wind farms that begin construction in year t ; (2) the average turbine price p_t^{tb} ; (3) the expected electricity price, the discounted sum over a 20-year contract of the expected retail electricity price of the state, $\sum_{s=t+1}^{t+20} \mathbb{E}_t \beta^{s-t} p_{ms}^e$, formed with the transition process of Supplemental Appendix Section SA.4.3; (4) the expected renewable credit price net of the standard, $\sum_{s=t+1}^{t+20} \mathbb{E}_t \beta^{s-t} p_{ms}^c (1 - z_{ms})$; (5) the inclusive value of the renewable portfolio gaps of the potential buyers Φ_{it} , defined in Section 4.3 and averaged over the wind farms of the state; (6) the cumulative wind capacity online in the state K_{mt} ; (7) the subsidy level d_t ; (8) the predicted utility demand, the part of the capacity equation (2) explained by the time-invariant shifters; (9) the non-utility demand shifter, the projection of the non-utility price p_i^{nu} on the demand shifters Z_i^{nu} in equation (4); (10) the matching shifter, the probability that no potential buyer is considered, $\prod_{j \in \mathcal{J}_i} (1 - q_{ij})$ from equation (5); (11) and the state-level land price W_{mt} , the entry cost shifter.

State variables (8)–(10) are time-invariant. The indicator for whether the state has a Renewable Portfolio Standard accompanies the state vector: the renewable credit price and the inclusive value are set to zero in states without a standard.

Basis functions I approximate the flow-profit surfaces and value functions using a polynomial basis $\{u_l(s_{it})\}_{l=1}^L$. The basis includes linear terms, squares, and pairwise cross terms for the four variables that evolve stochastically and enter the bargaining problem directly:

(1), (3), (4), and (5). Variables (2), (6), (8), (9), and (10) enter linearly. The subsidy level (7) enters the basis only in the post-2014 regime, when it follows the legislated phase-down. Before 2015, the subsidy level is constant in real terms, and policy status ω_t captures whether the subsidy is available. Land price (11) enters the value function basis linearly but is excluded from the flow-profit basis because it affects entry costs rather than flow profits. The approximated flow-profit surfaces with and without the subsidy are

$$\hat{\Pi}^1(\mathbf{s}_{it}) = \sum_{l=1}^L \hat{\gamma}_l^{\Pi,1} u_l(\mathbf{s}_{it}) \quad \text{and} \quad \hat{\Pi}^0(\mathbf{s}_{it}) = \sum_{l=1}^L \hat{\gamma}_l^{\Pi,0} u_l(\mathbf{s}_{it}),$$

respectively, and the value functions are approximated as

$$V(\mathbf{s}_{it}) = \sum_{l=1}^L \gamma_l^v u_l(\mathbf{s}_{it}).$$

Flow-profit surfaces I simulate the estimated static model on a grid of states constructed around the observed state-year panel for the fifteen sample states from 2007 to 2018. For each state-year cell, I draw 50 perturbations of the state-level primitives. Expected electricity prices, expected renewable credit prices, turbine prices, and turbine productivity receive independent normal shocks, each with a standard deviation equal to one quarter of the residual standard deviation of its corresponding transition process in Supplemental Appendix Table SA7. The renewable portfolio gaps of utilities within the state are scaled by a common log-normal factor with a mean of one and a log standard deviation of 0.5.

At each of the 9,000 grid points, I solve the bargaining problem with and without the subsidy for every wind farm–potential buyer pair in the cell. I then evaluate the consideration probabilities and the non-utility channel to compute the *ex-ante* profit of entry in equation (7) under each policy regime, along with expected contracted capacity. I project the simulated profits and capacity onto the polynomial basis using ordinary least squares, separately for the pre-2015 and post-2014 periods. Finally, I impose a lower bound of zero on predicted profits without the subsidy and require predicted profits with the subsidy to be at least as large as those without it.

C.2 Transition Dynamics and Solution of the Dynamic Problem

Transition dynamics Potential entrants perceive state variables as following first-order autoregressive processes. I estimate these processes using data for the fifteen sample states from 2007 to 2018 and report the results in Supplemental Appendix Table SA7. Turbine

productivity and turbine prices are national series, each modeled as an AR(1) process with a constant. Expected electricity prices and land prices follow AR(1) processes with state fixed effects. Expected renewable credit prices and the inclusive value of renewable portfolio gaps follow AR(1) processes that also depend on cumulative wind capacity in the state. I estimate these last two processes using states with a Renewable Portfolio Standard; both variables are zero in the remaining states. The subsidy level evolves deterministically along the legislated schedule: the dynamic state specifies \$22 per MWh through 2016, while the 2016, 2017, and 2018 investment cohorts receive 80%, 60%, and 40% of that amount, respectively. Innovations are independent normal shocks with standard deviations equal to the residual standard deviations of the transition regressions.

Cumulative wind capacity evolves endogenously. Potential entrants perceive its evolution as the AR(1) process in (12), whose coefficients (ρ_1^K, ρ_0^K) are determined by a fixed point rather than estimated from the data. For each year t from 2009 to 2014, I simulate the model over a six-year window beginning in t , using the model's predicted entry to update cumulative capacity. Renewable credit prices and the inclusive value evolve according to their transition processes, while the pool of potential entrants is held fixed at its year- t level. I then fit the AR(1) process to the simulated capacity path and update its coefficients with damping, repeating the procedure until convergence. For 2015–2018, I instead use an AR(1) process fitted to observed cumulative capacity from 2014 to 2018.

Value functions I solve the value functions on a collocation grid of 9,000 points, constructed by replicating the observed state-year panel 50 times and applying the same perturbations used for the flow-profit grid. Given a candidate coefficient vector γ^v , I evaluate the right-hand side of the Bellman equation in (10) or (11) at each grid point. The logistic entry cost shock allows this expectation to be evaluated in closed form. I then regress the resulting values on the polynomial basis to update the coefficients and repeat until the maximum absolute change in value across grid points falls below 10^{-3} .

For the pre-2015 period, I solve the stationary value function without the subsidy, V^0 , and each year's belief-weighted value function, $V^1(\cdot; b_t)$, which incorporates V^0 . These solutions use the belief-consistent transition process for cumulative capacity in the corresponding year. For 2015–2018, I first solve the stationary terminal value function for 2019, with a residual credit equal to 40% of the statutory level. I then solve the value functions for 2018, 2017, 2016, and 2015 by backward induction along the statutory phase-down. During estimation, I reject any parameter vector for which a value function fails to converge.

Predicted entry I evaluate the entry probability in (9) at the observed states in each state-year cell, using the value functions above to compute the value of waiting. The predicted number of entrants equals this probability multiplied by the pool of potential entrants, N_{mt} . Consistent with the equilibrium concept, I update the endogenous states during estimation using predicted entry. Cumulative wind capacity increases by the predicted number of entrants multiplied by expected capacity per entrant. The renewable credit price and the inclusive value evolve according to their transition processes, conditional on simulated cumulative capacity. The pool of potential entrants decreases by predicted entry and increases by the observed net inflow into the interconnection queue.

C.3 Moments, Optimization, and Inference

I minimize the sum of squared differences between observed and predicted entry across three sets of moments: national totals by year for 2009–2018, market totals over this period for MISO, SPP, PJM, CAISO, and ERCOT, and market totals in 2011. I jointly estimate market-specific entry costs evaluated at average land prices, $\kappa_{0,r} + \kappa_1 \bar{V}_r$, entry cost shock scale σ_ν , land price coefficient κ_1 , and renewal beliefs. I use particle swarm optimization followed by pattern search and retain the best solution across five starting points.

I compute standard errors using 100 state-block bootstrap draws, resampling states within SPP, MISO, and PJM while retaining California and Texas in every draw. For each draw, I re-estimate the transition processes, flow-profit surfaces, and dynamic parameters, holding market average land prices fixed at their sample values. Standard errors are the standard deviations of the bootstrap estimates and are conditional on the static estimates.

C.4 Simulation Details of the Counterfactual Analysis

The counterfactual analysis simulates the model forward from 2009 to 2018 using the same components as the estimation. Exogenous states, including turbine productivity, turbine prices, expected electricity prices, land prices, and time-invariant shifters, follow their observed paths. In the main counterfactual, only policy beliefs change. Endogenous states evolve with simulated entry. Cumulative capacity increases by the number of entrants multiplied by expected capacity per entrant, while renewable credit prices and the inclusive value follow their transition processes conditional on simulated cumulative capacity. The pool of potential entrants decreases by simulated entry and increases by the observed net

inflow into the interconnection queue:

$$N_{m,t+1} = \max\{0, N_{mt} - \hat{E}_{mt} + (N_{m,t+1}^{obs} - N_{mt}^{obs} + E_{mt}^{obs})\}.$$

Simulated entry $\hat{E}_{mt}$ equals the pool of potential entrants multiplied by the entry probability. For each scenario, I solve the perceived transition of cumulative capacity as a belief-consistent fixed point for each year before 2015 and solve the post-2014 value functions by backward induction, as in the estimation. The baseline uses the estimated beliefs, while the counterfactual without renewal uncertainty sets $b_t = 1$ in every year. At the point estimates, the baseline simulation reproduces the entry predicted during estimation.

Welfare accounting follows Table 3. A wind farm entering in year t with capacity k produces αk annually for 20 years. I discount its lifetime output to the entry year using the annuity factor $\sum_{s=1}^{20} \beta^s$ and discount all dollar values to 2009, with $\beta = 0.95$. Environmental benefits are calculated using the state's marginal operating emissions rate and a social cost of carbon of \$80 per ton. Other benefits are calculated using the wholesale market value of wind output, as described in Supplemental Appendix Section SA.6. Subsidies cover the first ten years of output. Turbine costs equal capacity multiplied by the turbine price in the entry year, and entry costs equal their deterministic component, $\kappa_{0,r} + \kappa_1 W_{mt}$. Wind farm profits are measured by the *ex-ante* profit of entry in equation (7), which is net of turbine costs, less the entry cost; net profit further subtracts the subsidy. Social surplus equals total benefits less turbine costs, entry costs, and subsidies.

C.5 Robustness of the Dynamic Estimates

This subsection presents the five robustness exercises described in Section 5.3 and the welfare effect of removing renewal uncertainty under each specification. All five use column (3) of Table 2, which estimates beliefs for 2009–2014, as the reference specification. Appendix Figure C1 also reports the welfare effects for all four specifications in Table 2.

Entry cost trend The baseline model allows entry costs to vary over time only through state-level land prices. I add a linear trend common to all markets to capture gradual changes in development frictions, such as growing siting resistance or longer permitting processes. Entry costs become $\kappa_{0,r(i)} + g(t - 2013.5) + \kappa_1 W_{it}$, where the market-specific parameters represent cost levels centered in 2013–2014 and $g = 0$ recovers the original specification. This extension adds one parameter and introduces calendar year as a state variable. Appendix Table C1 reproduces the original estimates in column (1) and reports estimates with

the trend in column (2). The estimated $g = 1.48$ million dollars per year implies that entry costs rise by approximately 13 million dollars from 2009 to 2018. Estimated renewal probabilities also increase, most notably from 0.40 to 0.59 in 2011. Because rising entry costs independently encourage earlier investment, the model attributes less of the observed early entry to the risk of subsidy expiration and therefore estimates less pessimistic renewal beliefs.

Post-2014 belief The baseline assumes that the subsidy follows the legislated phase-down with certainty. I relax this assumption by adding a single belief parameter, b_{post} , to the 2009–2014 specification. In each year after 2014, investors assign probability b_{post} to the phase-down continuing into the following year and $1 - b_{\text{post}}$ to permanent subsidy expiration; $b_{\text{post}} = 1$ recovers the baseline. Value functions are computed by backward induction, with continuation values weighted by these probabilities. Column (3) of Appendix Table C1 reports an estimate of $b_{\text{post}} = 0.94$, implying a perceived annual expiration probability of 6%. The estimate is not significantly different from one, consistent with the baseline certainty assumption. Other estimates change little: the 2011 renewal belief rises from 0.40 to 0.45, the remaining beliefs change by less than 0.01, and the land-price coefficient and entry cost shock scale remain stable.

Terminal subsidy The Consolidated Appropriations Act of 2015 extended the PTC through 2019 with a pre-announced phase-down. Projects starting construction by the end of 2016 receive the full subsidy, while those starting construction in 2017, 2018, and 2019 receive 80%, 60%, and 40%, respectively. I assume that developers treat this schedule as certain during 2015–2018. From 2019 onward, developers expect a permanent subsidy at 40% of the full rate, and the terminal value is the stationary value under that subsidy.

This terminal assumption is a simplification. Developers may have anticipated either further extensions or expiration, and the PTC was subsequently extended. The residual subsidy approximates these expectations. To assess sensitivity to this assumption, I re-estimate the model with terminal subsidies of 30%, 35%, 45%, and 50% of the full rate (columns (4)–(7) in Appendix Table C1). Entry cost parameters and renewal beliefs outside 2011 remain stable. The 2011 belief rises from 0.253 at a 30% terminal subsidy to 0.536 at 50%, compared with 0.395 in the baseline. Across all specifications, it remains well below the beliefs in adjacent years, which exceed 0.9. The finding of substantial perceived renewal uncertainty in 2011 therefore remains robust to the terminal assumption.

Reinstatement probability I use ω_t to indicate whether the subsidy is available in year

t . Ex post, $\omega_t = 1$ throughout because the PTC was repeatedly renewed despite temporary lapses. Assumption 1 treats expiration as permanent, making V^0 the value under permanent subsidy removal. I relax this assumption by allowing an expired subsidy to be reinstated with annual probability q . The value functions become

$$V^0(s_{it}) = \sigma_\nu \log \left(\exp \left[\frac{\Pi^0(s_{it})}{\sigma_\nu} \right] + \exp \left[\frac{\beta}{\sigma_\nu} \mathbb{E}_t [(1-q)V^0(s_{it+1}) + qV^1(s_{it+1}; b_t) \mid s_{it}] \right] \right),$$

$$V^1(s_{it}; b_t) = \sigma_\nu \log \left(\exp \left[\frac{\Pi^1(s_{it})}{\sigma_\nu} \right] + \exp \left[\frac{\beta}{\sigma_\nu} \mathbb{E}_t [b_t V^1(s_{it+1}; b_t) + (1-b_t)V^0(s_{it+1}) \mid s_{it}] \right] \right).$$

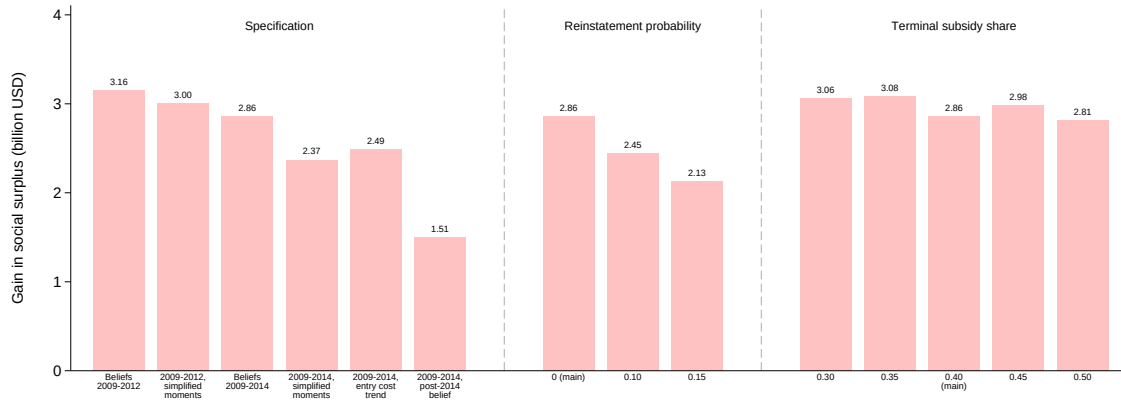
A higher q raises the value of waiting after expiration, requiring a lower estimated renewal probability b_t to rationalize the same early investment. Investment timing therefore identifies a combination of q and b_t , motivating the baseline normalization $q = 0$. Under this normalization, renewal beliefs are measured against permanent expiration; when $q > 0$, they are measured against a potentially temporary lapse. In the certain-renewal counterfactual, $b_t = 1$ throughout, so q does not directly affect investment decisions.

Columns (8)-(9) in Appendix Table C1 report estimates for $q = 0.1$ and $q = 0.15$. Under $q = 0.1$ in column (8), entry cost parameters remain close to the baseline, while renewal beliefs fall in every year. The 2011 belief declines from 0.395 to 0.058, but these estimates refer to different alternatives: permanent expiration in the baseline and a lapse with a 10% annual reinstatement probability in the extension. Because reinstatement makes expiration less costly, a lower renewal probability is needed to explain the observed rush to invest. Raising q to 0.15 in column (9) further lowers renewal beliefs, with the 2011 estimate reaching zero, alongside poorer model fit and shifts in estimated entry costs.

Welfare effects Appendix Figure C1 plots the gain in social surplus from removing renewal uncertainty, the difference between the scenario without renewal uncertainty and the baseline scenario in the accounting of Table 3, at every set of estimates: the four specifications of Table 2 and the specifications with an entry cost trend and a post-2014 belief in the left group, the reinstatement probabilities in the middle group, and the terminal subsidy levels in the right group. Removing renewal uncertainty raises the social surplus in every case, by 2.4 to 3.2 billion dollars across the columns of Table 2, by 2.5 and 1.5 billion dollars with an entry cost trend and a post-2014 belief, by 2.1 to 2.9 billion dollars across the reinstatement probabilities, and by 2.8 to 3.1 billion dollars across the terminal subsidy levels. The gain is smallest with a post-2014 belief because my sample period ends in 2018 and is not long enough to capture all the benefits of delayed entry, and the welfare gain varies little with the terminal subsidy level.

Discount factor and risk aversion Appendix Figure C2 examines how the estimated 2011 renewal belief varies with discounting and investor risk aversion. Panel (a) varies the discount factor from 0.80 to 0.97 and recovers the belief that matches model’s predicted entry in 2011, holding all other parameters at their base estimates. The belief falls as the discount factor declines, from 0.40 at 0.95 to 0.30 at 0.85 and 0.17 at 0.80. Renewal beliefs affect entry through the discounted value of waiting; in this exercise, heavier discounting requires more pessimistic beliefs to match the observed rush to invest. The moment fit is best at discount factors of 0.95–0.96. Panel (b) replaces the expected continuation value under the policy lottery with its certainty equivalent for an investor with constant absolute risk aversion a , measured per million dollars, and re-estimates the model. Risk aversion and renewal beliefs are not separately identified from entry timing because both risk aversion and pessimistic beliefs lower the value of waiting. The baseline beliefs should therefore be interpreted as risk-adjusted probabilities. As a increases from 0.01 to 0.10, the estimated 2011 belief rises from 0.48 to 0.61, while other beliefs and entry cost parameters change little and the objective improves slightly. The counterfactual eliminates the policy lottery, removing the direct role of aversion to policy risk. However, the welfare cost in Table 3 excludes the risk premium and therefore provides a lower bound on the cost of uncertainty for risk-averse investors.

Figure C1: Gain in Social Surplus from Removing Renewal Uncertainty by Specification



Notes: This figure plots the gains in social surplus from removing policy renewal uncertainty across alternative model specifications. Each bar is the social surplus in the scenario without renewal uncertainty minus the social surplus in the baseline scenario, in the accounting of Table 3, at one set of dynamic estimates, in billion dollars discounted to 2009 with a discount factor of 0.95. The left group covers the four columns of Table 2 and the specifications with an entry cost trend and a post-2014 belief in Supplemental Appendix Table SA8. The middle group varies the reinstatement probability (Supplemental Appendix Table SA9) and the right group the terminal subsidy share (Supplemental Appendix Table SA10).

Table C1: Robustness of the Dynamic Estimates to the Specification

	(1) Base	(2) Entry Cost Trend	(3) Post-2014 Belief	(4)	(5) Terminal Subsidy Share	(6)	(7)	(8) Reinstatement Probability	(9)
<i>Panel A: Specification</i>									
Entry Cost Trend	0	1.48 (1.89)	0	0	0	0	0	0	0
Policy Belief After 2014	1	1	0.940 (0.278)	1	1	1	1	1	1
Terminal Subsidy Share	0.40	0.40	0.40	0.30	0.35	0.45	0.50	0.40	0.40
Reinstatement Probability	0	0	0	0	0	0	0	0.10	0.15
<i>Panel B: Belief Parameters</i>									
Policy Belief 2009	0.808 (0.242)	0.925 (0.166)	0.812 (0.212)	0.743 (0.226)	0.781 (0.245)	0.840 (0.217)	0.867 (0.242)	0.402 (0.269)	0.285 (0.277)
Policy Belief 2010	0.969 (0.031)	0.997 (0.025)	0.967 (0.029)	0.953 (0.031)	0.959 (0.031)	0.974 (0.029)	0.983 (0.031)	0.907 (0.063)	0.897 (0.060)
Policy Belief 2011	0.395 (0.233)	0.585 (0.253)	0.448 (0.139)	0.253 (0.222)	0.361 (0.213)	0.499 (0.248)	0.536 (0.229)	0.058 (0.115)	0.001 (0.084)
Policy Belief 2012	0.959 (0.033)	0.981 (0.028)	0.953 (0.044)	0.940 (0.040)	0.949 (0.039)	0.964 (0.043)	0.973 (0.044)	0.884 (0.110)	0.861 (0.078)
Policy Belief 2013	0.949 (0.050)	0.973 (0.046)	0.945 (0.080)	0.935 (0.058)	0.944 (0.063)	0.957 (0.059)	0.963 (0.061)	0.891 (0.101)	0.860 (0.101)
Policy Belief 2014	0.959 (0.063)	0.977 (0.058)	0.953 (0.066)	0.950 (0.059)	0.957 (0.065)	0.968 (0.064)	0.972 (0.072)	0.926 (0.095)	0.902 (0.118)
Objective Value	181.54	145.60	122.09	240.84	206.08	153.78	133.18	176.52	203.65

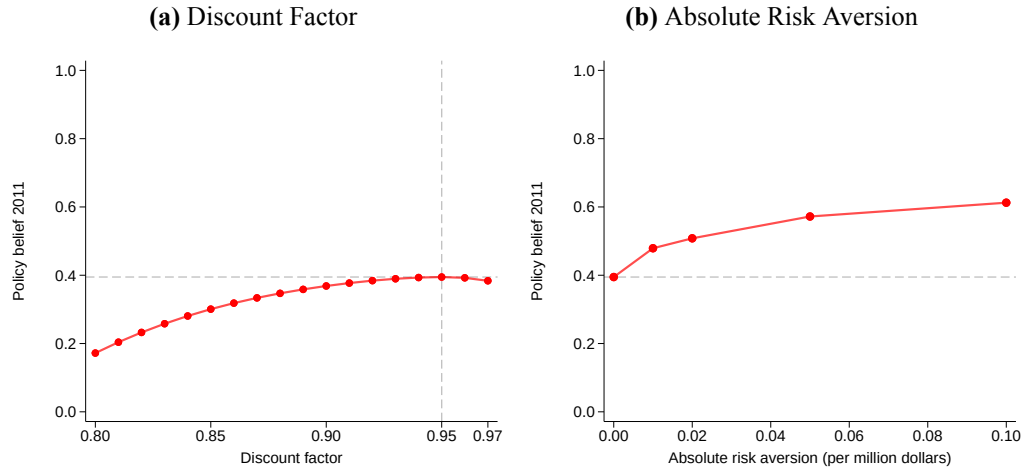
Notes: Column (1) is column (3) of Table 2. Column (2) adds a linear time trend to the entry costs, in million dollars per year and centered in 2013–2014. Column (3) adds the perceived probability that the legislated phase-down continues in the following year after 2014. Columns (4)–(7) evaluate the 2019 terminal value function at a residual credit equal to the stated share of the statutory level. Columns (8)–(9) allow an expired credit to be reinstated in the following year with the stated probability, in place of Assumption 1. The entry cost parameters are estimated in every column and are omitted here and reported in Supplemental Appendix Tables in Supplemental Appendix Table SA8-10. Standard errors are block-bootstrapped over states with 100 draws.

C.6 Queue Withdrawals and the Pool of Potential Entrants

The counterfactual analysis holds queue inflows and withdrawals at their observed levels in both scenarios, so the pool of potential entrants differs only through modeled entry. If renewal uncertainty also induced withdrawals, the pool without uncertainty would be larger than simulated. The data suggest this possibility: withdrawals peaked in 2012, the year of the lapse, when 42% of potential entrants at the start of the year withdrew, compared with annual withdrawal rates of 11% in 2009–2010 and 21% in 2015–2017.

To assess the implications, I allow a fraction λ of the 2012 withdrawals to remain in the queue in the scenario without renewal uncertainty. Setting $\lambda = 0$ recovers the main counterfactual, while $\lambda = 1$ attributes all 2012 withdrawals to renewal uncertainty. Intermediate

Figure C2: The 2011 Belief by Discount Factor and Risk Aversion



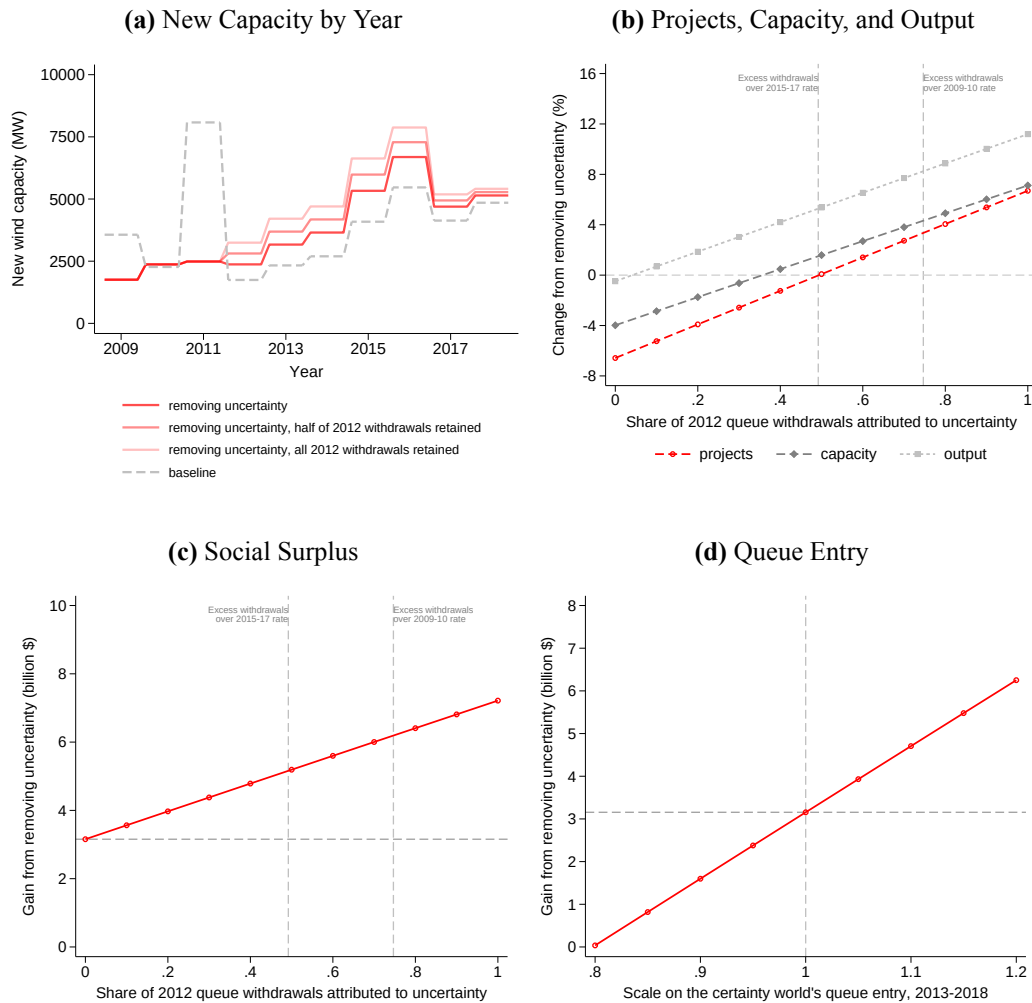
Notes: Panel (a) plots the 2011 belief that reproduces the model's predicted entry in 2011 for discount factors between 0.80 and 0.97, holding the other parameters at the estimates of column (3) of Table 2; the dashed lines mark the discount factor of 0.95 and the estimated belief. Panel (b) plots the estimated 2011 belief when the continuation value under the policy lottery is replaced by its certainty equivalent for an investor with constant absolute risk aversion, in units per million dollars; the dashed line marks the risk-neutral estimate.

values of $\lambda = 0.49$ and $\lambda = 0.75$ approximately restore the withdrawal rates observed in 2015–2017 and 2009–2010, respectively. Retained projects remain in the potential entrant pool in 2012, subsequently face the same entry probability as other projects in the pool, and withdraw at observed state-year rates from 2013 onward. The parameter estimates remain unchanged because estimation conditions on the observed pool.

Appendix Figure C3 reports the results. Panel (a) plots annual new capacity under the baseline and without renewal uncertainty for $\lambda = 0, 0.5$, and 1. Retention does not affect the initial rush to invest because it changes the pool only from 2012 onward. It instead increases subsequent entry in the scenario without uncertainty, when turbine technology has improved. Panels (b) and (c) show how the effects of removing uncertainty on project numbers, capacity, output, and social surplus vary with λ . All four increase monotonically and approximately linearly. Removing uncertainty generates more output than the baseline once λ exceeds approximately 0.04 and more projects once it exceeds 0.49, reversing the 6.6% decline in project numbers in the main counterfactual. The welfare cost of uncertainty rises from 3.2 billion dollars at $\lambda = 0$ to 5.2 billion dollars at $\lambda = 0.49$ and 6.2 billion dollars at $\lambda = 0.75$. Panel (d) examines queue inflows by scaling new applications in 2013–2018 by a factor between 0.8 and 1.2 in the scenario without uncertainty. The welfare gain from

removing uncertainty increases with this factor, from 0.04 billion dollars at 0.8 to 6.3 billion dollars at 1.2. Thus, if removing renewal uncertainty would have retained or attracted more projects than observed, the main counterfactual understates the welfare cost of uncertainty through these channels.

Figure C3: Removing Renewal Uncertainty under Alternative Assumptions about the Queue



Notes: This figure plots the results under alternative assumptions about the queue. Panel (a) shows the simulated new wind capacity by year in the baseline scenario (gray dashed line) and in the scenario without renewal uncertainty when $\lambda = 0, 0.5$, and 1 of the 2012 withdrawals from the interconnection queue are retained. Panel (b) shows the percentage changes in the number of projects, the total capacity, and the total output from removing renewal uncertainty as functions of λ , and Panel (c) shows the change in the social surplus in billion dollars. The two dashed vertical lines mark $\lambda = 0.49$ and $\lambda = 0.75$, which reduce the 2012 withdrawal rate to its 2015–2017 and its 2009–2010 levels, respectively. Panel (d) shows the change in the social surplus when the new queue applications of 2013–2018 in the scenario without renewal uncertainty are scaled by the factor on the horizontal axis; the dashed lines mark the main counterfactual.

Supplemental Appendix for The Dynamic Efficiency of Policy Uncertainty: Evidence from the Wind Industry

SA.1 PPA Data

The main data set I use for the static model is from the AWEA (American Wind Energy Association, now American Clean Power Association), which includes the Power Purchase Agreement (PPA) data in the U.S. wind industry. The wind capacity coverage is complete in the AWEA data, as the aggregate capacity aligns well with that from the EIA data across years (Panel (a) of Supplemental Appendix Figure [SA1](#)).

I keep the PPA data with utilities as the power purchasers from 2001 to 2019. The data is at the contract and purchaser level, and there are 721 observations in total. However, 13.5% of the observations do not have valid utility names and 4.7% of the observations miss valid wind farm IDs to be matched with the EIA data. Among observations without valid utility names, 20.6% only label the power purchasers as “City,” and 12.4% are flagged as “Undisclosed.” Among 34 wind farms without valid wind farm IDs, 64.7% have a total capacity of less than 5 MW. Otherwise, the missing pattern appears to be idiosyncratic. Comparing the total capacity and contract lengths between sub-samples with and without missing IDs as shown in Panels (c) and (d) of Supplemental Appendix Figure [SA1](#), the overall distributions resemble each other, although the contracts with missing IDs seem to have slightly smaller procured capacity.

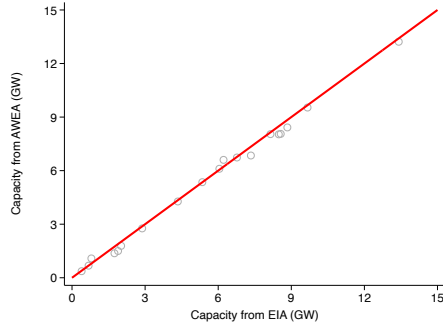
There are 36.3% of the contracts missing price information among all the contracts with valid utility names and wind farm IDs. I follow [Aldy, Gerarden, and Sweeney \(2023\)](#) and impute the missing PPA prices from the resale revenues and quantities reported in the EIA Form 923 from 2011 to 2019. By comparing the prices of wind farms whose price information is available both from EIA and AWEA as shown in Panel (b) of Supplemental Appendix Figure [SA1](#), I find they align well with each other.

SA.2 REC Price Data

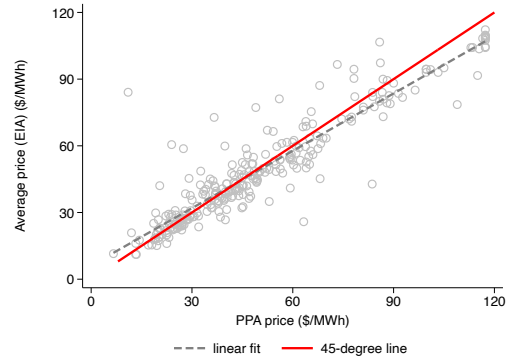
I obtain the Renewable Energy Credit (REC) price data between 2006 and 2019 from a financial service platform Marex. I calculate the REC price estimates in a given state and

Figure SA1: Data Description of the PPA Sample

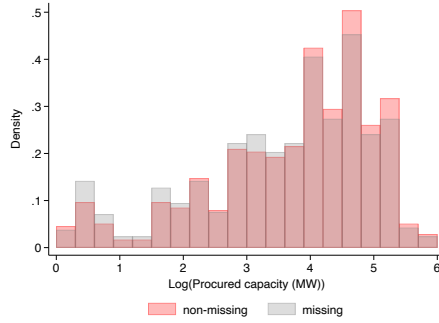
(a) Comparison: Aggregate Capacity



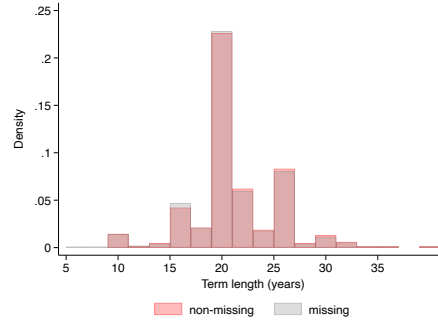
(b) Comparison: PPA Prices



(c) Log(Procured Capacity)



(d) Term Length



Notes: This figure presents the results of the data description for the PPA sample. Panels (a) and (b) show the results of the data quality cross-check between AWEA and EIA. Panel (a) plots the annual aggregate new capacity from EIA and AWEA. The red solid line denotes the 45-degree line. Panel (b) plots the PPA prices from EIA and AWEA for each wind farm. The gray dashed line denotes the linear fit, while the red solid line denotes the 45-degree line. I calculate the average price from the EIA-923 using the resale price in 2011-2019 for each wind farm following [Aldy, Gerarden, and Sweeney \(2023\)](#). Panels (c) and (d) show the distributions of the log procured wind capacity and the contract duration for two sub-samples respectively. The “non-missing” group denotes the AWEA sub-sample that matches both utility IDs and wind farm IDs with the EIA, and the “missing” group denotes the AWEA sub-sample with either unmatched utility IDs or unmatched wind farm IDs.

year by taking the average between bids and asks from all active REC markets following [Aldy, Gerarden, and Sweeney \(2023\)](#). However, only 14 states have available information from Marex and the time coverage also varies across states. I take two steps to impute REC prices for active REC states with missing data. First, for the 14 states covered by Marex, I

run the following regression to predict their REC prices in years with missing values.

$$y_{mt} = \beta_m^{rec} \times t + \xi_m + \epsilon_{mt}.$$

y_{mt} denotes the REC prices in state m and year t . ξ_m is the state fixed effect. I extrapolate the REC prices for those years with missing values from the estimated state-specific time trends β_m^{rec} .

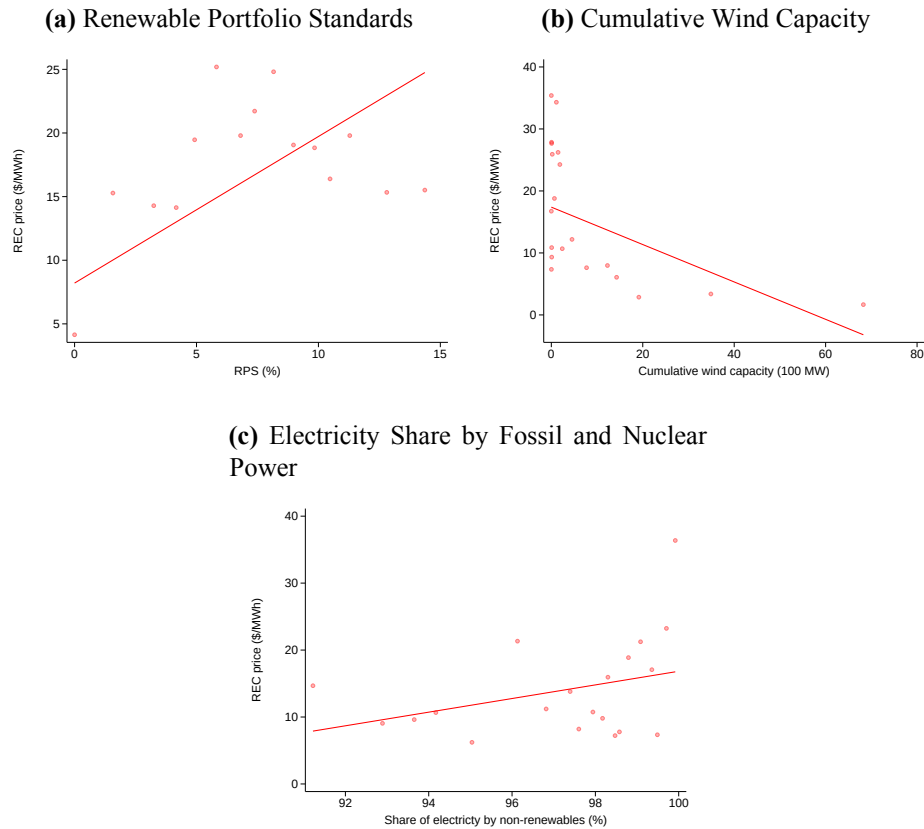
Second, I extrapolate the REC prices in other active REC states. State-level Renewable Portfolio Standards typically stipulate a minimum share of renewable-sourced electricity out of the total generation for each utility, and utilities need to purchase additional RECs if they fall short of the standards. The demand for the RECs is shifted by the stringency of the Renewable Portfolio Standards as well as the volume of electricity generated by non-renewable sources, while the supply of the RECs comes from new wind capacity addition and the entry of other renewable sources. Supplemental Appendix Figure SA2 demonstrates that the REC prices are positively correlated with the stipulated ratios in the Renewable Portfolio Standards, as well as the share of electricity generated from fossil fuels and nuclear energy, and they are negatively correlated with the amount of the existing wind capacity.

Moreover, the trading of the RECs is fragmented into different markets such that the credits are registered to be traded only in the corresponding tracking systems, as shown in Supplemental Appendix Table SA1 based on Table 1 of [Abito, Flores-Golfin, Hinchberger, van Benthem, and Vasey \(2025\)](#). The tracking system fixed effects could explain around 60% of the REC price variations. Therefore, I estimate the following regression and predict the REC prices for the rest of the active REC states.

$$y_{mt} = \beta \mathbf{X}_{mt} + \gamma_{kt} + \epsilon_{mt}. \quad (18)$$

y_{mt} denotes the REC prices in state m and year t . $\mathbf{X}_{mt}$ includes the RPS in year t , the cumulative wind capacity in state m and year t , as well as the share of electricity generated out of non-renewable sources. The corresponding tracking system of state m is denoted by k , and γ_{kt} is the tracking-system-by-year fixed effects. Therefore, I extrapolate the REC prices based on both observables and the time trend specific to the tracking system. For states where no price in the corresponding tracking system is available, I impute the REC prices with a national average in that year excluding the New England Power Pool (NEPOOL) because the REC prices in NEPOOL are an order of magnitude higher than the rest of the markets.

Figure SA2: Renewable Energy Credit Prices and Other Market Outcomes



Notes: This figure shows the relationships between state-level annual Renewable Energy Credit (REC) prices and state ratios of the renewable generation in the Renewable Portfolio Standards (Panel (a)), the amount of the cumulative wind capacity (Panel (b)), and the share of electricity generated by fossil fuels and nuclear energy (Panel (c)). The red circles are the binned scatter plot, while the red solid line is the linear fit.

SA.3 Interconnection Queue Data

I construct the pool of potential entrants from the project-level interconnection queue data compiled by Lawrence Berkeley National Laboratory (the “Queued Up” data file), which records for each interconnection request the date it entered the queue, the date it was withdrawn if applicable, its status, its resource type, its state, and its region.³⁷ I keep the requests whose resource type includes wind, with one row per request. Since I observe the time when a project entered the queue and withdrew from the queue, I define the former as entry and the latter as exit. I assume that on average wind projects stayed for two years in the queue

³⁷The data file is described at <https://emp.lbl.gov/queues>. I use the release covering requests through 2025 and keep the requests that entered the queue between 2004 and 2019.

Table SA1: REC Tracking System and Price Imputation

State	Established year	Tracking system	Imputation
Arizona	2006	None	national average
California	2002	WREGIS	no
Colorado	2004	WREGIS	regression
Connecticut	1998	NEPOOL-GIS	no
Delaware	2005	PJM-GATS	no
Hawaii	2001	None	national average
Illinois	2007	M-RETS, PJM-GATS	no
Indiana	2011	Not designated	national average
Iowa	1983	M-RETS	regression
Kansas	2015	NAR	national average
Maine	1999	NEPOOL-GIS	no
Maryland	2004	PJM-GATS	no
Massachusetts	1997	NEPOOL-GIS	no
Michigan	2008	MIRECS	no
Minnesota	2007	M-RETS	regression
Missouri	2007	NAR	national average
Montana	2005	M-RETS, WREGIS	regression
Nevada	1997	NVTREC, WREGIS	regression
New Hampshire	2007	NEPOOL-GIS	no
New Jersey	1991	PJM-GATS	no
New Mexico	2002	WREGIS	regression
New York	2004	NYGATS	national average
North Carolina	2007	NC-RETS	national average
North Dakota	2007	M-RETS	regression
Ohio	2008	M-RETS, PJM-GATS	no
Oklahoma	2010	None	national average
Oregon	2007	WREGIS	regression
Pennsylvania	2004	PJM-GATS	no
Rhode Island	2004	NEPOOL-GIS	no
South Carolina	2014	None	national average
South Dakota	2008	None	national average
Texas	1999	ERCOT	no
Utah	2008	WREGIS	regression
Vermont	2015	NEPOOL-GIS	regression
Washington	2006	WREGIS	regression
Wisconsin	1998	M-RETS	regression

Notes: This table documents the establishment year as well as the tracking system of the Renewable Energy Credit (REC) market for relevant states based on Table 1 of [Abito, Flores-Golfin, Hinchberger, van Benthem, and Vasey \(2025\)](#). The column “Imputation” documents how I impute missing REC prices in the corresponding states. “Regression” indicates that I impute REC prices following equation (18) with the stipulated ratios in the Renewable Portfolio Standards, the amount of the cumulative wind capacity, and the share of electricity generated from fossil fuels and nuclear energy, as well as time trends specific to the relevant tracking system. “National average” indicates that I impute the REC prices with a national average in that year excluding the NEPOOL when no price in the corresponding tracking system is available. “No” indicates that the data is not missing and no imputation is required.

before obtaining all the approvals and signing the interconnection agreements.³⁸ Another

³⁸ Anecdotes suggest that a typical project completed in 2008 spent fewer than two years in the queue for interconnection approval compared to three years in 2015, according to the [news](#). Although the backlog and congestion issues are salient in recent years, two-year waiting time might be a reasonable assumption because it is roughly the median in my sample period.

way to leave the queue is to successfully build a wind farm, which I back out using the EIA data.

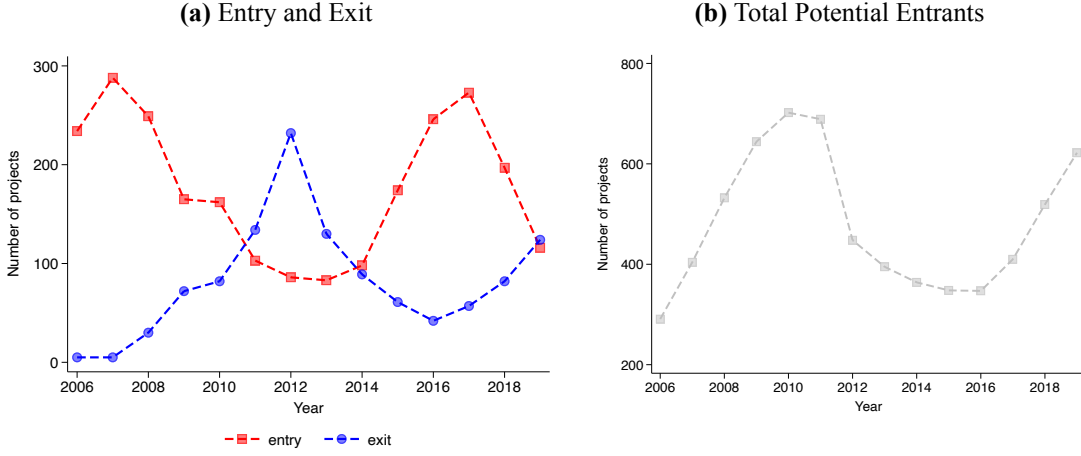
I calculate the number of potential entrants for the wind industry for each state as a cumulative number of projects that had entered the queue at least two years ago and had not built a wind farm or withdrawn from the queue. I denote the number of potential entrants in state m and year t as $\text{PotentialEntrants}_{mt}$. The number of projects that entered into the queue, withdrew from the queue and built a new wind farm as Entry_{mt} , Exit_{mt} and NewBuilt_{mt} , respectively. Therefore, $\text{PotentialEntrants}_{mt}$ can be recursively defined as follows.

$$\text{PotentialEntrants}_{mt} = \text{PotentialEntrants}_{mt-1} + \text{Entry}_{mt-2} - \text{Exit}_{mt} - \text{NewBuilt}_{mt-1}.$$

Consistent with the two-year tenure, projects that withdrew within two years of entering the queue never join the pool and are excluded from both the entry and the exit counts, and withdrawn requests without a withdrawal date are dropped. I define $\text{PotentialEntrants}_{m,2005}$ as the maximum of NewBuilt_{mt} in the state m over the sample period, serving as an initial value. I adjust $\text{PotentialEntrants}_{mt}$ to be equal to NewBuilt_{mt} if the former falls below the latter. I describe the time trend for Entry_{mt} , Exit_{mt} , and $\text{PotentialEntrants}_{mt}$ in Supplemental Appendix Figure SA3. The total number of projects that entered the queue peaked in 2007, fell to its lowest level in 2012 and 2013, and recovered after 2014. The total number of projects that withdrew from the queue experienced a peak in 2012 and displayed a hump shape. As a consequence of the time trend for entry, exit, and successful new builds, which peaked in 2011, the number of total potential entrants is also hump-shaped and peaked in 2010. The entry and withdrawal from the queue are both assumed to be exogenous to my model; Supplemental Appendix Section C.6 examines how this assumption affects the counterfactual results.

The fifteen states of the dynamic estimation all belong to an ISO or RTO, and their pools are constructed from the queue data as described above. Two complications remain. First, the withdrawal dates of the ERCOT queue are recorded only from 2018 onward, so the pool of Texas shrinks only through the construction of wind farms before 2018. Second, for the states with wind farms but without queue data, which are not part of the estimation sample, I assume that the number of projects that enter the queue or withdraw from the queue follows the aggregate time trend in the states with queue data, and the level is adjusted proportionally to the number of potential entrants in 2005.

Figure SA3: Entry, Exit, and Potential Entrants in Queues



Notes: This figure presents the aggregate time trends in the interconnection queues across MISO, CAISO, PJM, ISO-NE, NYISO, ERCOT, and SPP. “Entry” denotes the number of projects entering the queue, while “exit” denotes the number of projects withdrawing from the queue. The number of potential entrants in the wind industry in each state is defined as the cumulative number of projects that entered the queue at least two years earlier and have neither been built nor withdrawn from the queue.

SA.4 Additional Details for Bilateral Bargaining

SA.4.1 Derivation of the Capacity and Price Equations

Negotiated capacity equates the utility’s marginal value of wind energy with the wind farm’s marginal cost net of the subsidy. The negotiated price then divides the gains from trade according to the bargaining weight. To derive these two equations, I define the utility’s and wind farm’s gains from trade as

$$S_U = \pi_t^U(p_{ij}, k_{ij}) - \pi_t^U(p_{ij} = \infty), \quad S_W = \pi_t^W(p_{ij}, k_{ij}) - \pi_t^W(p_{ij} = \infty).$$

The Nash bargaining problem is $\max_{k_{ij}, p_{ij}} S_U^\rho S_W^{1-\rho}$. Because the contract price is a transfer between the parties,

$$\frac{\partial S_U}{\partial p_{ij}} = -\frac{\partial S_W}{\partial p_{ij}}.$$

For an interior agreement, the price first-order condition therefore implies

$$(1 - \rho)S_U = \rho S_W.$$

Thus, the utility receives a share ρ of the joint surplus. Substituting this relationship into the capacity first-order condition gives

$$\frac{\partial S_U}{\partial k_{ij}} + \frac{\partial S_W}{\partial k_{ij}} = 0,$$

so negotiated capacity maximizes joint surplus.

Define the annuity factors

$$A_T = \sum_{s=t+1}^{t+T} \beta^{s-t}, \quad A_{10} = \sum_{s=t+1}^{t+10} \beta^{s-t}.$$

Using the profit functions in Section 4.1, the marginal surpluses are

$$\begin{aligned} \frac{\partial S_U}{\partial k_{ij}} &= \alpha_i \sum_{s=t+1}^{t+T} \beta^{s-t} \mathbb{E}_t [p_s^e + p_s^c(1 - z_s) + \delta(\bar{k}_{ijs} - k_{ij})] - \alpha_i A_T p_{ij}, \\ \frac{\partial S_W}{\partial k_{ij}} &= \alpha_i [A_T p_{ij} + A_{10} d_t - \mathbf{X}'_{it} \boldsymbol{\gamma}_1 - \gamma_2 k_{ij} - \xi_{it}]. \end{aligned}$$

Adding these expressions cancels the price terms. Dividing by α_i gives equation (2).

To obtain the price equation, write the surpluses as

$$\begin{aligned} S_U &= \alpha_i k_{ij} \left\{ \sum_{s=t+1}^{t+T} \beta^{s-t} \mathbb{E}_t \left[p_s^e + p_s^c(1 - z_s) + \delta \left(\bar{k}_{ijs} - \frac{k_{ij}}{2} \right) \right] - A_T p_{ij} \right\}, \\ S_W &= \alpha_i k_{ij} \left[A_T p_{ij} + A_{10} d_t - \mathbf{X}'_{it} \boldsymbol{\gamma}_1 - \frac{\gamma_2}{2} k_{ij} - \xi_{it} \right] - \pi_t^W(p_{ij} = \infty). \end{aligned}$$

Substituting these expressions into $(1 - \rho)S_U = \rho S_W$ and dividing by $\alpha_i k_{ij}$ gives equation (3). The distinction between the two equations is that capacity depends on marginal values, whereas price divides total surplus and therefore depends on average values. This explains why the hassle cost contributes $\delta(\bar{k}_{ijs} - k_{ij})$ to the capacity equation but $\delta(\bar{k}_{ijs} - k_{ij}/2)$ to the price equation. Similarly, the turbine cost contributes $\gamma_2 k_{ij}$ to marginal cost but $\gamma_2 k_{ij}/2$ to average cost.

SA.4.2 Annualized Capacity Factor α_{it}

I parameterize wind power generation Q_{ijt}^w as a linear function of procured capacity k_{ij} , as I find that the annual total output on average is linearly increasing with nameplate capacity. I residualize both the annual total generation and the nameplate capacity on entry cohort dummies and age dummies and then plot the linear fit and local polynomial approximation

between these two variables. As shown in Supplemental Appendix Figure A9, the non-parametric relationship is very close to the linear fit, and the linear function has explanatory power as high as 0.83. Under the assumption of the linear production function, I define the annualized capacity factor $\alpha_{it} = \frac{Q_{ijt}^w}{k_{ij}}$.

I then explore how the annualized capacity factor evolves with age by estimating the model below, where age_{it} denotes the age of wind farm i in year t . I further control for the entry cohort of wind farms (cohort_i). I set the group of age one as the baseline group, and β_a measures the differences in capacity factors between other age groups and the baseline group within the cohort.

$$\alpha_{it} = \sum_{a=2}^{10} \beta_a \times \mathbb{1}(\text{age}_{it} = a) + \sum_{c=2004}^{2018} \beta_c \times \mathbb{1}(\text{cohort}_i = c) + \epsilon_{it}. \quad (19)$$

I plot the age effects β_a in Panel (a) of Supplemental Appendix Figure A9. The overall average capacity factor is relatively stable even for ten years after entry. The capacity factor peaks at age 5. However, the difference is only around 5% compared to the level of the baseline group. Moreover, I divide the sample into two groups: wind farms of age 1 and wind farms of age 2-5. I estimate equation (19) without age dummies and plot β_c for the two age groups in Panel (b). I find that capacity factors evolve systematically with the cohort, but display limited variation with respect to the age of wind farms. This is further supported by the fact that the cohort dummies alone explain 84.3% of the variations of the average capacity factor at the cohort-age level, while the age dummies alone explain 5.5% only. Therefore, I treat the annualized capacity factor to be constant as a wind farm ages and calculate it at the age of one for the best data coverage such that $\alpha_i = \alpha_{it}$ when $\text{age}_{it} = 1$.

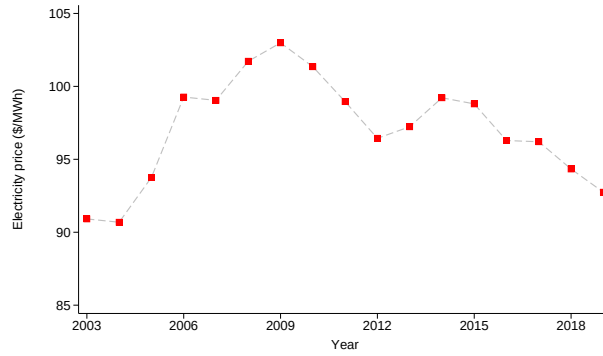
SA.4.3 Transition Dynamics of Market Prices

Utilities have a rational expectation of the future evolution of both retail electricity prices and renewable energy credit (REC) prices. I use the annual average retail electricity price p_{mt}^e in each state m from EIA-861. As shown in Supplemental Appendix Figure SA4, the average inflation-adjusted electricity price, weighted by the annual sales in each state, increased before 2009 but has declined since then due to plummeting natural gas prices. I model the evolution of electricity prices as an AR(1) process with state fixed effects ξ_m^p ,

$$p_{mt}^e = \gamma_1^p p_{mt-1}^e + \xi_m^p + \epsilon_{mt}^p, \quad (20)$$

and I use the estimates in column (1) of Supplemental Appendix Table SA2 to form the expected prices over the term of each contract. The R^2 of the regression is as high as 0.96. Columns (2) and (3) add a linear time trend and allow the AR(1) coefficient and the time trend to differ before and after 2009. The trend after 2009 is imprecisely estimated, and the persistence of the price is similar across columns, so I keep the AR(1) process with state fixed effects, which does not require utilities to anticipate the trend break in 2009.

Figure SA4: Time Trend of Aggregate Electricity Price



Notes: This figure shows the time trend of average electricity price. I measure the average electricity price with the state-level annual retail electricity price from EIA-861, weighted by the state-level annual electricity sales and adjusted by inflation.

Similarly, I estimate an AR(1) model for the renewable energy credit prices p_{mt}^c as shown in Supplemental Appendix Table SA3. I take the coefficient estimates from column (2), the AR(1) process with state fixed effects, and assume utilities have rational expectations with respect to the evolution of renewable energy credit prices and have perfect foresight with respect to the Renewable Portfolio Standards z_{mt} .

SA.4.4 Transition Dynamics of Renewable Portfolio Gaps

Utility j generates electricity using different fuel sources, including fossil fuels (f), procured wind (w), other renewable sources (or), and other sources (o). I denote generation capacity as k_{jt}^a for utility j , year t , and type a , and the corresponding electricity generation as Q_{jt}^a . The total electricity generation Q_{jt} can be expressed as $Q_{jt} = Q_{jt}^w + Q_{jt}^f + Q_{jt}^{or} + Q_{jt}^o$. Utilities are faced with the Renewable Portfolio Standards requirement z_{mt} in state m and year t . I define renewable portfolio gap Q_{jt}^{gap} as follows.

$$Q_{jt}^{gap} = z_t(Q_{jt}^f + Q_{jt}^{or} + Q_{jt}^o) - Q_{jt}^{or}.$$

Table SA2: Transition Dynamics of Electricity Prices

	Electricity Price		
	(1)	(2)	(3)
Lagged Electricity Price	0.709*** (0.052)	0.721*** (0.059)	
Time Trend		-0.100 (0.071)	
Lagged Electricity Price $\times \mathbb{1}(\text{Year} \leq 2009)$			0.702*** (0.097)
Lagged Electricity Price $\times \mathbb{1}(\text{Year} > 2009)$			0.690*** (0.050)
Time Trend $\times \mathbb{1}(\text{Year} \leq 2009)$			0.842*** (0.277)
Time Trend $\times \mathbb{1}(\text{Year} > 2009)$			-0.190 (0.148)
$\mathbb{1}(\text{Year} > 2009)$			6.424** (2.440)
Observations	816	816	816
Adjusted R^2	0.964	0.964	0.965
State Dummies	✓	✓	✓

Notes: This table shows the transition dynamics of electricity prices at the state and yearly levels. Column (1) is the empirical model in equation (20), which I use to form expectations. Column (2) adds a linear time trend, and column (3) allows the AR(1) coefficient and the time trend to differ before and after 2009. Standard errors are clustered at the state level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table SA3: Transition Dynamics of Renewable Energy Credit Prices

	REC Price			
	(1)	(2)	(3)	(4)
Lagged REC Price	0.885*** (0.019)	0.613*** (0.035)	0.880*** (0.021)	0.577*** (0.043)
Time Trend			-0.111*** (0.029)	-0.203*** (0.060)
Observations	445	445	445	445
Adjusted R^2	0.832	0.843	0.833	0.848
State Dummies		✓		✓

Notes: This table shows the transition dynamics of renewable energy credit (REC) prices at the state and yearly levels. Column (2) is the AR(1) process with state fixed effects, which I use to form expectations. Columns (3) and (4) add a linear time trend. Standard errors are clustered at the state level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

I first describe the overall time trend of electricity generation by energy source, separately for all the utilities and utilities in the Power Purchase Agreement (PPA) sample. The

share of coal-fired electricity is decreasing over time, while the share of gas-fired electricity is increasing at the national level as shown in Supplemental Appendix Figure SA5. Despite limited volumes, procured wind and other renewables (including solar, biomass, geothermal, and utility-owned wind) are both increasing. Meanwhile, total generation from nuclear, petroleum, hydroelectric, and other energy sources is mostly stable. Compared to the entire sample of utilities, those in the Power Purchase Agreement sample have a much larger coal power share and a smaller natural gas power share.

I next estimate the transition process of electricity output portfolios at the utility level. I categorize different energy sources into four types: coal, natural gas, other non-renewables (including nuclear, petroleum, and others), and other renewables (including solar, biomass, geothermal, and wind directly owned by utilities). I exclude hydroelectric power following Hollingsworth and Rudik (2019), as many Renewable Portfolio Standards excluded hydroelectric power built before the implementation. I use the AR(1) model to capture the evolution process of net generations from these four different energy sources. As the capacity investment is lumpy, I exclude utilities that have never used a certain fuel type from the regression. I take the coefficient estimates from the AR(1) model with utility dummies and a time trend. The results are shown in Supplemental Appendix Table SA4.

I assume utilities have rational expectations with respect to the evolution of their own electricity generation from each type of fuel source, and they have perfect foresight with respect to the Renewable Portfolio Standards. If a utility has never used a certain fuel type during the sample period, I assume that its expectation of future usage remains zero.

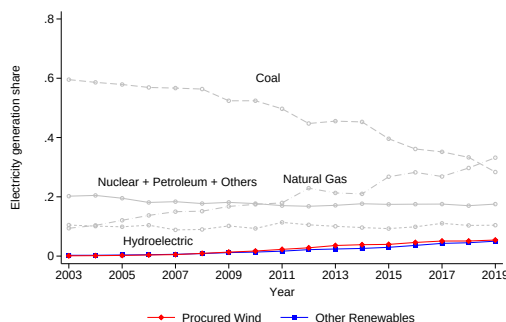
SA.5 Estimation Details for Demand of Non-Utility Buyers and Buyer Choice

SA.5.1 Demand of Non-Utility Buyers

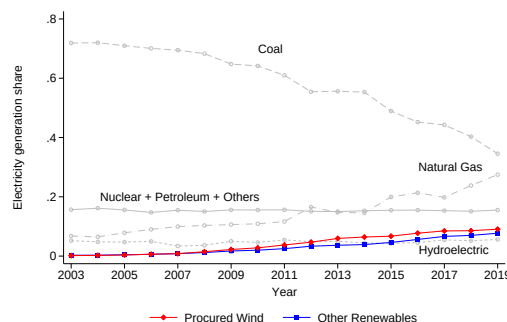
Panel B of Table 1 reports the linear demand function (4), estimated on the 233 wind farms that sold capacity to non-utility buyers or signed merchant hedge contracts, with the three supply-side instruments described in Section 5.2: an indicator for state industry recruitment support, the distance from the wind farm to the nearest substation, and the number of wind component manufacturing sites in the state. Supplemental Appendix Table SA5 tests the

Figure SA5: Time Trend of Output Share by Energy Source

(a) All Utilities



(b) PPA Sample



Notes: This figure shows the time trend of the shares of electricity generated by different energy sources. Panel (a) displays the time trend for all utilities, while Panel (b) shows the time trend for utilities from the Power Purchase Agreement sample. Other renewables include solar, biomass, geothermal, and utility-owned wind.

robustness of the estimate to the choice of instruments. It reports the demand function in logs, so that the price coefficient is the price elasticity of demand. Column (1) uses the three baseline instruments and gives an elasticity of -1.78 (0.30); the first-stage Kleibergen–Paap F statistic is 15.0. Columns (2)–(4) add the state-level agricultural land price, the state sales tax and property tax incentive dummies, or both, as additional instruments. The elasticity ranges from -1.85 to -1.97 across these columns and stays within one standard error of the baseline estimate.

SA.5.2 Buyer Choice

Potential buyers For each wind farm i , the set of potential buyers $\mathcal{J}_i$ in Section 4 consists of the utilities within 300 miles of the wind farm that signed at least one power purchase agreement for wind capacity during the sample period, together with the utility that wind farm i contracts with when that utility is located farther away. The distance between a wind farm and a utility is calculated using the coordinates of the wind farm and the closest power plant owned by the utility in EIA-860. The estimation sample of the buyer choice model consists of 635 wind farms, 402 of which sell capacity to a utility and 233 through the non-utility channel, and 8,413 wind farm and utility pairs, so a wind farm has 13 potential buyers on average.

Descriptive patterns Supplemental Appendix Figure SA6 summarizes the matching pat-

Table SA4: Transition Dynamics of Electricity Generation by Source

Net Generation				
<i>Panel A: Coal and Natural Gas</i>				
	Coal		Natural Gas	
	(1)	(2)	(3)	(4)
Lagged Variable	0.853*** (0.093)	0.947*** (0.014)	0.956*** (0.018)	1.043*** (0.006)
Time Trend	-0.082*** (0.020)		0.010*** (0.003)	
Observations	2615	2615	7950	7954
Adjusted R^2	0.961	0.961	0.978	0.978
Utility Dummies	✓		✓	
State Dummies $\times$ Time Trend		✓		✓
<i>Panel B: Other Renewable and Non-Renewable Sources</i>				
	Other non-Renewables		Other Renewables	
	(5)	(6)	(7)	(8)
Lagged Variable	0.692*** (0.083)	0.992*** (0.008)	1.045*** (0.038)	1.109*** (0.026)
Time Trend	0.000 (0.002)		0.001 (0.001)	
Observations	10845	10847	2734	2738
Adjusted R^2	0.985	0.983	0.977	0.975
Utility Dummies	✓		✓	
State Dummies $\times$ Time Trend		✓		✓

Notes: This table shows the transition dynamics of net electricity generation using coal, natural gas, other non-renewable sources (including nuclear, petroleum, and others), and other renewable sources (including solar, biomass, geothermal, and wind directly owned by utilities) at the utility and yearly levels. The odd columns include utility dummies and a linear time trend, and the even columns include state-specific time trends. Standard errors are clustered at the state level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

terms between utilities and wind farms, using all pairs of a wind farm and a utility in the data before the restriction to the potential buyers. Panel (a) shows the raw distribution of the geographical distance between the matched utility and the focal wind farm. The distribution is truncated at 600 miles. The distribution displays a long tail, but most matched pairs are within 400 miles of each other. Panel (b) shows the distribution of the relative distance of the matched utility and the focal wind farm, which measures how far away the matched utility is compared to the rest of the utilities in the buyer pool. This variable takes the value zero if the matched utility is the closest option, while it takes the value one if the furthest. Panel (b) shows that a wind farm tends to match with a utility that is closer geographically. Panel (c) explores whether a matched utility is likely to be in the same state as the focal

Table SA5: Robustness Checks: Demand for Non-Utility Buyers

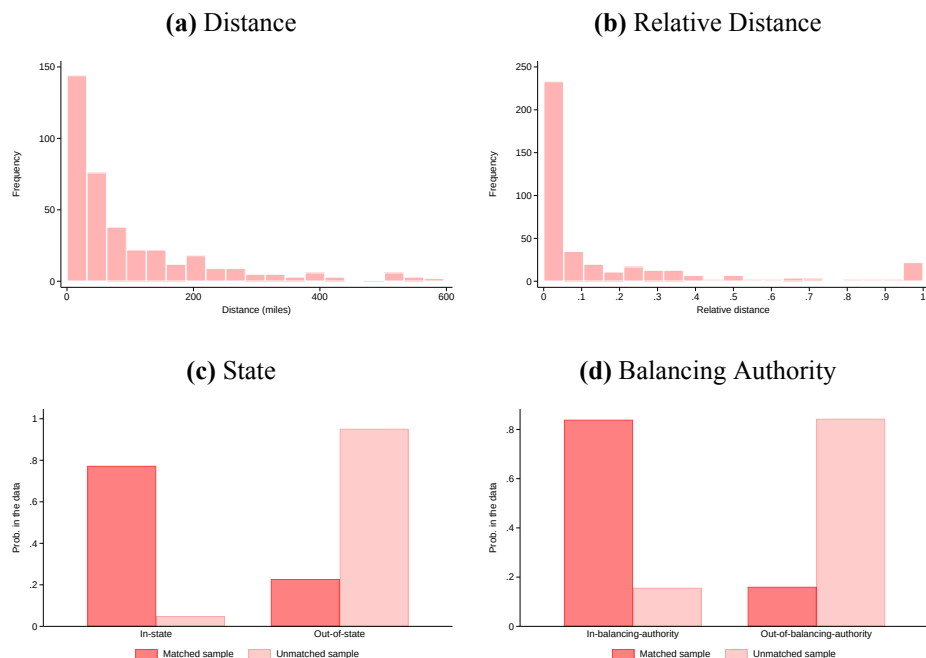
		log(Capacity)		
	(1)	(2)	(3)	(4)
log(Price)	-1.775 (0.302)	-1.845 (0.225)	-1.967 (0.277)	-1.916 (0.194)
Observations	233	233	233	233
R^2	0.358	0.341	0.308	0.323
Balancing-Authority Dummies	✓	✓	✓	✓
Contract-Type Dummies	✓	✓	✓	✓
<i>Instruments:</i>				
Baseline Instruments	✓	✓	✓	✓
Land Price		✓		✓
State Policies			✓	✓

Notes: This table shows the estimation results of the demand function for non-utility buyers (equation (4)) in logs, by two-stage least squares. The baseline instruments are the three supply-side instruments of Section 5.2: an indicator for state industry recruitment support, the distance from the wind farm to the nearest substation, and the number of wind component manufacturing sites in the state. Columns (2)–(4) add the annual agricultural land price at the state level, the state sales tax and property tax incentive dummies, or both. All columns control for the capacity factor, the turbine price, and the turbine brand dummies. Robust standard errors are in parentheses.

wind farm. Around 77% of the pairs of a wind farm and its matched utility are from the same state, while fewer than 5% of the pairs of a wind farm and an unmatched utility are from the same state. Panel (d) presents a similar pattern for whether a utility and wind farm pair is in the same balancing authority: 84% of the matched pairs, against 16% of the unmatched pairs, are in the same balancing authority. Overall, a wind farm is more likely to be matched to a utility that is geographically closer and within its own state or balancing authority, which motivates the consideration probability in equation (5).

Determinants of the utility choice Supplemental Appendix Table SA6 relates the identity of the matched utility to the characteristics of the potential buyers within the pool. The dependent variable is a dummy variable that takes the value one if the utility is the chosen buyer for the wind farm, and the sample consists of the 10,134 pairs of a wind farm with a utility agreement and a potential buyer within 300 miles. I find utilities that are in the same state as the focal wind farm, in the same balancing authority as the focal wind farm, or that are closer, are more likely to be chosen. The renewable portfolio gap, which measures the compliance demand and thus proxies potential profit, has a positive effect on the choice probability but statistically insignificant and economically small. This pattern moti-

Figure SA6: Matching Patterns between Utilities and Wind Farms



Notes: This figure summarizes the matching pattern between utilities and wind farms, using all pairs of a wind farm and a utility in the data. Panel (a) shows the raw distribution of the geographical distance between the matched utility and the focal wind farm. The distribution is truncated at 600 miles. Panel (b) shows the distribution of the relative distance of the matched utility and the focal wind farm, which measures how far away the matched utility is compared to the rest of the utilities in the buyer pool. This variable takes the value zero if the matched utility is the closest option, while it takes the value one if the furthest. Panel (c) explores whether a matched utility is likely to be in the same state as the focal wind farm and Panel (d) explores whether a utility and wind farm pair is in the same balancing authority.

vates using location shifters, whether the buyer is in the same state and the same balancing authority as the wind farm, and the distance between them, as the key determinants of the probability that a buyer enters the wind farm's consideration set. Negotiated profits matter only at the next step when the wind farm chooses the most profitable buyer among those it considers. The parameters are estimated by maximum likelihood as described in Section 5.2, and Panel C of Table 1 reports the estimates.

SA.6 Calculation of Social Surplus of Wind Energy

I evaluate the benefits of wind energy following Callaway, Fowle, and McCormick (2018). I assume wind farms operate for 20 years and calculate total benefits from their twenty-year

Table SA6: Determinants of the Utility Choice

	$\mathbb{1}(\text{Utility Choice})$		
	(1)	(2)	(3)
Renewable Portfolio Gap (10^{-6} MWh)	0.001*** (0.000)	0.000 (0.000)	0.001 (0.000)
Distance (100 Miles)	-0.002 (0.005)	-0.016*** (0.005)	-0.023*** (0.004)
Same State	0.067*** (0.005)	0.056*** (0.005)	0.051*** (0.006)
Same Balancing Authority	0.035*** (0.003)	0.037*** (0.003)	0.043*** (0.005)
Observations	10134	10073	10071
Adjusted R^2	0.049	0.109	0.101
State-Year Dummies		✓	✓
Wind Farm Dummies			✓

Notes: This table explores the determinants of utility choice of wind farms if they sell capacity through utility Power Purchase Agreements. The dependent variable is a dummy variable that takes the value one if the utility is the chosen buyer for the wind farm. The sample consists of the pairs of a wind farm with a utility agreement and a potential buyer within 300 miles. Standard errors are clustered at the wind farm level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

operations. Wind energy substitutes for fossil fuels in generating electricity and thus there are two sources of benefits from more wind energy on the grid: reducing carbon emissions, and displacing the energy and the capacity that the system would otherwise procure. I value the second source with the wholesale market value of wind output, and the first with the marginal operating emissions rate and the social cost of carbon.

Environmental benefits I estimate the average marginal operating emissions rate (MOER) of coal- or gas-fueled power plants in each state, which is defined as the marginal response in the system-wide emissions with respect to the total production change from generators due to more renewable energy, as [Callaway, Fowle, and McCormick \(2018\)](#) find that regional average MOERs offer a useful means of “calibrating regional policy incentives to compensate for external emissions benefits.” I access the data of total electricity production and carbon emission for each state at the hourly level between January 1, 2004, and December 31, 2018, from the Clean Air Markets Program Data (formerly, Continuous Emissions Monitoring Systems Database). Following [Callaway, Fowle, and McCormick \(2018\)](#), I first cluster the days according to their hourly load profiles and their peak loads using a k-means clustering algorithm. I categorize all observations into eight markets according

to their ISOs or RTOs, including CAISO, ERCOT, ISO-NE, MISO, PJM, SPP, NYISO, and non-ISO states, and all dates into two seasons: summer/fall (May to October) and winter/spring (November to April). I generate 12 clusters of days within each market and season. The MOER is estimated using the equation below for each state m , cluster k , season s , and hour-of-the-day h , where E_{mkt} and G_{mkt} represent emissions and electricity generations in hour t .

$$E_{mkt} = \alpha_{mkhs}^E + \phi_{mkhs}^E G_{mkt} + e_{mkt}.$$

ϕ_{mkhs}^E is the estimated MOER for each state m , season s , hour-of-the-day h , and cluster k . As I calculate the total benefits from twenty-year operations of wind farms, I take an average ϕ_m^E as the mean MOER for state m , and I value the avoided emissions at a social cost of carbon of 80 dollars per ton.

Wholesale market value The avoided fuel costs and the capacity value of wind output are valued jointly with the wholesale market value of wind, which is the generation-weighted average of the energy and the capacity prices that wind output earns in the wholesale market of each region. I take the market value by region from the data file of the Land-Based Wind Market Report of Lawrence Berkeley National Laboratory ([Wiser et al., 2025](#)), whose construction follows [Millstein et al. \(2021\)](#), and I average it over 2009–2018 and express it in the constant dollars of the rest of the welfare accounting. The values are 18.6 dollars per MWh in MISO, 24.9 in SPP, 31.9 in PJM, 32.5 in CAISO, and 20.5 in ERCOT.

SA.7 An Alternative Dynamic Model

There is an alternative dynamic model for the evolving policy beliefs, which preserves the stationarity of the problem.³⁹ The notation is the same as in Section 4.3. ω_t represents the policy status in year t , which could take three values: (1) $\omega_t = H$, which indicates that the federal subsidy is enacted in year t and the probability of policy renewal is 1; (2) $\omega_t = L$, which indicates that the federal subsidy is enacted in year t , but the probability of policy renewal is only $b < 1$; (3) $\omega_t = 0$, which indicates that the federal subsidy is terminated. In each period, the *ex-ante* likelihood of $\omega_t = H$ conditional on policy renewal is equal to ρ_H .

I maintain Assumption 1 from Section 4.3 that policy elimination will be perceived as

³⁹I thank Kenneth Hendricks and Jean-François Houde for bringing up this modeling option and for the extensive discussion of its feasibility.

perpetual. The option value when the realized state variable is s_{it+1} conditional on different policy statuses ω_t can be written as follows.

$$\begin{aligned}\mathbb{E}[V(s_{it+1}, \omega_{t+1}) | \omega_t = H] &= V(s_{it+1}, \omega_{t+1} = H) \times \rho_H \\ &+ V(s_{it+1}, \omega_{t+1} = L) \times (1 - \rho_H).\end{aligned}$$

$$\begin{aligned}\mathbb{E}[V(s_{it+1}, \omega_{t+1}) | \omega_t = L] &= V(s_{it+1}, \omega_{t+1} = H) \times \rho_H \times b \\ &+ V(s_{it+1}, \omega_{t+1} = L) \times (1 - \rho_H) \times b \\ &+ V(s_{it+1}, \omega_{t+1} = 0) \times (1 - b).\end{aligned}$$

$$\mathbb{E}[V(s_{it+1}, \omega_{t+1}) | \omega_t = 0] = V(s_{it+1}, \omega_{t+1} = 0).$$

The advantage of this model is to preserve the stationarity of the dynamic problem and use two parameters b and ρ_H to capture evolving policy beliefs. ρ_H can be identified from the frequency of investment spikes, while b can be identified from the magnitude of investment spikes. However, the stationarity of the problem conflicts with the data pattern of jumping investment spikes across years such that the model fails to predict when the investment spikes will occur. The only solution is to index ρ_H and b by t , but the model will be isomorphic to my baseline model.

SA.8 Proof of Identification

The main identification challenge is to separate entry costs from beliefs about policy renewal. The argument has two steps. During the certainty window, renewal beliefs are known, so entry probabilities identify the entry cost parameters. Given these parameters and the state transitions, higher renewal beliefs make waiting more attractive and reduce current entry. Each cohort's belief can therefore be recovered from its entry rate, provided that rate lies within the model's predicted range. Below, I state and prove this result, then discuss its application. For simplicity, I use a common entry cost intercept (the empirical model allows it to vary across wholesale markets). The two steps describe where identification comes from; in estimation, costs and beliefs are estimated jointly.

Setup. Each year, a potential entrant chooses whether to enter or wait. The state vector s contains the determinants of wind farm profits and the land price W . Policy status is $\omega \in \{0, 1\}$, where one means that the policy is available and zero means that it has expired.

Entry yields expected profit $\pi^\omega(\mathbf{s})$, measured in dollars and obtained from the static model. Entry costs are $\kappa_0 + \kappa_1 W + \nu$, where ν is an independent logistic shock with location zero and scale $\sigma_\nu > 0$, drawn each year. Define

$$\Pi^\omega(\mathbf{s}) = \pi^\omega(\mathbf{s}) - \kappa_0 - \kappa_1 W.$$

The payoff from entry is $\Pi^\omega(\mathbf{s}) - \nu$, while waiting yields $\beta C(\mathbf{s})$. Here, $\beta \in (0, 1)$ is the discount factor and C is the expected value next year.

I use Mf to denote the conditional expectation of f next year and $\Lambda(z) = e^z/(1 + e^z)$. Suppressing the policy superscript and state arguments, the value before the shock is drawn and the entry probability are

$$V = \sigma_\nu \log(e^{\Pi/\sigma_\nu} + e^{\beta C/\sigma_\nu}), \quad P = \Lambda\left(\frac{\Pi - \beta C}{\sigma_\nu}\right). \quad (21)$$

These imply

$$V = \Pi - \sigma_\nu \log P, \quad \sigma_\nu \log \frac{P}{1 - P} = \Pi - \beta C. \quad (22)$$

Because entry is a terminal decision, the first identity expresses the value function in terms of the entry payoff and entry probability (Hotz and Miller, 1993). The second identity gives the log odds of entry.

Policy beliefs. During the uncertainty window, investors believe that an expired credit is permanently terminated. Each cohort also applies a constant annual renewal probability b_t to all future years. The value without the credit satisfies

$$V^0 = \sigma_\nu \log(e^{\Pi^0/\sigma_\nu} + e^{\beta MV^0/\sigma_\nu}). \quad (23)$$

When the credit is available, the value under belief b satisfies

$$\begin{aligned} V^1(\mathbf{s}; b) &= \sigma_\nu \log(e^{\Pi^1(\mathbf{s})/\sigma_\nu} + e^{\beta C(\mathbf{s}; b)/\sigma_\nu}), \\ C(\mathbf{s}; b) &= b(MV^1(\cdot; b))(\mathbf{s}) + (1 - b)(MV^0)(\mathbf{s}). \end{aligned} \quad (24)$$

Thus, waiting yields the value with the policy next year for probability b , and the value without the policy for probability $1 - b$. Entry occurs with probability

$$P^E(\mathbf{s}; b) = \Lambda\left(\frac{\Pi^1(\mathbf{s}) - \beta C(\mathbf{s}; b)}{\sigma_\nu}\right).$$

During the certainty window, investors face a known subsidy path. I write the continuation value as $C_t = M_t V_{t+1}$, allowing transitions and values to vary with the legislated phase-

down.

Data and notation. The argument takes conditional entry probabilities and state transitions as identified from the data. In the main specification, the certainty window is 2013–2018 and the uncertainty window is 2009–2012. Let $\mathcal{P}$ contain the consecutive-year pairs in the certainty window. For a sequence of functions f_t , define

$$\Delta_t^\beta f = f_t - \beta M_t f_{t+1}, \quad \ell_t = \log \frac{P_t}{1 - P_t} - \beta M_t \log P_{t+1}.$$

Profits π_t^1 reflect the subsidy available in year t .

Proposition 1 *Suppose that:*

- (a) *The static model identifies profits in dollars without using entry data.*
- (b) *The discount factor is known, and investors' perceived transitions match those identified from the state panel.*
- (c) *Entry cost parameters are constant across the two windows, and land prices affect only entry costs.*
- (d) *For each $(t, t + 1) \in \mathcal{P}$, renewal is perceived as certain, $C_t = M_t V_{t+1}$, and P_{t+1} is identified at all states reachable from year t .*
- (e) *$\pi^1 > \pi^0$ throughout a set containing all cohort states and all states reachable from them.*

Assume bounded profits on the relevant state space, so that the Bellman contraction arguments apply.

- (i) *Entry costs. For every $(t, t + 1) \in \mathcal{P}$,*

$$\Delta_t^\beta \pi^1 = \sigma_\nu \ell_t + (1 - \beta) \kappa_0 + \kappa_1 \Delta_t^\beta W. \quad (25)$$

If ℓ_t , the constant, and $\Delta_t^\beta W$ are linearly independent across the relevant state-year observations, then $(\kappa_0, \kappa_1, \sigma_\nu)$ is uniquely identified.

- (ii) *Beliefs. Hold costs and transitions fixed. For cohort t with state distribution μ_t , the predicted entry rate*

$$m_t(b) = \int P^E(\mathbf{s}; b) d\mu_t(\mathbf{s})$$

is continuous and strictly decreasing in b . Thus, an observed rate $\bar{p}_t \in [m_t(1), m_t(0)]$ identifies a unique belief satisfying $m_t(b_t) = \bar{p}_t$. Outside this range, no admissible

belief exactly matches the observed rate.

Proof. (i) *Entry costs.* Substitute $V_{t+1} = \Pi_{t+1}^1 - \sigma_\nu \log P_{t+1}$ into the log-odds identity at t :

$$\sigma_\nu \log \frac{P_t}{1 - P_t} = \Delta_t^\beta \Pi^1 + \beta \sigma_\nu M_t \log P_{t+1}.$$

Since

$$\Delta_t^\beta \Pi^1 = \Delta_t^\beta \pi^1 - (1 - \beta)\kappa_0 - \kappa_1 \Delta_t^\beta W,$$

rearranging gives equation (25). Since all terms except the three cost parameters are identified, linear independence gives a unique solution.

(ii) *Beliefs.* Let $\mathcal{T}_b$ be the Bellman operator in equation (24). It is increasing in its value-function argument and is a contraction with modulus at most βb .

First, $\pi^1 > \pi^0$ implies $\mathcal{T}_b V^0 > V^0$. Iterating from V^0 therefore gives $V^1(\cdot; b) > V^0$. Second, for $b' > b$ and any $f \geq V^0$, I have $\mathcal{T}_{b'} f \geq \mathcal{T}_b f$. Iterating $\mathcal{T}_{b'}$ from $V^1(\cdot; b)$ then gives $V^1(\cdot; b') \geq V^1(\cdot; b)$.

Together, these results imply

$$C(\mathbf{s}; b') - C(\mathbf{s}; b) \geq (b' - b)M[V^1(\cdot; b) - V^0](\mathbf{s}) > 0.$$

A higher belief therefore raises the value of waiting without changing current entry profits. Entry probabilities, and hence $m_t(b)$, strictly decrease.

Continuity follows because the Bellman operators are continuous in b and share a contraction bound below one. Their fixed points, entry probabilities, and aggregate entry rates are therefore continuous in b . This establishes the unique inversion on $[m_t(1), m_t(0)]$. ■

Interpretation. During the certainty window, variation in profits and land prices identifies entry costs. Profits measured in dollars provide the monetary scale needed to identify σ_ν (Magnac and Thesmar, 2002). Land prices shift entry costs conditional on profits, while the level of entry helps identify the intercept. These sources of variation must jointly satisfy the rank condition in part (i). With market-specific intercepts, the same argument applies using market indicators.

During the uncertainty window, higher renewal beliefs make waiting more attractive. Conditional on costs and transitions, greater entry ahead of an expiration date therefore indicates a lower renewal belief. This is the bunching argument made precise.

The identification result holds transitions fixed. In estimation, I update endogenous state transitions to be consistent with predicted entry. The proof does not establish monotonicity after this equilibrium adjustment, and the numerical exercises below assess the resulting relationship. The discount factor is calibrated rather than identified by the proposition.

Application and numerical evidence. I first check whether the sufficient conditions from Proposition 1 are satisfied in my setting. The static model estimates profits from contracts and buyer choices, without exploiting entry data (condition (a)). I calibrate $\beta = 0.95$ and estimate transitions from the state panel (condition (b)). Land prices enter entry costs but not the bargaining equations (condition (c)). I assume entry cost parameters are constant across the two windows. Moreover, I allow a linear trend in entry costs in a robustness check. The trend is \$1.5 million per year and imprecisely estimated, and the 2011 belief rises from 0.40 to 0.59.

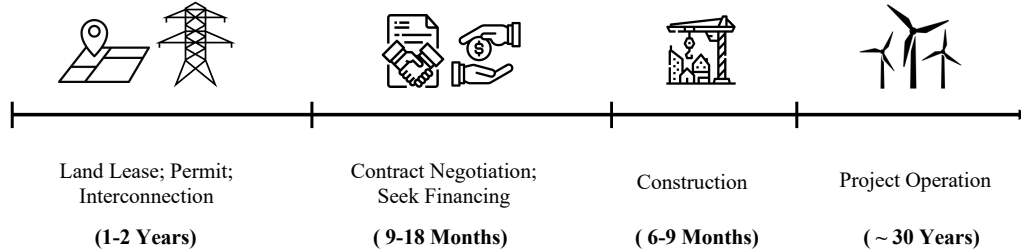
The safe harbor rule introduced in 2013 and the legislated phase-down in 2015 motivate the certainty window from 2013 to 2018 (condition (d)). I implement a robustness check by estimating the 2013 and 2014 beliefs instead of fixing them at one. The belief parameters in 2013 and 2014 are 0.95 and 0.96, respectively, not statistically different from 1. I also allow uncertainty about continuation of the phase-down after 2014, which gives a post-2014 belief of 0.94. These estimates support the certainty approximation.

The subsidy raises estimated profits in all 90 observed state-year cells in 2009–2014, by a minimum of \$10 million and a median of \$38 million. The median is 1.7 times the shock scale in the main specification and 2.9 times that scale in the specification estimating beliefs for 2009–2014. This supports the profit condition at observed states (condition (e)).

To assess the range of entry predictions, I set each cohort's belief near zero and at one, hold the other parameters fixed, and re-solve the model with equilibrium transitions. In the specification estimating beliefs for 2009–2014, every observed count lies between these predictions. In 2011, the model predicts 117 entrants near certain termination and 15 under certain renewal, compared with 96 observed. In 2012, the predictions are 88 and 10, compared with 18 observed. Predictions at the estimated beliefs also lie between the endpoints. These comparisons are consistent with decreasing entry as beliefs rise.

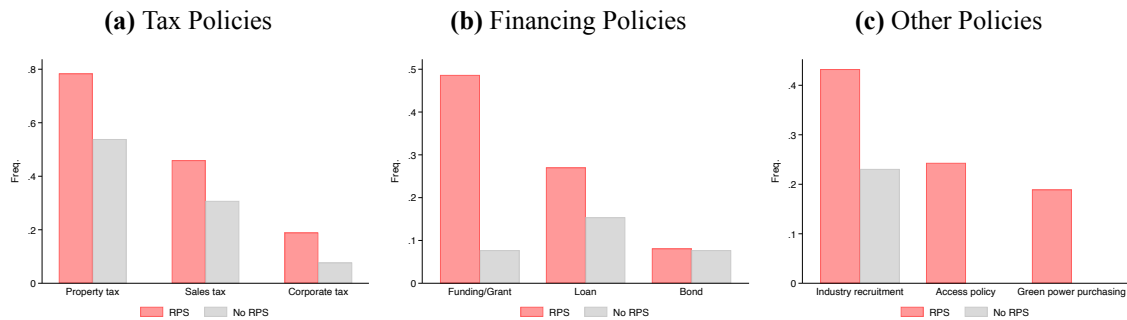
SA.9 Other Figures and Tables

Figure SA7: Timeline of Building a Wind Farm



Notes: The main source of the time statistics is the Wind Powers America Annual Report 2019 by AWEA (AWEA, 2020).

Figure SA8: State-level Policies



Notes: This figure shows the frequencies of different types of state policies for states with or without the RPS. State policies, including the RPS, are hand-collected by the author from DSIRE (<https://www.dsireusa.org>).

Table SA7: Transition Dynamics of the State Variables in the Dynamic Model

	Lagged variable	Lagged K_{mt}	Constant	State FE	Residual s.d.	Obs.
Turbine productivity $\bar{\alpha}_t$	0.907 (0.144)		0.359 (0.448)		0.160	11
Turbine price p_t^{tb}	1.015 (0.139)		-0.774 (1.765)		1.076	11
Expected electricity price	0.800 (0.042)			✓	0.061	180
Expected renewable credit price	0.804 (0.055)	-0.005 (0.002)	0.095 (0.029)		0.106	168
Inclusive value Φ_{it}	1.025 (0.044)	-0.150 (0.108)	0.749 (0.935)		8.315	168
Land price W_{mt}	0.935 (0.024)			✓	0.102	165

Notes: This table reports the AR(1) transition processes of the exogenous state variables and of the two endogenous state variables that respond to the cumulative wind capacity, estimated on the fifteen sample states between 2007 and 2018. Standard errors, in parentheses, are clustered at the state level for the state-level processes with fixed effects and robust otherwise. The residual standard deviations are the scales of the perceived shocks.

Table SA8: Dynamic Estimates with an Entry Cost Trend and a Post-2014 Belief

Parameter and Notation	(1) Beliefs 2009–2014	(2) with Entry Cost Trend	(3) with Post-2014 Belief
<i>Panel A: Entry Cost Parameters</i>			
Entry Cost, MISO	68.78 (23.86)	71.08 (20.92)	68.53 (16.95)
Entry Cost, SPP	28.46 (20.38)	36.24 (22.56)	34.05 (19.89)
Entry Cost, PJM	83.67 (18.42)	84.05 (21.08)	80.60 (13.89)
Entry Cost, CAISO	216.75 (81.02)	213.48 (77.65)	202.56 (51.15)
Entry Cost, ERCOT	56.43 (30.97)	69.51 (31.24)	58.06 (27.13)
Land Price	20.09 (9.66)	20.98 (8.59)	21.02 (9.77)
Scale of Entry Cost Shock	13.13 (5.47)	12.27 (5.77)	11.70 (4.26)
Entry Cost Trend	0	1.48 (1.89)	0
<i>Panel B: Belief Parameters</i>			
Policy Belief 2009	0.808 (0.242)	0.925 (0.166)	0.812 (0.212)
Policy Belief 2010	0.969 (0.031)	0.997 (0.025)	0.967 (0.029)
Policy Belief 2011	0.395 (0.233)	0.585 (0.253)	0.448 (0.139)
Policy Belief 2012	0.959 (0.033)	0.981 (0.028)	0.953 (0.044)
Policy Belief 2013	0.949 (0.050)	0.973 (0.046)	0.945 (0.080)
Policy Belief 2014	0.959 (0.063)	0.977 (0.058)	0.953 (0.066)
Policy Belief After 2014	1	1	0.940 (0.278)
Annual aggregate moments	✓	✓	✓
Market aggregate moments	✓	✓	✓
Market × 2011 Moments	✓	✓	✓
Objective Value	181.54	145.60	122.09

Notes: This table presents dynamic parameter estimates with an entry cost trend or a post-2014 belief parameter. Column (1) is column (3) of Table 2. Column (2) adds a linear time trend to the entry costs, in million dollars per year and centered in 2013–2014. Column (3) adds the perceived probability that the legislated phase-down continues in the following year after 2014. Entry costs are in million dollars at the average land price of each market. Standard errors are block-bootstrapped over states with 100 draws.

Table SA9: Dynamic Estimates by Reinstatement Probability

Parameter and Notation	(1)	(2)	(3)
	Reinstatement Probability 0 (Main)	0.10	0.15
<i>Panel A: Entry Cost Parameters</i>			
Entry Cost, MISO	68.78 (23.86)	64.48 (18.21)	63.02 (19.49)
Entry Cost, SPP	28.46 (20.38)	23.49 (16.88)	20.70 (16.45)
Entry Cost, PJM	83.67 (18.42)	80.38 (20.87)	80.07 (17.04)
Entry Cost, CAISO	216.75 (81.02)	215.72 (74.06)	209.79 (75.85)
Entry Cost, ERCOT	56.43 (30.97)	54.55 (27.72)	55.29 (29.59)
Land Price	20.09 (9.66)	19.04 (9.13)	18.83 (9.54)
Scale of Entry Cost Shock	13.13 (5.47)	12.64 (5.30)	12.90 (5.27)
<i>Panel B: Belief Parameters</i>			
Policy Belief 2009	0.808 (0.242)	0.402 (0.269)	0.285 (0.277)
Policy Belief 2010	0.969 (0.031)	0.907 (0.063)	0.897 (0.060)
Policy Belief 2011	0.395 (0.233)	0.058 (0.115)	0.001 (0.084)
Policy Belief 2012	0.959 (0.033)	0.884 (0.110)	0.861 (0.078)
Policy Belief 2013	0.949 (0.050)	0.891 (0.101)	0.860 (0.101)
Policy Belief 2014	0.959 (0.063)	0.926 (0.095)	0.902 (0.118)
Annual aggregate moments	✓	✓	✓
Market aggregate moments	✓	✓	✓
Market × 2011 Moments	✓	✓	✓
Objective Value	181.54	176.52	203.65

Notes: This table presents dynamic parameter estimates with different assumptions on reinstatement probabilities. Each column allows an expired credit to be reinstated in the following year with the stated probability, in place of Assumption 1. Column (1) is column (3) of Table 2. Entry costs are in million dollars at the average land price of each market. Standard errors are block-bootstrapped over states with 100 draws.

Table SA10: Dynamic Estimates by Terminal Subsidy Level

Parameter and Notation	(1)	(2)	(3)	(4)	(5)
	0.30	0.35	0.40	0.45	0.50
<i>Panel A: Entry Cost Parameters</i>					
Entry Cost, MISO	71.85 (20.25)	69.25 (18.24)	68.78 (23.86)	64.68 (19.68)	63.31 (19.51)
Entry Cost, SPP	35.11 (21.06)	31.43 (20.07)	28.46 (20.38)	24.06 (18.60)	20.53 (16.58)
Entry Cost, PJM	82.69 (18.82)	81.59 (19.20)	83.67 (18.42)	80.86 (19.46)	81.99 (19.88)
Entry Cost, CAISO	226.39 (67.87)	219.87 (78.44)	216.75 (81.02)	208.85 (84.74)	205.01 (88.40)
Entry Cost, ERCOT	69.19 (30.82)	61.37 (30.28)	56.43 (30.97)	45.99 (29.16)	38.93 (28.49)
Land Price	19.50 (9.27)	19.55 (9.13)	20.09 (9.66)	20.06 (9.54)	20.86 (9.51)
Scale of Entry Cost Shock	12.69 (5.17)	12.31 (5.69)	13.13 (5.47)	12.06 (5.31)	12.47 (5.34)
<i>Panel B: Belief Parameters</i>					
Policy Belief 2009	0.743 (0.226)	0.781 (0.245)	0.808 (0.242)	0.840 (0.217)	0.867 (0.242)
Policy Belief 2010	0.953 (0.031)	0.959 (0.031)	0.969 (0.031)	0.974 (0.029)	0.983 (0.031)
Policy Belief 2011	0.253 (0.222)	0.361 (0.213)	0.395 (0.233)	0.499 (0.248)	0.536 (0.229)
Policy Belief 2012	0.940 (0.040)	0.949 (0.039)	0.959 (0.033)	0.964 (0.043)	0.973 (0.044)
Policy Belief 2013	0.935 (0.058)	0.944 (0.063)	0.949 (0.050)	0.957 (0.059)	0.963 (0.061)
Policy Belief 2014	0.950 (0.059)	0.957 (0.065)	0.959 (0.063)	0.968 (0.064)	0.972 (0.072)
Annual aggregate moments	✓	✓	✓	✓	✓
Market aggregate moments	✓	✓	✓	✓	✓
Market × 2011 Moments	✓	✓	✓	✓	✓
Objective Value	240.84	206.08	181.54	153.78	133.18

Notes: This table presents dynamic parameter estimates with different assumptions on terminal subsidy levels. Each column evaluates the 2019 terminal value function at a residual credit equal to the stated share of the statutory level. Column (3) is column (3) of Table 2. Entry costs are in million dollars at the average land price of each market. Standard errors are block-bootstrapped over states with 100 draws.